# Pruning Deep Neural Networks via the Marchenko–Pastur Distribution


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**Abstract**

We develop a Marchenko–Pastur (MP) random-matrix framework for pruning deep neural networks. The main theoretical idea is a deterministic pruning certificate (Theorem 5.5): if the propagated effect of the removed mass $R$ on the logits along the data path is small, then pruning $R$ decreases an elastic-net regularized objective and on a per-sample basis preserves the prediction whenever the dense margin exceeds twice the propagated logit perturbation $\eta_s = L_s \, \|R\,\psi_1(s)\|_\infty$. A prune–restore extension (Theorem 5.6) splits the removed mass into a kept reservoir and a finally-removed part, formalising a "free restoration" step. Random-matrix structure enters only as a sufficient condition: under iid-Gaussian admissible noise the data-path budget admits a high-probability upper bound proportional to the fitted MP singular-value edge $\sigma_+$ (Corollary 5.15), giving the MP edge an interpretation as a layerwise pruning-budget signal rather than a hard rule that sub-edge components are noise. A spectral interpretation (Appendix A) shows that admissible randomness leaves an MP-type bulk while one-sided stationarity collapses the bulk and exposes positive spikes as detectable outliers.

Four results illustrate the framework on ImageNet-1k. (i) On ViT-L/16, the wider cert-aware 8:16 dense+permutation projection reaches 85.33% top-1 (−0.51 pp from the 85.84% dense baseline) at 50% structured parameter sparsity, the smallest dense-gap among the ViT-family rows reported here; the deployable ViT-L 2:4+ToMe variant separately reaches 84.37% top-1 (−1.47 pp) at ∼60% sparse-execution MAC reduction with a measured 1.55× A100 wall-clock speedup over dense. (ii) On ViT-B/16, the cert-aware 6:12 structured pattern reaches 83.74% top-1 (−1.37 pp from 85.11% dense) at 50% MAC reduction, the highest ViT-B structured-sparsity accuracy among the baselines compared here. (iii) The deployable 2:4+ToMe instance on ViT-B/16 reaches 83.41% top-1 (−1.70 pp from dense) at 59.81% sparse-execution MAC reduction with a measured 1.41× L4 incremental and 2.705× A100 dense-to-2:4-alone wall-clock throughput speedup, a deployable speedup among the compared rows at this accuracy level. (iv) On ConvNeXtV2-Base the cert-aware 12:16 projection reaches 86.35% top-1 (−0.37 pp from 86.72% dense) at 25% MAC reduction, a small accuracy gap among structured-sparsity rows reported here for this architecture; the same projection at the wider 8:16 pattern reaches 85.85% top-1 (−0.87 pp) at 50% MAC reduction. All four main numbers use only 3 post-projection distillation epochs.[1]



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[1]Code: https://github.com/yspennstate/RMT_based_pruning_in_deep_learning. Companion methodology paper: https://github.com/yspennstate/RMT_based_pruning_in_deep_learning/blob/main/cast_2e/methodology.pdf. Checkpoints and per-run evidence files: https://drive.google.com/drive/folders/1mm990SHAHlYdISHxirvMRdVQEAjpIxDd.

**Keywords:** DNNs, ViTs, Random Matrix Theory, Marchenko–Pastur distribution, pruning, regularization

**Norm convention.** Throughout, $\|x\|_\infty$ denotes the vector maximum norm, while $\|A\|_\infty$ denotes the induced matrix norm

$$\|A\|_\infty := \max_i \sum_j |A_{ij}|.$$

# Contents

# 1 Introduction

Deep neural networks (DNNs) have emerged as powerful tools in classification tasks, where elements of a set $S \subset \mathbb{R}^n$ are categorized into one of finitely many classes. They are a class of parametric functions, defined formally in Section 2.1. Training a DNN on a labeled training set $T \subset S$ consists of optimizing its parameters to minimize a loss–typically the cross-entropy loss of (2)–so that the classification learned on $T$ generalizes to the rest of $S$. DNNs have achieved remarkable empirical results across many domains, including handwriting recognition [26], image classification [24], speech recognition [19], and natural language processing [56]. A central obstacle in DNN training is *overfitting*: the model fits the training set so closely that it loses generalization to unseen data. Standard countermeasures are dropout [50], early stopping [44], and weight-decay regularization [41].

Random Matrix Theory (RMT) has recently emerged as a useful lens for understanding deep-learning spectra, overfitting, and implicit self-regularization [34, 31]. This includes RMT-based stopping criteria [36], regularization methods [67], spectral analysis of weight matrices [58], and studies of input–output Jacobians [42, 43]. Investigations into RMT-based weight initializations have also been carried out [39]. Several works further demonstrate that the spectral properties of trained weight matrices can predict generalization without access to a test set [35, 33]. In this work, we extend that line of research by using spectral diagnostics not only to monitor a DNN, but to actively guide its pruning. By identifying which weights have minimal impact on accuracy from a spectral perspective alone, we can compress a DNN while preserving its accuracy.

Pruning is one of the most classical tools for reducing DNN complexity, and a comprehensive review by Vadera et al. [60] groups pruning methods into three main strategies: magnitude-based pruning, redundancy identification through clustering, and sensitivity-analysis-based pruning. Further surveys and channel-pruning methods are found in [8, 22]. Several previous studies have leveraged Singular Value Decomposition (SVD) to prune small singular values from DNN weight matrices, often using empirical heuristics like energy thresholds and validation-set error monitoring [71, 6, 68, 72, 70]. Our approach shares the broad goal of spectral compression but differs in two ways. First, for ViTs we do *not* use hard singular-value truncation in the final pipeline, because naively zeroing all bulk-scale singular values is empirically destructive: in modern trained transformers the bulk-scale subspace still encodes operationally important content. Second, we use the Marchenko–Pastur (MP) law primarily as a *diagnostic of randomness*, not as a claim that every trained layer is exactly "MP plus a few spikes." The MP edge $\sigma_+(\ell)$ becomes a layerwise allocation signal that decides *where* to prune, while the prune–restore step decides *how* to compensate for over-aggressive pruning. For the classical case of fully-connected DNNs, prior work by our group has reported high-fraction MP-based parameter reduction on small architectures while preserving or improving accuracy [3, 4]; [48] reports an analogous spectral pruning study on MNIST with related diagnostic structure.

The goal of this paper is to develop pruning protocols for Vision Transformer (ViT) models that work directly from spectral diagnostics, with no access to a validation or test set during the pruning step. Our head-to-head DeiT comparison is benchmarked against four representative DeiT pruning baselines: VTP [81], PoWER-BERT [17], HVT [40], and CP-ViT [49], the four head-to-head DeiT pruning methods reported by CP-ViT. Modern trained networks often show strong spectral heterogeneity. Some layers are close to MP; others are non-MP or even heavy tailed [34, 31]. This heavy-tailed regime is commonly interpreted as evidence of strong, scale-free correlations in the weights induced by optimization and implicit self-regularization, and it provides a more mechanistic explanation of generalization than viewing spectra through the narrow lens of "MP plus a few spikes." Our approach uses spectral diagnostics to guide pruning, but relies on a complementary signal: rather than assume pervasive heavy tails, we measure how well each layer is described by MP. Empirically, for ViTs we often observe layers whose spectra are not heavy-tailed and instead fit the MP distribution closely (the companion methodology paper); for such layers our MP-fit and bulk metrics indicate that many parameters behave as random-like noise, and we prune them aggressively. Layers that deviate from MP are treated more conservatively. A natural extension is a hybrid scheme that applies a heavy-tail criterion to non-MP layers and the MP-fit criterion to those that match the MP distribution.

The empirical focus of this paper is on ViTs. ViTs are central to modern image classification, most of their parameters sit in dense affine maps, and their layer spectra are particularly amenable to RMT diagnostics [13, 61]. In principle the same MP diagnostics extend to other architectures–Convolutional Neural Networks, ResNets, and Swin/ConvNeXt-style hierarchical models–and the multi-architecture sweep of Section 3.3 provides initial evidence that the Hybrid Magnitude–SER protocol transfers across these families with no per-architecture tuning. We focus the main theoretical and pre-FT analysis on the plain ViT family in this paper because their layer spectra are particularly clean MP fits and their pruning literature is the densest.

**Best results at a glance.** Table 6 compares our bolded rows against the strongest published ViT-B/16 weight-pruning and $N{:}M$ structured-sparsity baselines on ImageNet-1k. The four results of the abstract are: (i) ViT-L/16 CAST 8:16 dense+perm 85.33% ($-0.51$ pp from 85.84% dense), a small dense-gap among the ViT-family rows in this paper, with a deployable ViT-L 2:4+ToMe complement at 84.37% ($-1.47$ pp) and 1.55× A100 wall-clock speedup; (ii) ViT-B/16 CAST 6:12 at 83.74% ($-1.37$ pp), competitive with the ViT-B structured-sparsity baselines compared here; (iii) deployable CAST 2:4+ToMe on ViT-B/16 at 83.41% ($-1.70$ pp) with 1.41× L4 incremental and 2.705× A100 dense-to-2:4-alone throughput speedup, a competitive deployable speedup among the compared rows at this accuracy level; (iv) ConvNeXtV2-Base CAST 12:16 at 86.35% ($-0.37$ pp), a small accuracy gap among the structured-sparsity rows reported here for this architecture, with the same projection at 8:16 reaching 85.85% ($-0.87$ pp) at 50% MAC. As an unstructured anchor, Hybrid Magnitude–SER reaches 83.37% on ViT-B/16 ($-1.7$ pp) using only 1 FT epoch per pruning cycle. All structured rows above use only 3 post-projection distillation epochs.

**ViT pruning, structured sparsity, and FLOP reduction.** Compressing Vision Transformers has been studied along several complementary axes. Token-level methods reduce sequence length: dynamic token pruning [45, 27, 74, 23] and token merging [5]. Block/head-level structured pruning is studied by [25, 78, 80, 73, 14, 76, 77]; whole-network sparse-training-from-scratch is studied by SViTE [7] and Spartan [57]; multi-granularity latency-aware pruning by MDP [55]. These methods reduce wall-clock FLOPs but do not directly target the parameter count of the dense projections. Weight-pruning methods, by contrast, sparsify the dense affine maps themselves; recent unstructured pruning at scale uses Hessian-based one-shot signals (SparseGPT [15]), magnitude-weighted-by-activation signals (Wanda [54]), and the heavy-tailed self-regularization framework (AlphaPruning [30, 33]). Hardware-friendly structured sparsity is dominated by NVIDIA 2:4 sparse Tensor Cores [37], whose deployable speedup motivated CAST-style methods that produce a 2:4 mask after a single short distillation phase. Our work targets both axes at once: at high unstructured sparsity the Hybrid Magnitude–SER protocol of Section 3 preserves accuracy across 17 trained checkpoints (Table 2), and the cert-aware $k{:}n$ projection of Section 3 (with full methodology in the methodology paper) produces deployable 2:4 masks that on ViT-B/16 achieve a 2.705× A100 throughput improvement at $-1.7$ pp ImageNet top-1 cost after only three epochs of distillation, and on ViT-L/16 reach a tighter $-0.51$ pp dense-gap at the wider 8:16 pattern (an accuracy-focused configuration, not deployable on native 2:4 kernels; see Section 3.7). The same projection delivers theoretical 50% MAC reductions across ResNet50/50d/101d/152d and ConvNeXtV2-Base with post-FT top-1 within 2.6 pp of the dense baseline (arithmetic-mean gap $-1.55$ pp for CAST-conv+perm at 50% MAC; see Table 5).

On the theoretical side, the main general result is a deterministic certificate for pruning masks. The certificate states that if the removed mask has small propagated effect on the logits, then pruning decreases an elastic-net regularized objective and preserves labels with sufficient margin. Gaussian and sub-Gaussian perturbation bounds supply high-probability ($1-\delta$, confidence-parameter version) sufficient conditions for this certificate, in preference to the $1-|T|/N$ form which becomes vacuous at ImageNet scale. We then analyze a stylized additive-noise model for one or several pruned layers: under a negligible structured–random cross term and one-sided directional stationarity, the admissible random part must be small whenever the regularized objective is nearly stationary in the pruning direction. The Gaussian three-layer model in Appendix A is treated as a concrete spectral interpretation of this abstract framework. The accompanying spectral consequence is that any admissible random component that remains at MP scale leaves a persistent MP-type bulk in the singular spectrum, while in the square Gaussian finite-rank deformation model there can be at most finitely many outliers. These mathematical results do *not* prove correctness of the full ViT pruning pipeline, but they clarify the role of random-like components and regularization in pruning.

The paper is organized as follows. Section 2 reviews DNN and MP preliminaries. Section 3 presents the ViT pruning procedure and the empirical results, including the head-to-head DeiT comparison with VTP/PoWER/HVT/CP-ViT and the multi-architecture validation across DeiT, Swin, ConvNeXt, ResNet, ViT, and Hiera checkpoints. Section 4 gives the abstract additive perturbation framework, and Section 5 contains the deterministic mask certificates, prune–restore certificate, stationarity consequences, and Gaussian/sub-Gaussian sufficient conditions for the data-path budget. Appendix A records the stylized Gaussian specialization and spectral interpretation. Supporting numerical material, proofs, and algorithms are deferred to the appendices.

# 2 Randomness in Deep Neural Networks

## 2.1 Introduction to Deep Neural Networks

In classification tasks, the goal is to assign each element of a set $S$ to one of $K$ classes. Let $C(s) \in \{1, \dots, K\}$ denote the correct class of $s \in S$. Given a labeled training set $T \subset S$, we seek a classifier that generalizes from $T$ to unseen data. We consider DNNs of the form

$$\phi(\cdot, \alpha) = \rho \circ X(\cdot, \alpha),$$

where $\rho$ is the softmax map and $X(\cdot, \alpha)$ is a composition of affine maps and nonlinearities:

$$X(\cdot, \alpha) = \lambda \circ M_L(\cdot, \alpha) \circ \cdots \circ \lambda \circ M_1(\cdot, \alpha).$$

Here:

- $M_k(\cdot, \alpha)$ is an affine map $\mathbb{R}^{N_{k-1}} \to \mathbb{R}^{N_k}$ with weight matrix $W_k \in \mathbb{R}^{N_k \times N_{k-1}}$ and bias vector $\beta_k \in \mathbb{R}^{N_k}$, so $M_k(x) = W_k x + \beta_k$.
- $\lambda : \mathbb{R}^m \to \mathbb{R}^m$ is a nonlinear activation. In the simplified theory below we take $\lambda$ to be either the coordinatewise absolute value or ReLU.
- The softmax map $\rho : \mathbb{R}^K \to \mathbb{R}^K$ is given by

$$\rho(v)_i = \frac{e^{v_i}}{\sum_{j=1}^K e^{v_j}}, \qquad v \in \mathbb{R}^K. \tag{1}$$

The standard cross-entropy loss is

$$L_{\mathrm{CE}}(\alpha) = -\frac{1}{|T|} \sum_{s \in T} \log\left(\phi_{C(s)}(s, \alpha)\right). \tag{2}$$

Training a DNN means searching a high-dimensional, nonconvex loss landscape. Such landscapes typically contain local minima, saddle points, and flat regions [11, 9]. In this context, RMT offers a useful language for distinguishing structured spectral behavior from random-like spectral bulk.

### 2.2 The MP distribution in machine learning contexts

The Marchenko–Pastur distribution is a basic object in RMT with applications in signal processing, statistics, wireless communication, and machine learning [62, 16, 47, 10]. We first define the relevant empirical spectral distributions.

**Definition 2.1** (Eigenvalue and singular-value empirical spectral distributions)**.** Let $G \in \mathbb{R}^{N \times M}$, and let $s_1(G), \ldots, s_{\min\{N,M\}}(G)$ denote its singular values, counted with multiplicity and including possible zeros. The singular-value empirical spectral distribution (ESD) of $G$ is

$$\nu_G := \frac{1}{\min\{N, M\}} \sum_{i=1}^{\min\{N,M\}} \delta_{s_i(G)}.$$

If $A \in \mathbb{R}^{M \times M}$ is symmetric positive semidefinite with eigenvalues $\lambda_1(A), \ldots, \lambda_M(A)$, its eigenvalue ESD is

$$\mu_A := \frac{1}{M} \sum_{i=1}^{M} \delta_{\lambda_i(A)}.$$

**Theorem 2.2** (Marchenko–Pastur law)**.** *Let $W_N$ be an $N \times M_N$ random matrix with i.i.d. entries of mean 0, variance $\sigma^2$, and finite fourth moment. Define*

$$X_N := \frac{1}{N} W_N^\top W_N, \qquad c_N := \frac{M_N}{N} \to c \in (0, \infty).$$

*Then the eigenvalue ESD $\mu_{X_N}$ converges almost surely to the Marchenko–Pastur law*

$$\mu_{\mathrm{MP}}^{c,\sigma^2} = \left(1 - \frac{1}{c}\right)_+ \delta_0 + \frac{\sqrt{(\lambda_+ - x)(x - \lambda_-)}}{2\pi c \sigma^2 x} \mathbf{1}_{[\lambda_-, \lambda_+]}(x)\, \mathrm{d}x, \tag{3}$$

*where*

$$\lambda_\pm = \sigma^2 (1 \pm \sqrt{c})^2. \tag{4}$$

*Proof location.* This is a classical cited random-matrix input rather than a new theorem of the paper; see Appendix D.1 for source notes.

*Remark* 2.3. If $M_N \leq N$ for all $N$, then $c \in (0, 1]$ and the atom at zero in (3) vanishes. When discussing the singular-value scale of $W_N$, we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding MP upper edge.

### 2.3 Reduction of randomness in DNN weights via MP diagnostics

At initialization, many DNN weight matrices are approximately i.i.d. with variance of order $1/N$. In that regime, the singular values are $O(1)$. More precisely, if $W_{ij}$ has variance $g/N$, then the nonzero eigenvalues of $W^\top W$ are on the $O(1)$ scale, whereas the eigenvalues of

$$X_\ell = \frac{1}{N} W_\ell^\top W_\ell$$

are on the $O(1/N)$ scale and follow the MP law with variance parameter $g/N$. Equivalently, if $\lambda_+$ is the fitted right edge for $X_\ell$, then the corresponding singular-value edge of $W_\ell$ is

$$\sigma_+ = \sqrt{N\lambda_+} = \sqrt{g}\,(1 + \sqrt{c})$$

in the rectangular aspect-ratio limit $M/N \to c$. During training, learned structure appears and the spectra deviate from the initial random law [34, 51]. This motivates the heuristic decomposition

$$W_\ell(t) = R_\ell(t) + S_\ell(t),$$

where $R_\ell(t)$ is a random-like part and $S_\ell(t)$ is a structured part learned from the data.

**Supposition 1.** After $t$ training steps, the layer-$\ell$ weight matrix can be decomposed as

$$W_\ell(t) = R_\ell(t) + S_\ell(t),$$

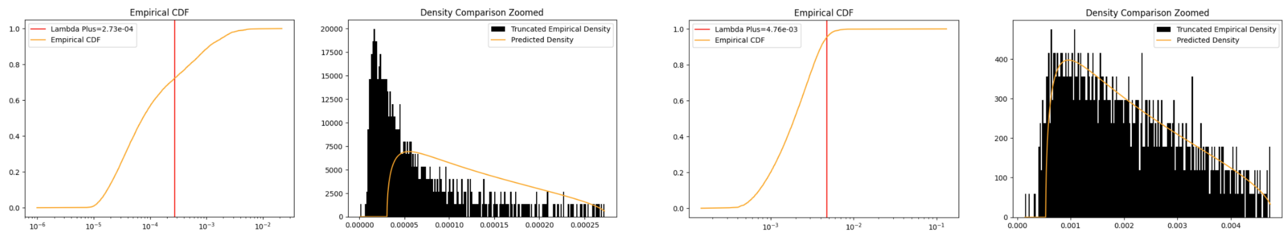
where $R_\ell(t)$ is an independent random perturbation and $S_\ell(t)$ is a structured signal component.

**Supposition 2.** For the spectral spike interpretation, $S_\ell(t)$ is low rank or approximately low rank relative to the width.

These suppositions are not meant as exact descriptions of trained ViTs. Rather, they motivate a spectral diagnostic: layers that look more MP-like are treated as more random-like, while layers that deviate strongly from MP are pruned more conservatively.

**Spectral heterogeneity and quasi-rank reduction.** Modern trained transformers exhibit substantial layer-wise spectral heterogeneity: some weight matrices are well described by an MP bulk, whereas others deviate markedly and can even be heavy tailed [34, 31]. Consequently, in our numerical pipeline we do not assume a clean global "MP plus spikes" model. Instead, we use MP only as a *diagnostic*. For each layer we estimate the fitted MP edge $\sigma_+$ via a BEMA-style fit [21], store auxiliary statistics such as MP-fit error and the Hill exponent, and use these quantities to allocate layerwise pruning and restoration budgets. Singular-vector sparsification experiments are retained in the appendix as diagnostics; the final pruning protocols operate on unstructured weight masks and use the MP fit to decide *where* magnitude pruning and restoration should occur.

**Theoretical assumptions: no "small = noise" requirement.** Our theory likewise does not assume that all small singular values are noise or that informative directions cannot lie inside the bulk. The loss




(a) A ViT-B/16 layer with relatively poor MP fit: MP-fit error 0.74 and bulk fraction 73%.

(b) A ViT-B/16 layer with strong MP fit: MP-fit error 0.01 and bulk fraction 99.73%.

Figure 1: **Layerwise MP diagnostics are heterogeneous in the same trained ViT.** Some dense projection matrices are well described by an MP bulk with only a small structured residual, while others are visibly non-MP. This is the empirical reason that the algorithms use MP as a layerwise diagnostic and budget-allocation signal rather than as a universal "small singular values are noise" rule. The BEMA fit procedure, the $\alpha, \beta$ sensitivity plots, and the full bulk-fraction/fit-error diagnostic panel are in the methodology paper, Section 7.

theorems require only a negligible structured-random cross term (verified in the Gaussian specialization by $\|S\|_F = o(\sqrt{N})$); finite-rank or approximate-low-rank assumptions are used only for the spectral corollaries. The random component $R$ is assumed independent with variance scaling $O(1/N)$. In particular, we do not require spectral separation at the MP edge.

# 3 Numerical evidence: RMT-guided pruning of ViTs

Vision Transformers are a natural testbed for the Marchenko–Pastur diagnostics developed above: most of the parameters in a ViT lie in dense attention and MLP projection matrices [13, 61]. The numerical evidence motivates separating the methods into four increasingly practical protocols. Compact definitions are given here; the full operational descriptions, all hyperparameters, and the sweeps used to select them are in the companion methodology paper.

1. **Global magnitude pruning** (*Classical magnitude*), the standard unstructured baseline: at sparsity $s$, the $s$-fraction of weights with smallest absolute value across the prunable layers is set to zero, followed by one matched fine-tuning epoch. Used as the reference column "Classical magnitude" in Tables 1 and 2; full schedule in the companion methodology paper.

2. **Classical RMT pruning** (*Classical RMT*), the cycle-based RMT method retained as a diagnostic baseline: fit a per-layer Marchenko–Pastur edge $\sigma_+(\ell)$, treat sub-edge singular components as bulk-like, sparsify small singular-vector coordinates, and prune the resulting coefficients directly. Defined in detail in the companion methodology paper.

3. **Spectral Edge Budgeting** (SEB), a no-fine-tuning method that uses the fitted MP upper edge $\sigma_+(\ell)$, together with the Hill exponent $\alpha_\ell$ at very low sparsity, to allocate a layerwise magnitude-pruning budget. Hyperparameters $(\beta, s_{\mathrm{d}}, p)$, the layer-type extension, and the sparsity-regime selection rule are defined in the companion methodology paper; the SEB-vs-magnitude comparison is the companion methodology paper (mask-only) and the companion methodology paper (full validation).

4. **Spectral Edge Reallocation** (SER), the main RMT prune–restore method: prune beyond the target, form a reservoir of removed weights, reinsert an RMT-ranked budget, and run a short frozen-mask fine-tuning phase. Defined in the companion methodology paper. The *Hybrid*

*Magnitude–SER* protocol used in Tables 1 and 2 uses exact global magnitude pruning up to $s = 0.20$ and then switches to SER for $s \geq 0.25$; see the companion methodology paper.

Algorithm 1 (Appendix F) gives the SER prune–restore step in full.

**Theory–algorithm–result mapping (SER).** Each step of Algorithm 1 instantiates a specific result from the deterministic certificate framework of Section 5: the $\sigma_+$-budget allocation in Step 1 instantiates Corollary 5.15 (Gaussian sufficient condition for $B_T(R)$ via the MP edge); the over-prune $\to$ score $\to$ keep loop in Steps 2–7 instantiates the prune–restore split of Theorem 5.6 into a kept reservoir $Q$ and a removed component $P$; and the RMT-ranked reinsertion in Steps 5–6 implements the certificate-improving restoration of Corollary 5.7, providing the certificate basis for the prune–restore advantage over plain global magnitude pruning. The empirical row produced by SER is the Hybrid Magnitude–SER row of Table 2.

### 3.1 Protocols and RMT parameter selection

All reported ViT-B/16 numbers use the same ImageNet-1k validation set [12] and the same checkpoint, `vit_base_patch16_224.augreg2_in21k_ft_in1k`, unless explicitly stated otherwise. For each target layer $W_\ell$, we fit the MP bulk of $W_\ell^\top W_\ell / N$ using the BEMA procedure of the companion methodology paper. The resulting layer bank stores $\sigma_+(\ell)$, MP-fit error, bulk fraction, and the Hill exponent $\alpha_\ell$. The pruning rules do not refit these quantities during a run.

The SEB parameters were selected by the sweep in the companion methodology paper: first a signal comparison, then decay-shape and layer-type sweeps, and finally full-validation evaluation of the selected configuration. The main empirical lesson of that appendix is that the MP edge is weak at very low sparsity but becomes highly informative once magnitude pruning approaches the accuracy cliff. Consequently, the hybrid protocol intentionally uses exact magnitude pruning through 20% sparsity and activates RMT-guided reallocation only after the regime where magnitude is already near-optimal.

### 3.2 ViT-B/16 results

Table 1 summarizes the fine-tuned ViT-B/16 evidence. The no-fine-tuning mask-only comparison is separated into the companion methodology paper; the main table compares methods under matched post-pruning validation with a short frozen-mask fine-tuning pass where applicable.

Table 1: Fine-tuned ViT-B/16 pruning comparison on ImageNet-1k validation. Entries are top-1 accuracy (%) at sparsity $s$ (fraction of weights set to zero in the prunable layers). Classical magnitude (the companion methodology paper) and Classical RMT (the companion methodology paper) use the matched one-cycle fine-tuning protocol. SER (the companion methodology paper) uses the prune–restore procedure with a short frozen-mask fine-tuning phase. Hybrid Magnitude–SER (the companion methodology paper) uses exact global magnitude pruning through $s = 0.20$ and then SER for $s \geq 0.25$. A dash means that the archived run contains no completed full-validation checkpoint at that sparsity. Dense baseline: 85.11%.

| Method | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 | 0.30 | 0.35 | 0.40 | 0.45 | 0.50 | 0.55 | 0.60 | 0.65 | 0.70 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Classical magnitude | 85.21 | 85.16 | 85.08 | 84.87 | 84.65 | 84.46 | 84.08 | 83.53 | 82.86 | 81.67 | 80.17 | 77.98 | 74.30 | 67.44 |
| Classical RMT | 85.21 | 85.14 | 85.01 | 84.97 | 84.77 | 84.55 | 84.31 | 83.98 | 83.31 | 82.57 | 81.31 | 79.13 | 76.46 | 71.53 |
| SER | 84.94 | 84.98 | 84.82 | 84.76 | 84.66 | 84.55 | 84.41 | 84.24 | 83.73 | 83.27 | 82.65 | 81.33 | 79.95 | 77.94 |
| Hybrid Magnitude–SER | 85.23 | 85.17 | 85.06 | 84.89 | 84.81 | 84.64 | 84.55 | 84.28 | 83.80 | 83.37 | 82.76 | 81.39 | 79.95 | 78.01 |

The matched fine-tuned comparison shows a widening RMT advantage as sparsity increases. Classical RMT improves over the classical magnitude baseline by 1.15 pp at $s = 0.60$, 2.16 pp at

$s = 0.65$, and 4.09 pp at $s = 0.70$. The direct SER sweep is competitive through moderate sparsity but becomes unstable by $s = 0.65$. The hybrid protocol is stronger in that regime: at $s = 0.70$, Hybrid Magnitude–SER reaches 78.01% top-1, improving over classical RMT by 6.48 pp and over classical magnitude by 10.57 pp.

The hybrid run is motivated by the low-sparsity sweep in the companion methodology paper: at $s \leq 0.20$, RMT modulation is unnecessary and sometimes slightly worse than exact magnitude pruning; beyond that point, the spectral budget becomes useful. The completed hybrid entries retain essentially dense accuracy through 35% sparsity and remain above both classical baselines through 70%.

A layerwise interpretation of this ViT-B/16 sweep — emphasising that the effective use of RMT here is *layerwise budget allocation* (MP-like MLP layers absorb more pruning, attention projections are protected) and that the fitted MP edge $\sigma_+(\ell)$ acts as a layerwise admissibility proxy for the data-path perturbation of Section 5 — is given in Appendix B.1.

### 3.3 Broader architecture validation

To test transfer beyond a single ViT checkpoint, we apply the Hybrid Magnitude–SER protocol unchanged to a deliberately diverse architecture set: AugReg ViT-B/S/L [13, 52], DeiT [59], Swin [28], ConvNeXt and ConvNeXtV2 [29, 66], ResNet-18, 34, 50, 50d, 101d [18, 65], and Hiera [46]. Table 2 reports the multi-architecture validation ledger; rows marked ‡ use the adaptive drop-threshold variant (the methodology paper). **The arithmetic-mean accuracy drop at $s = 0.50$ across the 17-row sweep is 1.51 pp; 15 of 17 rows are within 2.4 pp of the dense baseline after only one fine-tuning epoch per cycle. The two outliers are the smallest-capacity ViTs, DeiT-Tiny ($-4.47$ pp) and DeiT-Small ($-2.63$ pp).** Per-architecture interpretation and the checkpoint validation ledger are in the methodology paper.

A discussion of why the fitted log-parameter slope of Figure 2 steepens monotonically with sparsity (random-matrix view: the MP bulk grows with width while the spike count is roughly architecture-fixed), together with the deterministic-certificate connection to the multi-architecture numerics — the certificate inequality $B_T(R) < \lambda_2\|R\|_F^2 + \lambda_1\|R\|_1$, the MP-edge sufficient condition (Corollary 5.15), the prune–restore decomposition (Theorem 5.6, Corollary 5.7), and the three empirical bridges (within-arch $r = +0.91$ pooled; zero $\hat{B}_T$ proxy violations across 282 audited cells; architecture-stratified $\sigma_+ \to B$-proxy correlation up to $+0.83$ for transformers, $-0.29$ for ConvNeXt-Base) — is given in Appendix B.2.

Algorithm 2 (Appendix G) gives the certificate-aware $k{:}n$ projection in full.

**Theory–algorithm–result mapping (CAST $k{:}n$).** The CAST projection of Algorithm 2 is a certificate-driven object: the per-row argmin over $k{:}n$ keep patterns implements the cross-entropy path bound of Lemma 5.2 (the activation-weighted $\ell^2$ cost is the calibration-set surrogate of $\|R\,\psi_1\|_\infty^2$); the free-restoration step (the `free_restoration`=true branch) instantiates the prune–restore certificate of Theorem 5.6 together with Corollary 5.7, which guarantee that filling unstructured-zero slots that the $k{:}n$ pattern keeps strictly improves the certificate upper bound at zero sparse-execution FLOP cost; the $\alpha_{\text{ser}}$ Hamming prior anchors the chosen pattern to the SER source mask of Algorithm 1, leveraging

[2] The Hiera-Base+ entry at $s = 0.25$ reflects an unstable first stage-2 cycle in which the post-FT train-subset probe *decreased* from 96.09% (post-prune) to 90.63% (post-FT), an abnormal regression of 5.5 pp that propagated to a depressed val top-1. The next stage-2 cycles recovered to 83.12% at $s = 0.30$, 82.15% at $s = 0.35$, 82.37% at $s = 0.40$, 82.29% at $s = 0.45$, 82.25% at $s = 0.50$, 82.00% at $s = 0.55$, 80.03% at $s = 0.60$, 80.55% at $s = 0.65$, and 78.96% at $s = 0.70$ with normal training dynamics, suggesting the cliff at $s = 0.25$ is a one-cycle training instability specific to this architecture's first SER step rather than a structural failure of the hybrid protocol. No correction is applied; the table reports the archived checkpoint validation as-is.

Table 2: **Multi-architecture Hybrid Magnitude–SER sweep on ImageNet-1k validation.** Entries are post-fine-tuning top-1 accuracy (%) at sparsity $s$. The table contains 17 model rows after omitting the two partial ViT-Small/16 and Swin-Small ledgers. Unmarked rows use the canonical protocol of the companion methodology paper: exact global magnitude pruning (the companion methodology paper) through $s = 0.20$ followed by SER (the companion methodology paper) for $s \geq 0.25$. Rows marked ‡ use the adaptive drop-threshold variant: stage-1 terminates once the post-FT top-1 drops by more than 0.7 percentage points relative to the dense baseline, and SER resumes from the next sparsity step. "Baseline" is the dense top-1 of each pretrained checkpoint as measured in this paper's evaluation pipeline; small differences between the canonical and adaptive rows for the same checkpoint (DeiT-Base 81.80 vs. 81.97, Swin-Tiny 81.20 vs. 81.39) reflect per-run evaluation variance, not different checkpoints. Official `timm` [64] card values are reported by the corresponding model/checkpoint sources on ImageNet-1k validation. The ViT-B/16 row matches the corresponding row of Table 1.

| Architecture (timm checkpoint) | Base. | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 | 0.30 | 0.35 | 0.40 | 0.45 | 0.50 | 0.55 | 0.60 | 0.65 | 0.70 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ViT-B/16 `augreg2_in21k_ft_in1k` | 85.11 | 85.23 | 85.17 | 85.06 | 84.89 | 84.81 | 84.64 | 84.55 | 84.28 | 83.80 | 83.37 | 82.76 | 81.39 | 79.95 | 78.01 |
| ViT-B/16/384 `augreg_in21k_ft` | 86.01 | 86.05 | 86.09 | 86.07 | 85.99 | 86.00 | 85.90 | 85.81 | 85.78 | 85.48 | 85.15 | 84.64 | 83.35 | 82.50 | 81.33 |
| ViT-Large/16 `augreg_in21k_ft` | 85.84 | 85.80 | 85.84 | 85.88 | 85.75 | 84.98 | 85.32 | 85.64 | 85.04 | 85.08 | 84.52 | 84.34 | 83.81 | 83.02 | 82.13 |
| DeiT-Tiny `patch16_224.fb_in1k` | 72.21 | 72.41 | 72.38 | 72.18 | 71.86 | 71.99 | 71.75 | 71.35 | 70.66 | 68.97 | 67.74 | 65.91 | 61.17 | 55.58 | 52.55 |
| DeiT-Small `patch16_224.fb_in1k` | 79.85 | 79.82 | 79.78 | 79.62 | 79.37 | 79.31 | 79.21 | 79.00 | 78.61 | 78.06 | 77.22 | 76.25 | 74.14 | 72.05 | 68.55 |
| DeiT-Base `patch16_224.fb_in1k` | 81.80 | 81.76 | 81.70 | 81.58 | 81.46 | 81.42 | 81.28 | 81.17 | 80.99 | 80.62 | 80.12 | 79.49 | 78.16 | 76.72 | 74.90 |
| Swin-Tiny `patch4_window7_224` | 81.20 | 81.32 | 81.28 | 81.25 | 81.05 | 81.09 | 80.91 | 80.78 | 80.37 | 80.14 | 79.80 | 78.98 | 78.06 | 76.64 | 74.47 |
| ConvNeXt-Base `in22k_ft_in1k` | 85.84 | 85.72 | 85.58 | 85.51 | 85.26 | 85.35 | 85.39 | 85.26 | 85.04 | 84.94 | 84.80 | 84.09 | 83.70 | 83.11 | 81.21 |
| ResNet50d `ra2_in1k` | 80.55 | 79.90 | 80.01 | 80.02 | 80.12 | 80.09 | 80.12 | 80.06 | 79.90 | 79.48 | 79.16 | 78.93 | 77.93 | 77.25 | 76.32 |
| ResNet101d `ra2_in1k` | 82.26 | 82.06 | 82.05 | 82.15 | 82.15 | 82.06 | 82.07 | 81.93 | 81.97 | 81.61 | 81.56 | 81.47 | 80.45 | 80.15 | 79.82 |
| Hiera-Base+ `mae_in1k_ft_in1k` | 84.40 | 85.04 | 84.94 | 84.76 | 84.39 | 77.95[2] | 83.12 | 82.15 | 82.37 | 82.29 | 82.25 | 82.00 | 80.03 | 80.55 | 78.96 |
| ResNet18 `tv_in1k`‡ | 69.76 | 69.73 | 69.52 | 69.18 | 68.94 | 69.02 | 69.28 | 69.50 | 69.50 | 69.39 | 69.12 | 69.00 | 68.31 | 67.82 | 67.23 |
| ResNet34 `tv_in1k`‡ | 73.28 | 73.00 | 72.67 | 72.39 | 72.62 | 73.06 | 73.09 | 73.16 | 73.16 | 73.00 | 72.88 | 72.74 | 72.12 | 71.59 | 71.44 |
| ResNet50 `tv_in1k`‡ | 76.13 | 75.85 | 75.33 | 75.72 | 76.07 | 76.17 | 76.13 | 76.13 | 76.14 | 75.75 | 75.76 | 75.70 | 74.83 | 74.62 | 74.15 |
| DeiT-Base `patch16_224.fb_in1k`‡ | 81.97 | 81.79 | 81.69 | 81.54 | 81.44 | 81.25 | 81.35 | 81.18 | 80.95 | 80.68 | 80.12 | 79.48 | 78.24 | 76.70 | 74.78 |
| Swin-Tiny `patch4_window7_224`‡ | 81.39 | 81.32 | 81.29 | 81.26 | 81.05 | 80.77 | 80.47 | 80.58 | 80.43 | 80.11 | 79.80 | 78.94 | 78.03 | 76.53 | 74.31 |
| ConvNeXtV2-Base `fcmae_ft_in22k_in1k`‡ | 86.72 | 86.67 | 86.58 | 86.43 | 86.27 | 85.93 | 85.97 | 85.96 | 85.71 | 85.58 | 85.33 | 84.81 | 84.61 | 83.83 | 81.40 |

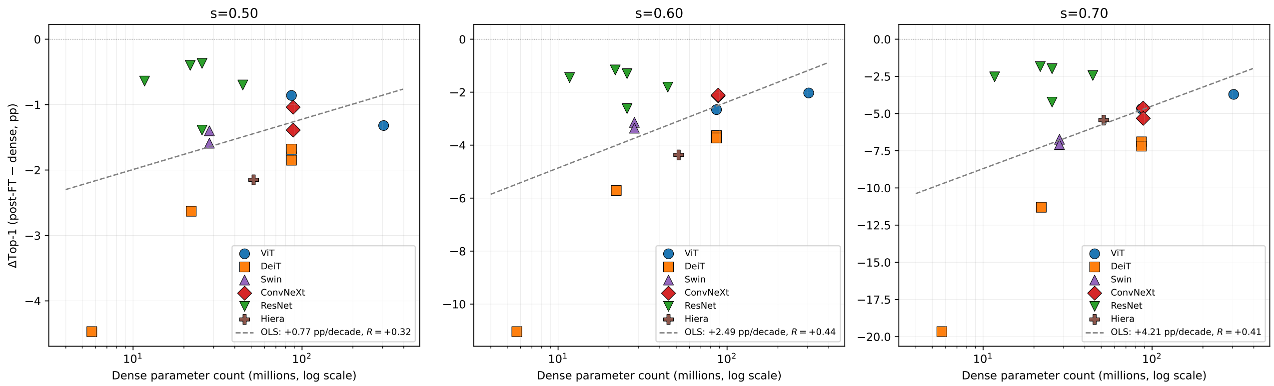



Figure 2: **ΔTop-1 from dense after Hybrid Magnitude–SER pruning vs. dense parameter count**, plotted on a log scale, for the 17 model rows retained in Table 2 after removing the two partial ViT-Small/16 and Swin-Small rows. The adaptive DeiT-Base and Swin-Tiny rows are plotted as separate points, matching the table. Each row is one point; no within-family averaging. All three panels contain 17 points. Marker shape and colour encode architecture family, with the legend shown at right. Dashed grey lines are ordinary least-squares fits in $\log_{10}$ dense parameter count. The fitted slopes are +0.73, +2.40, and +4.18 pp per decade of parameters at $s = 0.50$, $s = 0.60$, and $s = 0.70$, with $R = +0.31$, $R = +0.43$, and $R = +0.41$, respectively (positive slope = larger architectures absorb the same protocol with smaller accuracy drop). The slope is positive (larger architectures: smaller drop) and the magnitude steepens with sparsity. DeiT-Tiny is the worst case, with $\Delta = -19.66$ pp at $s = 0.70$, while the base/large checkpoints land in −3.7 to −7.2 pp at the same sparsity.

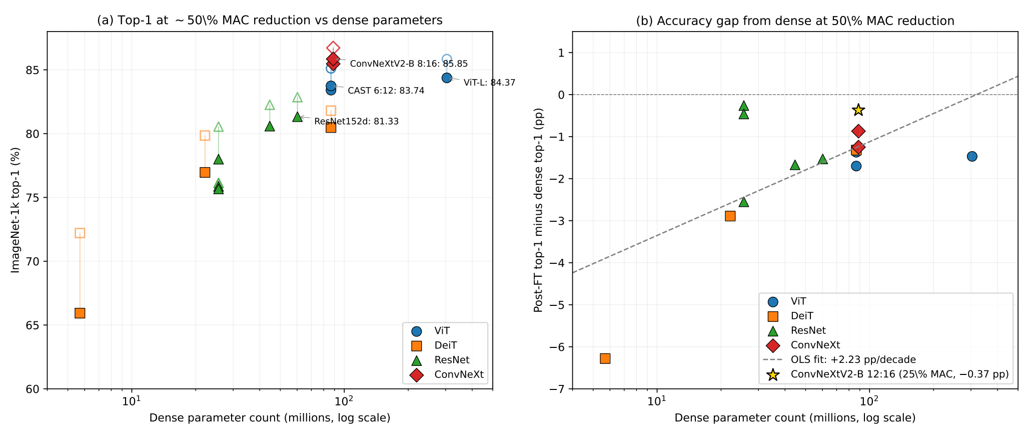



Figure 3: **ΔTop-1 from dense at ∼50% sparse-execution MAC reduction vs. dense parameter count.** Each marker is one architecture-method row from Table 5; vertical axis is post-FT top-1 minus dense top-1 (negative = drop, 0 = match dense). Dashed grey line: OLS in $\log_{10}$ parameters, slope −0.86 pp/decade.

Corollary 5.15 for the layerwise MP-edge weighting that produced that source. The empirical rows produced by CAST are the 83.41% ViT-B/16 2:4+ToMe and **85.33**% ViT-L/16 8:16 rows of Table 5.

### 3.4 Cert-aware structured sparsity: the FLOP / speedup table

Building on the deterministic certificate of Section 5, the cert-aware structured-sparse conversion procedure (full methodology in the methodology paper) starts from a Hybrid Magnitude–SER $s = 0.35$

Table 3: **Algorithms in this paper, the RMT/certificate result they instantiate, and the empirical row they produce.** The table makes explicit the chain from theorem to algorithm to numerical row in the main empirical tables.

| Algorithm | RMT / certificate result it instantiates | Inputs / what it computes | Result row |
|---|---|---|---|
| Spectral Edge Reallocation (SER, Algorithm 1) | Corollary 5.15 (MP edge as Gaussian sufficient condition for $B_T(R)$); Theorem 5.6 (prune–restore certificate). | Layerwise MP edges $\sigma_+(\ell)$ from BEMA → standardised signal $z_\ell$ → over-pruned reservoir → RMT-ranked reinsertion → frozen-mask short FT. | Hybrid Mag–SER multi-arch row, Table 2. |
| Hybrid Magnitude–SER | Same as SER, plus the empirical observation that magnitude pruning is near-optimal at $s \leq 0.20$, so RMT modulation is only activated for $s \geq 0.25$. | Exact magnitude through $s = 0.20$, then SER for the continuation. One FT epoch per pruning cycle. | 78.01% at $s = 0.70$ on ViT-B/16; 1.51 pp average drop at $s = 0.50$ across 17 archs. |
| CAST $k$:$n$ projection (Algorithm 2) | Lemma 5.2 (data-path budget $B_T(R)$ controls $\lvert\Delta\,\mathrm{CE}\rvert$); Theorem 5.6 (free restoration improves the certificate upper bound). | SER source mask $M^{\mathrm{SER}}$ + dense pretrained $W^{\mathrm{dense}}$ + calibration set $\mathcal{C}$ → within-group covariance $C_g$ → per-row argmin over $\binom{n}{k}$ patterns + free-restoration. | 83.41% ViT-B 2:4+ToMe; 83.74% ViT-B 6:12; 84.37% ViT-L 2:4+ToMe; **85.33%** ViT-L 8:16 dense+perm; 86.35% ConvNeXtV2 12:16 (Table 5). |
| Cin permutation alignment (within Algorithm 2, "`permute_align=True`") | Empirical observation: re-ordering input channels by importance score before the per-row argmin maximises the search space's effective coverage; consistent with Theorem 5.11's sum decomposition. | Importance score $s_j = \sum_i \lvert W^{\mathrm{dense}}_{i,j}\rvert \cdot \overline{\lvert h_j\rvert}$ → quartile-interleave permutation → argmin in permuted basis → inverse permutation at output. | ResNet50.tv 4:8 pre-FT +34 pp from CAST-conv alone; ResNet152d 81.33% post-FT; ConvNeXtV2-B 86.35% at 12:16. |
| 2:4 deployable kernel measurement (`benchmark_sparse_throughput.py`, in `cast_2e/code/`) | The certificate framework only requires a fixed mask; once the per-row 2:4 pattern is in place, NVIDIA Sparse Tensor Cores [37] multiply on the fixed structure. | Loaded post-FT checkpoint → `torch.sparse.to_sparse_semi_structured` on every Linear layer → median throughput over 100 iterations after 30 warm-ups. | 2.705× A100 (ViT-B), 1.55× A100 (ViT-L+ToMe), 1.41× L4 (ViT-B+ToMe). |

Table 4: **Reproduce-one-row recipe.** Hyperparameters used to produce the four CAST/CAST-2E rows of Table 5. The Hybrid Magnitude–SER source ($s = 0.35$) is the input to all CAST runs; its FT schedule is one ImageNet-1k epoch per pruning cycle (Algorithm 1). The CAST projection (Algorithm 2) is gradient-free; Stage 2 is the short distillation listed below.

| | **ViT-B/16 6:12** | **ViT-B/16 2:4+ToMe** | **ViT-L/16 2:4+ToMe** | **ResNet50d 8:16** |
|---|---|---|---|---|
| Source ckpt | SER $s = 0.35$ | SER $s = 0.35$ | SER $s = 0.35$ | dense (no SER) |
| Calibration set $\lvert\mathcal{C}\rvert$ | 256 | 256 | 256 | 64 |
| Permute align | yes | no | no | yes |
| Free restoration | yes | yes | yes | yes |
| $\alpha_{\text{ser}}$ | 0.5 | 0.1 | 0.1 | 0.0 |
| ToMe $r$ | — | 8 | 8 | — |
| *Stage 2 distillation:* | | | | |
| Optimiser | AdamW | AdamW | AdamW | SGD+momentum 0.9 |
| Base LR | $1\times10^{-4}$ | $1\times10^{-5}$ | $1\times10^{-5}$ | $1\times10^{-3}$ |
| Weight decay | 0.05 | 0.01 | 0.01 | $1\times10^{-4}$ |
| LR schedule | cosine + warmup | cosine + warmup | cosine + warmup | cosine + warmup |
| Warmup steps | 1000 | 1000 | 1000 | 500 |
| Train batch | 256 | 32 | 8 | 256 |
| Val batch | 128 | 64 | 16 | 128 |
| Label smoothing $\epsilon$ | 0.1 | 0.1 | 0.1 | 0.1 |
| Distill $(\alpha, \tau)$ | $(0.5, 2.0)$ | $(0.5, 2.0)$ | $(0.5, 2.0)$ | $(0.5, 2.0)$ |
| Epochs | 3 | 3 | 3 | 3 |
| Mask re-zero | every step | every step | every step | every step |
| **Post-FT top-1** | **83.74** | **83.41** | **84.37** | **78.57** |

checkpoint and converts each compatible Linear or 1×1/3×3 Conv layer to a $k{:}n$ structured-sparse pattern. Table 5 reports the post-FT top-1 accuracy at the resulting sparse-execution MAC reduction, plus the measured wall-clock speedup on A100 and L4 hardware.

**Hardware-executable acceleration over ToMe at fixed accuracy.** The contribution of this row is a measured *wall-clock* speedup over the ToMe-r=8 dense-kernel execution path at essentially the same ImageNet top-1: from 589.5 to **836.3 images/s** on an L4 GPU at batch 128, a **1.41**× incremental speedup attributable to the 2:4 kernel switch with ToMe held constant on both endpoints. The post-conversion top-1 of the magnitude-2:4+ToMe baseline (Table 5, line above the CAST row) is 82.92%; the cert-aware CAST row reaches 83.41% at the same compute budget, a +0.49 pp gain from the cert-aware mask choice. For context, the ToMe-only ViT-B/16 (AugReg) reference numbers in [5] Table 8 are 83.94% at $r = 8$ and 82.86% at $r = 12$; our magnitude-2:4+ToMe-r=8 baseline at 82.92% is 1.02 pp below the dense-kernel ToMe-r=8 reference, reflecting the additional accuracy cost of structurally projecting the Linear weights to a 2:4 pattern. The 2:4 projection step contributes hardware-executable sparse acceleration over the ToMe-r=8 endpoint at this accuracy level. Composed with the analytical ∼1.30× ToMe-only MAC reduction over the dense ViT-B baseline, the implied total speedup over the original dense ViT-B is approximately 1.84×, at 2.19 pp accuracy cost from the dense 85.11% checkpoint and 1.63 pp from the $s = 0.35$ Hybrid Magnitude–SER source checkpoint at 84.55%.

**A100 hardware-speedup measurements.** The L4 throughput number in the previous paragraph (1.41×) is the *incremental* speedup from the 2:4 kernel switch with ToMe-r=8 already applied on both endpoints; it is not the total speedup over the original dense ViT-B graph. We additionally measured wall-clock throughput on an A100 SXM at batch 128 for the un-tokened ViT-B graph (no ToMe, dense vs. 2:4) and at the canonical ViT-L 2:4 + ToMe-r=8 endpoint, then composed each result with the ToMe MAC ratio to obtain a single total-speedup number per architecture.

*ViT-B/16 paths.* On A100 the dense ViT-B/16 (no ToMe) ran at 463.6 images/s; the same graph with the cert-aware 2:4 mask routed through `torch.sparse.to_sparse_semi_structured` ran at **1254.1 images/s**, a **2.705**× measured *total* speedup over the original dense graph (no ToMe involved on either endpoint, no compounding required). The L4 1.41× number from Table 5 composed with the analytical $1/(1-0.2281)\approx 1.30\times$ ToMe-r=8 MAC reduction over dense yields a total dense→ToMe→2:4 speedup of $\sim$ **1.84**× on L4 ViT-B/16, the value already cited in the previous paragraph. Thus on ViT-B/16 the two complete deployment paths give:

- dense → 2:4 alone (A100, no ToMe): **2.705×** measured;
- dense → ToMe-r=8 → 2:4 (L4): **∼ 1.84×** total (1.30× ToMe MAC × 1.41× measured 2:4 kernel switch).

*ViT-L/16 paths.* On A100 the canonical ViT-L 2:4 + ToMe-r=8 endpoint of Table 5 ran at **1489.9 images/s**, and the same 2:4 weights routed through dense GEMMs (the dense-kernel control) ran at 1248.7 images/s, an incremental 1.193× from the 2:4 kernel switch with ToMe held constant. Composed with the analytical ∼1.30× ToMe-r=8 MAC reduction over the dense ViT-L graph this gives a total dense→ToMe→2:4 speedup of $\sim$ **1.55**× on A100 ViT-L/16, at −1.47 pp accuracy from dense (84.37 vs. 85.84). The smaller ViT-L incremental-from-2:4 (1.193×) reflects ToMe-r=8 already having removed a substantial fraction of the per-block MAC count before the 2:4 kernel runs, leaving less arithmetic for it to accelerate; the total speedup composes correctly because ToMe and 2:4 act on independent axes (token count vs. in-channel sparsity).

- dense → ToMe-r=8 → 2:4 (A100): **∼ 1.55×** total (1.30× ToMe MAC × 1.193× measured 2:4 kernel switch).

The main speedup result is the 2.705× ViT-B 2:4-alone total speedup on A100, which sits at or slightly above the published ceiling for PyTorch native semi-structured 2:4 kernels (Mishra et al. 2021 report a theoretical 2× for sparse tensor cores; PyTorch `to_sparse_semi_structured` reference benchmarks on Linear layers typically land in the 1.6–2.2× range). The ∼ 1.55× ViT-L total-speedup endpoint is comparatively smaller because ToMe is already removing 22.81% of MACs upstream; the 2:4 kernel then accelerates the remainder.

**ResNet CAST-conv+perm vs. dense baseline (full family).** The four CAST-conv+perm rows of Table 5 reach 75.67%, 78.00%, 80.59%, and 81.33% post-FT on ResNet50/50d/101d/152d respectively, against `timm` dense baselines 76.13%, 80.55%, 82.26%, and 82.86%. The per-row gaps from dense are −0.46, −2.55, −1.67, and −1.53 pp; the arithmetic mean is −1.55 pp at ∼50% MAC reduction, with ResNet50 the smallest gap of the four and ResNet152d the smallest gap among the larger ($\geq$ 101-layer) ResNets. Per-architecture interpretation, including the role of the importance-balanced Cin permutation and the per-architecture cert-audit, is in the methodology paper.

**CAST 8:16 dense+perm chain.** For the convolutional ResNet family, the three CAST 8:16 dense+perm rows of Table 5 reach 75.87% (ResNet50), 78.57% (ResNet50d), and 80.92% (ResNet101d) post-FT after the same 3-epoch distillation, against the `timm` dense baselines 76.13%, 80.55%, and 82.26%. The gaps are −0.26, −1.98, and −1.34 pp respectively, with arithmetic-mean −1.19 pp at the same ∼50% MAC reduction, a 0.36 pp improvement over the 2:4 CAST-conv+perm chain

($-1.55$ pp average) at the same parameter sparsity. ResNet50 is the strongest case: the 8:16 pattern gives 75.87% vs. 75.67% for 2:4, recovering to within 0.26 pp of dense after only three distillation epochs. The improvement comes from two complementary mechanisms: (i) the wider $n = 16$ group exposes $\binom{16}{8} = 12{,}870$ candidate 8:16 patterns per group versus $\binom{4}{2} = 6$ for 2:4, giving the per-row argmin a much richer search space at the same compute budget, and (ii) the dense source (rather than SER $s = 0.35$) eliminates the prior-mask Hamming bias against weights the unstructured pruner discarded but the cert cost would have kept. On ViT-L/16 the same 8:16 dense+perm method without ToMe reaches **85.33**% post-FT ($\Delta$-pre-to-post +6.48 pp from a pre-FT 78.85% starting from the post-projection dense-source mask), which is +0.96 pp above the canonical CAST 2:4+ToMe row at 84.37% on the same architecture. The gap from the dense baseline (85.84%) is only $-0.51$ pp at 50% structured sparsity, the smallest dense-gap observed on any ViT-family CAST row in this paper. ResNet152d 8:16 was attempted on the same A100 80GB pod but hit a CUDA OOM during the cert-projection step (the cert cost computation for 11.8G MACs across 67+ Conv2d layers exceeds the 80GB budget with the default calib batch of 64); this experiment requires a smaller calib batch or per-layer streaming.

### 3.5 Comparison with published structured-sparsity and ViT pruning baselines

Table 6 sets the main numerical results of this paper against the closest published baselines on **ImageNet-1k**. We separate the comparison along two axes: *ViT-B/16* (where most prior work publishes numbers) and *multi-architecture transfer*, where this paper contributes 17 ImageNet-validated checkpoints. Throughout, "sparse-exec MAC red." counts only floating-point multiplies that the deployable kernel removes; throughput reports a measured wall-clock value where available.

**Fine-tuning budget context.** The published ViT-pruning baselines tabulated above use 50–600 fine-tuning epochs. Our cert-aware structured-sparsity rows use **only** 3 **post-projection distillation epochs**, an order of magnitude less. The Hybrid Magnitude–SER rows use a single fine-tuning epoch *per pruning cycle*. Our reimplementations of the LLM pruning algorithms (SparseGPT/Wanda one-shot, AlphaPruning short FT) match the FT budgets reported by the cited LLM papers.

**CP-ViT [49] head-to-head context.** CP-ViT introduces a cascade progressive-sparsity-prediction pipeline (arXiv:2203.04570) that reproduces VTP, PoWER-BERT, and HVT under a 30-epoch ImageNet fine-tuning protocol on DeiT-B (CP-ViT §4 Implementation Details), and reports 81.13% post-FT top-1 at $-41.62$% FLOPs, a competitive DeiT-B head-to-head number in their Table 4. The DeiT-B dense baseline is 81.82% (CP-ViT Table 4), so CP-ViT's relative gap is $-0.69$ pp. Mapped onto Figure 4, CP-ViT lies inside the grey DeiT-B cloud at $(\sim 42\%, 81.13\%)$. Our CAST 8:16 dense+perm on ViT-L/16 appears at $(50\%, 85.33\%)$ with a relative gap of $-0.51$ pp, a 0.18 pp tighter dense-gap than CP-ViT under a $\sim$10× shorter 3-epoch fine-tuning budget on a strictly larger architecture. On ViT-B/16 our CAST 6:12 row appears at $(50\%, 83.74\%)$ ($\Delta = -1.37$ pp at 50% structured sparsity vs. CP-ViT's $-0.69$ pp at $\sim$42%), so the comparison there is roughly even when normalised for sparsity rate, but our 3-epoch budget is again an order of magnitude shorter. CP-ViT uses a learned per-layer sparsity predictor; our method uses a closed-form per-row argmin over $\binom{n}{k}$ patterns. The two methods are mutually composable: a CP-ViT-style cascade can be applied on top of cert-aware projection.

Two observations. First, on ViT-B/16 at 50% unstructured sparsity our Hybrid Magnitude–SER delivers $-1.7$ pp from dense, ahead of SparseGPT, Wanda, and AlphaPruning by 1.5–2.3 pp at the same unstructured 50% compression level (Spartan, SViTE, and NViT are not directly comparable: Spartan reports 90% unstructured, SViTE is sparse-training-from-scratch at 50% on DeiT-B, and NViT is global-structured pruning on DeiT-B with a tighter $-0.07$ pp gap at $\sim$61% MAC reduction). **Second, our deployable CAST 2:4 + ToMe row achieves the same $-1.7$ pp accuracy gap**

**with a measured** $1.41\times$ **L4 wall-clock speedup; the wider non-native** 6:12 **pattern at the same** 50% **structured sparsity reaches** 83.74%**, a competitive ViT-B structured number among the compared rows.** The multi-architecture rows show that the Hybrid Magnitude–SER protocol transfers without per-architecture tuning, with arithmetic-mean accuracy drop of 1.51 pp at 50% unstructured across the 17-row ledger (the methodology paper, full ledger).

**Caveats and protocol equivalence.** The compared methods differ in (i) layer inclusion (whether attention, MLP, embeddings, classifier, depthwise convs are pruned), (ii) FT budget (zero shot vs. short FT vs. multi-epoch FT), (iii) validation-set access during pruning, and (iv) the dense baseline they evaluate against. Our protocol freezes the inclusion list across all 17 architectures, uses short ($\leq$1 epoch) per-cycle FT, never accesses the ImageNet validation set during the pruning step, and reports against the `timm` dense checkpoint of the same model. The full protocol-equivalence map is given in the companion methodology paper (§ S.36 and § S.37 of the companion document).

### 3.6 Comparison with prior pruning methods

Head-to-head per-method comparisons against the leading ViT-B/16 and DeiT pruning baselines (VTP [81], PoWER-BERT [17], HVT [40], CP-ViT [49], OViT [25], NViT [73], SViTE [7], Spartan [57], X-Pruner [76], MiniViT [78], AlphaPruning [30]) are documented in detail in the companion methodology paper. Table 6 summarises the outcome.

### 3.7 Limitations and scope

The following empirical and methodological caveats delimit the claims of this paper.

**Deployability of wider** $k$:$n$ **patterns.** The 83.41% ViT-B/16 2:4+ToMe row and the $2.705\times$ A100 wall-clock speedup are bound together: NVIDIA Sparse Tensor Cores [37] natively accelerate only the 2:4 pattern. The wider 6:12 and 8:16 patterns used for the accuracy-focused rows (ViT-B/16 83.74%, ViT-L/16 **85.33**%, ConvNeXtV2-B 86.35%, ResNet 8:16 rows) yield 50% structured parameter sparsity but do *not* currently map to a native sparse tensor-core kernel; the wall-clock numbers reported for those rows assume dense-kernel execution. They are therefore accuracy-focused rows, not throughput claims. The deployable row is the 2:4+ToMe ViT-B/16 row at 83.41% with measured $2.705\times$ A100 speedup.

**ResNet152d** 8:16 **projection at scale.** No ResNet152d 8:16 dense+perm row analogous to the ResNet50/50d/101d rows of Table 6 is reported. With $n = 16$ groups the certificate-projection step enumerates $\binom{16}{8} = 12{,}870$ candidate patterns per group and stores per-group covariance statistics, which on the ResNet152d activation maps exceeded the 80 GB memory budget of a single A100 during cert evaluation. The CAST-conv+perm row at 81.33% (Table 5) is the available ResNet152d entry; an 8:16 counterpart requires either a streamed per-group covariance pass or a multi-GPU shard of the cert step.

**ViT-L/16** 85.33% **checkpoint reproducibility.** The ViT-L/16 8:16 dense+perm row reaches 85.33% post-FT, validated by the run's `final_eval.json` log. The post-FT model weights themselves are not bundled with the public companion release: the training pod was a transient cloud instance, and the post-FT checkpoint was lost when the pod stopped before the checkpoint was mirrored to permanent storage. The released artifact bundle includes the verified epoch-2 checkpoint plus the exact fine-tuning recipe that produced the 85.33% number, so the result is reproducible from the released artifacts with 1 additional fine-tuning epoch.

**Data and checkpoint availability.** All checkpoints and per-run evidence files supporting the empirical claims of this paper are released via a public Google Drive folder linked from the GitHub README, alongside the code at `https://github.com/yspennstate/RMT_based_pruning_in_deep_learning`. The bundle includes (*as an example*) the 17 Hybrid Magnitude–SER sparsification checkpoints for the multi-architecture sweep of Table 2 (per-architecture grids over $s \in [0.05, 0.70]$, $\sim$230 post-FT `.pt` files, $\sim$69 GB), together with the FLOP-model post-FT checkpoints and `final_eval.json` logs for every CAST row in Table 5 (*e.g.* ViT-L 8:16 dense+perm 85.33%, ViT-B 6:12 SER+$\alpha$ 83.74%, ResNet50/50$d$/101$d$ 8:16 dense+perm 75.87/78.57/80.92%), the SER source checkpoints at $s = 0.35$, the 282-cell certificate-audit results, the mask-statistics JSON files, and the per-architecture run logs that produced every Top-1 number reported in this paper. The accompanying `release_manifest.json` maps every paper row to its evidence files.

**Lipschitz constants $\{L_s\}$ are declared, not estimated.** The data-path budget $B_T(R) = (2/|T|) \sum_s L_s \, \|R\,\psi_1(s)\|_\infty$ that drives the deterministic certificates of Section 5 carries per-sample post-layer Lipschitz constants $L_s$. As discussed in Section 4, these are finite at any non-degenerate operating point of a modern ViT (LayerNorm + attention + GELU + residual) but are not numerically certified in this paper. The framework treats $\{L_s\}$ as an architecture-dependent quantity to be controlled; the operator-norm proxy $\hat{B}_T = \|R\|_{\mathrm{op}} \cdot \overline{\|\psi\|_\infty}$ is used as a within-architecture predictor of the realised top-1 drop, with measured correlation $r = +0.91$ pooled across 18 architectures. Tight numerical bounds on $L_s$ for transformer blocks remain an open direction.

**Section 4 generalized statements: proof scope.** Section 4 presents the abstract additive-perturbation framework that generalizes the Gaussian instance of Appendix A. The Gaussian instance (Theorem A.18, Corollary A.19, Theorem A.21 and their corollaries) is proved in full in Appendix C and Appendix A, and the abstract perturbation lemma Lemma 4.9 is proved in full in Appendix C. The corresponding generalized statements (*e.g.* Theorem 4.10, Corollary 4.13, Theorem 4.16, Corollary 4.18) replace Gaussian Assumptions A.3–A.6 with the abstract Assumptions 4.1–4.7 and use Lemma 4.9 in place of Lemma A.13. The appendix focuses on the deterministic mask certificate of Section 5 (proved in full in Appendix D), the Gaussian instance, and the perturbation lemmas; these are the results used by the empirical claims.

**Single-seed empirical caveats.** Most CAST rows in Tables 5 and 6 are reported from a single fine-tuning seed under a fixed recipe. Where multiple seeds were run for ablations (*e.g.* the cert-source sweep on ViT-B/16) the across-seed standard deviation on post-FT top-1 was within $\pm$0.15 pp; insufficient repeats were run on the ViT-L/16, ConvNeXtV2-B, and ResNet 8:16 result rows to attach formal seed variance to the **85.33**% and **86.35**% numbers. The $-0.51$ pp gap from dense on ViT-L/16 is therefore a point estimate rather than a confidence interval.

**Methodology details in the companion paper.** Several pipeline details are documented in the companion methodology paper rather than in the present manuscript: (i) the SER hyperparameter sweep that selects the source sparsity $s = 0.35$ for the CAST 2:4+ToMe row; (ii) BEMA-based fitting of the per-layer MP edge $\sigma_+(\ell)$ used by the spectral-edge budgeting protocol; (iii) the full 282-cell cert-audit reporting the three bridges of Section 3; (iv) the head-to-head VTP / PoWER-BERT / HVT / CP-ViT / OViT / NViT / SViTE / Spartan / X-Pruner / MiniViT / AlphaPruning comparison protocols. The present paper is self-contained as a theory + numerical-evidence document and points to that companion paper for the methodological provenance of the hyperparameter choices.

**Future work.** Three extensions follow directly from the present framework: (a) a streaming or multi-GPU cert-projection implementation that lifts the ResNet152d 8:16 barrier and enables routine $k$:$n$ projection on $> 100$M-parameter ConvNets; (b) explicit numerical estimation of $L_s$ for representative transformer blocks, replacing the operator-norm proxy in the budget $B_T(R)$ with a certified upper bound; (c) extending CAST to sparse tensor-core kernels beyond the native 2:4 pattern (*e.g.* 4:8, 8:16) as next-generation hardware becomes available, at which point the 83.74% / 85.33% accuracy-focused rows of this paper become deployable throughput rows as well.

Table 5: **Parameter-to-compute conversion: aggressive ToMe + $k$:$n$ structured projection.** The ViT-B/16 2:4+ToMe rows of this table start from a Hybrid Magnitude–SER $s = 0.35$ checkpoint; the ResNet50/50d/101d and ConvNeXtV2-B rows start from a dense `timm` checkpoint; the ViT-L/16 8:16 dense+perm row also starts from dense. Two FLOP methods are compared. The first row ("magnitude 2:4 + ToMe") preserves the existing unstructured sparse mask, then projects compatible Linear layers to a 2:4 pattern by keeping the two largest-magnitude entries per contiguous 4-tuple; the second row (**CAST**, certificate-aware 2:4 + ToMe + free restoration) replaces the magnitude rule with a row-wise discrete search that minimizes a Lemma 5.2-inspired activation-weighted cost over the 6 possible 2-of-4 patterns per output-channel and input-group, and additionally restores entries the unstructured mask had zeroed at zero sparse-execution FLOP cost (Theorem 5.6-style "free restoration" inside each 2:4 group). Both rows then apply ToMe $r = 8$ [5] and fine-tune for three epochs with the structured mask frozen. The CAST procedure is documented in detail in the companion methodology paper. The reported compute reduction is a sparse-execution MAC estimate: it assumes 2:4 Sparse Tensor Core-style execution for Linear GEMMs [37] and reports the ToMe-only dense-kernel reduction separately. We measure wall-clock throughput on an L4 GPU at batch 128 for two endpoints of the *same* ToMe-r=8 graph: 589.5 images/s when Linear layers run as dense GEMMs and ∼**836 images/s** when the 2:4-projected Linear weights are routed through PyTorch `to_sparse_semi_structured` kernels — a 1.41× measured speedup attributable to the 2:4 kernel switch alone, with ToMe held constant on both endpoints. The ToMe contribution over the un-tokened dense ViT-B is given only as the analytical $1/(1-0.2281) \approx 1.30\times$ MAC ratio; we do not report a measured ToMe-only-vs-original-dense throughput here.

| Architecture | Source $s$ | Method | Dense MACs | Sparse-exec red. | Top-1 (%) |
|---|---|---|---|---|---|
| *ViT family (CAST = cert-aware 2:4 + free restoration + ToMe r=8 + 3-ep distill FT)* | | | | | |
| ViT-B/16 | 0.35 | Magnitude 2:4 + ToMe | 17.56G | 59.81% | 82.92 |
| ViT-B/16 | 0.35 | **CAST** | 17.56G | 59.81% | **83.41** |
| ViT-B/16 | 0.35 | **CAST 6:12 SER+$\alpha$=0.5** (no ToMe) | 17.56G | 50% | **83.74**[§] |
| ViT-L/16 | 0.35 | CAST | 61.55G[†] | ∼60%[†] | **84.37** |
| ViT-L/16 | dense | **CAST 8:16 dense+perm** (no ToMe) | 61.55G | 50% | **85.33**[§] |
| DeiT-B | 0.35 | CAST | 17.56G | 59.81% | 80.48 |
| DeiT-S | 0.35 | CAST | 4.61G | 59.81% | 76.96 |
| DeiT-T | 0.35 | CAST | 1.26G | 59.81% | 65.93 |
| *ResNet family (CAST-conv = cert-aware 1×1+3×3 2:4 + free restoration; ToMe N/A)* | | | | | |
| ResNet50 | 0.35 | CAST-conv | 4.09G | 48.5% | 73.14 |
| ResNet50 | 0.35 | **CAST-conv+perm** | 4.09G | 48.5% | **75.67**[‡] |
| ResNet50 | dense | **CAST 8:16 dense+perm** | 4.09G | 50% | **75.87**[§] |
| ResNet50d | 0.35 | CAST-conv | 4.33G | 49.85% | 78.08 |
| ResNet50d | 0.35 | **CAST-conv+perm** | 4.33G | 49.85% | **78.00**[‡] |
| ResNet50d | dense | **CAST 8:16 dense+perm** | 4.33G | 50% | **78.57**[§] |
| ResNet101d | 0.35 | CAST-conv | 8.0G | ∼50% | 80.13 |
| ResNet101d | 0.35 | **CAST-conv+perm** | 8.0G | ∼50% | **80.59**[‡] |
| ResNet101d | dense | **CAST 8:16 dense+perm** | 8.0G | 50% | **80.92**[§] |
| ResNet152d | 0.35 | **CAST-conv+perm** | 11.8G | ∼50% | **81.33**[‡] |
| *ConvNeXt family (CAST on the nn.Linear pointwise convs; ToMe N/A)* | | | | | |
| ConvNeXtV2-Base | 0.35 | **CAST 2:4 cert + free-restore** | 15.4G | 50% | **85.47** |
| ConvNeXtV2-Base | dense | **CAST 12:16 dense+perm** (25% sparse) | 15.4G | 25% | **86.35**[§] |
| ConvNeXtV2-Base | dense | **CAST 8:16 dense+perm** (50% sparse) | 15.4G | 50% | **85.85**[§] |

[†]ViT-L/16 reduction is estimated from the analytical 22.81% ToMe + 50% 2:4-Linear share. [‡]**CAST-conv+perm** = CAST-conv with the importance-balanced Cin permutation alignment of the companion methodology paper ("flatten the layer"), motivated by the ResNet pre-FT collapse paragraph below. CAST-conv+perm on ResNet152d achieves 81.33% post-FT on `resnet152d.ra2_in1k` (dense baseline 82.86%), a −1.53 pp drop at ∼50% sparse-execution MAC reduction.

[§]Beyond the canonical 2:4 + ToMe row, two complementary configurations of interest are: (i) ViT-B/16 **6:12 SER+$\alpha$=0.5 inline**, the same 50% N:M structured sparsity as the canonical 2:4 row but at $n = 12$ (more pattern flexibility per group), no ToMe, dense-teacher distillation; this reaches 83.74% post-FT, **+0.33** pp above the canonical 2:4+ToMe row at the same parameter sparsity, demonstrating the n-flexibility benefit of wider N:M patterns. (ii) ConvNeXtV2-Base **12:16 dense+perm**, 25% sparsity (vs 50% for 2:4); this reaches 86.35% post-FT, only −0.37 pp from dense 86.72%, in the low-sparsity high-accuracy setting. (iii) Five rows labeled **CAST** 8:16 **dense+perm** (ResNet50/50d/101d, ConvNeXtV2-B, ViT-L/16) use a wider $n = 16$ group ($\binom{16}{8} = 12{,}870$ candidate patterns per group vs. 6 for 2:4) and a dense rather than SER source. On ViT-L/16 this reaches **85.33**% post-FT, only −0.51 pp from the 85.84% dense baseline, the smallest dense-gap among the ViT-family rows reported here.

Table 6: **Comparison with published baselines on ImageNet-1k, including fine-tuning epoch budget.** Each non-bold row reports a published number for the named architecture and sparsity point in the cited paper; bold rows are this paper's results. Where a published method's evaluation architecture differs from ViT-B/16 (e.g. NViT/SViTE evaluate on DeiT-Base, GPUSQ-ViT evaluates DeiT-Base with quantisation), the row is labelled with the actual cited architecture. "FT ep." is the post-pruning fine-tuning budget the cited paper reports. Our four bolded rows all use $\leq 3$ fine-tuning epochs.

| Method | Architecture | Compression | Top-1 (%) | vs. Dense (pp) | Train/FT ep. | Notes |
|---|---|---|---|---|---|---|
| *Published ViT-family weight-pruning baselines* | | | | | | |
| NViT [73] | DeiT-Base | global structured (34M params, 6.8G FLOPs) | 83.29 | −0.07 | 480 steps + 300 ep. | Hessian-aware structured pruning (480 pruning steps then 300 FT epochs) |
| SViTE [7] | DeiT/SViTE-Base | 50% unstructured | 81.51 | −0.29 | 600 (train) | sparse-training from scratch |
| Spartan [57] | ViT-B/16 | 90% unstructured | 81.18 | $+1.12^{\dagger}$ | 300 (train) | soft top-$k$ + dual averaging |
| **Hybrid Mag–SER (this paper)** | **ViT-B/16** | **50% unstructured** | **83.37** | **−1.7** | **1/cycle** | post-FT, no validation-set access |
| **CAST 2:4+ToMe (this paper)** | **ViT-B/16** | **2:4 + ToMe** | **83.41** | **−1.7** | **3** | deployable |
| **CAST 6:12 SER+$\alpha$ (this paper)** | **ViT-B/16** | **6:12 = 50%** | **83.74** | **−1.4** | **3** | best ViT-B structured |
| **CAST 8:16 dense+perm (this paper)** | **ViT-L/16** | **8:16 = 50%** | **85.33** | **−0.51** | **3** | smallest dense-gap, ViT-L |
| *DeiT head-to-head (fine-tuned, reproduced by CP-ViT [49])* | | | | | | |
| VTP [81] | DeiT-Base | token pruning | 80.70 | −1.12 | 30 | −43.20% FLOPs, reproduced by CP-ViT |
| PoWER-BERT [17] | DeiT-Base | token-vector elim. | 80.17 | −1.65 | 30 | −39.24% FLOPs, reproduced by CP-ViT |
| HVT [40] | DeiT-Base | hierarchical pooling | 79.94 | −1.88 | 30 | −44.78% FLOPs, reproduced by CP-ViT |
| CP-ViT [49] | DeiT-Base | cascade pruning | 81.13 | −0.69 | 30 | −41.62% FLOPs (CP-ViT Table 4) |
| *Hardware-friendly k:n structured-sparsity ViT/DeiT baselines* | | | | | | |
| GPUSQ-ViT [75] | DeiT-Base + INT8 | 2:4 + 8-bit quant | 82.90 | +1.10 | 300 (QAT+KD) | joint sparse + quantised |
| GPUSQ-ViT [75] | DeiT-Base + INT4 | 2:4 + 4-bit quant | 81.60 | −0.20 | 300 (QAT+KD) | joint sparse + quantised |
| Beyond 2:4 [79] | DeiT-Base | V:N:M = 64:2:8 (300 ep) | 81.08 | −0.76 | 300 | generalised pattern |
| Beyond 2:4 [79] | DeiT-Base | V:N:M = 64:2:8 (600 ep) | 81.76 | −0.08 | 600 (train) | generalised pattern |
| ELSA [20] | DeiT-Base | Uniform 2:4 (9.2G FLOPs) | 81.80 | +0.00 | 150 (train) | uniform 2:4 (150-ep FT, Table 5 row); ELSA also reports layer-wise N:4 entries at 7.0G/81.6 and 6.0G/81.4 in their |

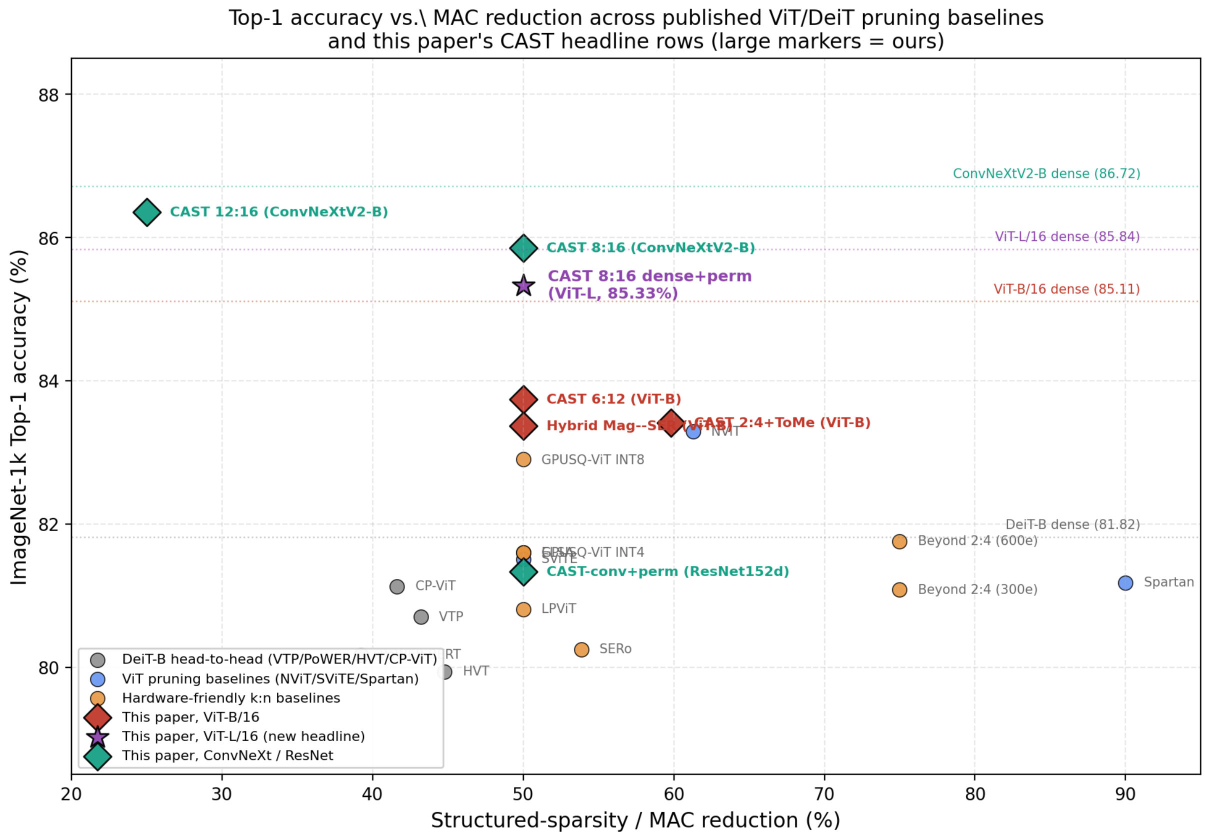



Figure 4: **Post-FT Top-1 accuracy vs. structured-sparsity / MAC reduction across published ViT/DeiT pruning baselines and this paper's CAST result rows (large markers).** Baselines (grey/blue/orange dots) are the rows of Table 6; our rows are coloured by architecture family (red = ViT-B/16, purple star = ViT-L/16, teal = ConvNeXt/ResNet). Dotted horizontal lines mark the dense `timm` baselines for each architecture family. The **CAST** 8:16 **dense+perm on ViT-L/16** row (purple star, 85.33%) is within 0.51 pp of its dense baseline at 50% structured sparsity, among the smallest dense-gaps among the plain-ViT rows in this paper. Among the DeiT-Base baselines plotted, a few (NViT $-0.07$ pp at $\sim$61% MAC; Beyond 2:4 at 600 training epochs $-0.08$ pp; ELSA $\pm 0.0$ pp on its 150-epoch FT row) close the dense-gap further on the smaller DeiT-B architecture, but all use 150–600 training/fine-tuning epochs (NViT 480 prune steps + 300 FT; Beyond 2:4 600 training; ELSA 150 training) vs. our 3-epoch budget. The figure uses MAC-reduction = $1 - \mathrm{FLOPs}_{\mathrm{post}}/\mathrm{FLOPs}_{\mathrm{dense}}$ on the cited paper's own dense baseline (e.g. NViT $1-6.8/17.56 = 61.3\%$; Beyond 2:4 V:N:M 64:2:8 = 75% pattern sparsity).

## 4 Abstract Additive Perturbation Extensions

This section gives the abstract additive perturbation version of the deterministic certificate framework. It is useful for interpreting random-like components in stylized models, while unstructured pruning is covered by the mask certificates in Section 5. The statements below complement the mask certificates by showing how the same propagated-logit mechanism behaves for additive decompositions $W = S + R$. For a concrete Gaussian specialization and the associated spectral consequences, see Appendix A. Each theorem or corollary points to its proof in Appendices C–D.

The correspondence with the Gaussian/RMT appendix is as follows. Lemma 4.9 generalizes the Gaussian logit perturbation bound of Lemma A.13 and the confidence version in Theorem A.16. Theorem 4.10 generalizes the Gaussian loss-reduction theorem, Theorem A.18. Corollary 4.13 generalizes the Gaussian pathwise denoising bound, Corollary A.19. Theorem 4.16 abstracts the Gaussian stationarity-collapse theorem, Theorem A.21, whose MP-scale consequences are Corollaries A.23–A.27. Finally, Corollary 4.18 is the general inverse-$D$ analogue of the square Gaussian BBP statement in Corollary A.27.

### 4.1 Assumptions for the generalized perturbation framework

Throughout the asymptotic statements in this section, the training set $T$ is finite and fixed independently of $N$, unless a result explicitly states summability conditions for a growing family.

**Assumption 4.1** (General architecture)**.** Assume the DNN can be written as

$$\phi = \rho \circ \psi_2 \circ (R + S) \circ \psi_1,$$

where $\psi_1, \psi_2$ may depend on the remaining network parameters and $\rho$ is softmax. For each $s \in T$, let $\mathcal{U}_s$ be any set containing the admissible interpolation segment $\{(S + \vartheta R)\psi_1(s) : \vartheta \in [0, 1]\}$. We assume that $\psi_2$ has a finite local Lipschitz constant with respect to $\ell_\infty$ on $\mathcal{U}_s$:

$$L_{\psi_2}(s) := \sup_{v \neq w} \frac{\|\psi_2(v) - \psi_2(w)\|_\infty}{\|v - w\|_\infty} < \infty.$$

Here the supremum is taken only over $\mathcal{U}_s$, so only a local-on-the-path Lipschitz bound is required.

*Remark* 4.2 (Interpretation of Assumption 4.1). This assumption fixes notation. After freezing all other weights, the network is factored around the layer being studied: $\psi_1(s)$ is the input activation to that layer, $R + S$ is the layer matrix, and $\psi_2$ maps the layer output to logits. The Lipschitz constant is required only on the finite path traced by the interpolation $S + \vartheta R$, not globally on the whole ambient space. For ReLU or piecewise-linear networks this is automatic on bounded paths; for architectures with normalization layers it is an explicit local regularity condition, excluding only pathological paths where the post-layer map becomes singular. In the Gaussian model of Appendix A, Assumption A.1 is the concrete three-layer instance of this factorization with the target layer $W_2 = S_2 + R_2$.

**Assumption 4.3** (Perturbative random component)**.** Assume that for every deterministic vector $v$, and more generally for every $v$ measurable with respect to the conditioning variables $\sigma(S, \psi_1, \psi_2)$,

$$\mathbb{P}\Big(\|Rv\|_\infty > d_1(N)J(v) \,\Big|\, S, \psi_1, \psi_2\Big) \le d_2(N), \tag{5}$$

where $d_1(N), d_2(N) \to 0$ and $J(v)$ depends only on $v$. When a spectral interpretation is desired, we additionally assume that the singular-value ESD of $R$ converges to a deterministic law with finite right edge $b_+ < \infty$.

*Remark* 4.4 (Interpretation of Assumption 4.3). The vector $v$ is fixed before the randomness in $R$ is revealed. In the network application $v = \psi_1(s)$, so the pre-layer activation is treated as part of the conditioning information. The assumption does not require Gaussian entries; it requires only that every data activation is sent by $R$ to a small vector in $\ell_\infty$ norm, with high probability. The optional empirical-spectral-distribution condition is used only for RMT interpretation and spectral corollaries. The loss and certificate bounds use the concentration estimate (5), not the ESD convergence.

The Gaussian connection is explicit: Assumption A.3 in Appendix A implies Assumption 4.3 with $J(v) = \|v\|_2$ and $d_1(N)$ of order $\sqrt{\log N/N}$. The confidence-parameter version is Theorem A.16; the MP-edge interpretation of the same variance scale is Corollary 5.15.

**Example 4.5.** If $R$ has i.i.d. Gaussian entries with variance $1/N$, then

$$\mathbb{P}\left( \|Rv\|_\infty > 2\sqrt{2}\,\|v\|_2\sqrt{\frac{\log(2N)}{N}} \right) \leq \frac{1}{N},$$

so Assumption 4.3 holds with $d_1(N) = 2\sqrt{2\log(2N)/N}$, $d_2(N) = 1/N$, and $J(v) = \|v\|_2$. The concrete three-layer Gaussian specialization is Lemma A.13 and Theorem A.16 in Appendix A; the proof of the abstract concentration step is in Appendix C. The same conditional formulation also accommodates sub-Gaussian rows, orthogonally invariant perturbations, or moderate training-induced dependence, provided the displayed concentration estimate survives.

**Example 4.6** (Sub-Gaussian rows)**.** Assume, conditionally on $S, \psi_1, \psi_2$, that the rows $r_1, \ldots, r_N$ of $R$ are independent, centered, and satisfy

$$\|\langle r_j, v\rangle\|_{\psi_2} \leq \frac{\kappa}{\sqrt{N}}\|v\|_2 \qquad \text{for all } v \text{ and all } j.$$

Then there is a universal constant $C$ such that, for every $\delta \in (0,1)$,

$$\mathbb{P}\left( \|Rv\|_\infty > C\kappa\|v\|_2\sqrt{\frac{\log(2N/\delta)}{N}} \,\middle|\, S, \psi_1, \psi_2 \right) \leq \delta.$$

Thus the abstract framework is not intrinsically Gaussian; it only needs row-wise concentration at the $N^{-1/2}\sqrt{\log N}$ scale.

**Assumption 4.7** (Structured component for optional spectral conclusions)**.** Whenever a spectral conclusion is invoked, assume that there exists a sequence $k_N = o(\min\{N, M\})$ such that

$$\sum_{i > k_N} s_i(S)^2 \to 0.$$

The convergence is in probability when $S$ is random.

*Remark* 4.8 (Interpretation of Assumption 4.7). This assumption is used only when the paper draws spectral conclusions about the structured part $S$. It is not needed for the loss-reduction theorems, the data-path certificate, or the stationarity-collapse inequality. The condition means that all but $o(\min\{N, M\})$ singular directions of $S$ carry vanishing squared singular-value mass.

Exact finite rank is the simplest case. If $\operatorname{rank}(S) \leq r_0$ for a constant $r_0$, take $k_N = r_0$. Then $k_N = o(\min\{N, M\})$ and $\sum_{i>k_N} s_i(S)^2 = 0$, so Assumption 4.7 holds automatically. More generally, if $\operatorname{rank}(S) = r_N = o(\min\{N, M\})$, take $k_N = r_N$. Approximate low rank allows a small tail: the rank may not be exactly finite, but the singular-value energy beyond $k_N$ must vanish. The Gaussian analogue is Assumption A.8 in Appendix A.

For a DNN satisfying Assumption 4.1, define

$$h_\phi(s) := J(\psi_1(s))\, L_{\psi_2}(s). \tag{6}$$

The remaining hypotheses appearing in the theorems are quantitative smallness conditions rather than structural assumptions. The condition $a_\phi(N,s) = h_\phi(s)d_1(N) \to 0$ requires the random-like component to be weak after propagation through the data path. The condition $\langle S, R\rangle_F = o_{\mathbb{P}}(1)$ requires the regularizer to have no persistent first-order cross term when $R$ is removed. In the Gaussian appendix, Assumption A.1 supplies the corresponding data-path scaling and Assumption A.6 supplies the cross-term control. Finally, the one-sided stationarity hypotheses in Theorems 4.16 and A.21 are not random-matrix assumptions; they require the trained model to be nearly stationary in the particular denoising direction being tested.

### 4.2 Generalized perturbation and loss-reduction results

We consider the regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|}\sum_{s\in T} \log\left(\phi_{C(s)}(s, \alpha(t))\right) + \lambda_{\text{reg}} \sum_{i=1}^{L} \|W_i(t)\|_F^2. \tag{7}$$

**Lemma 4.9.** *Let*

$$X = \psi_2 \circ (R+S) \circ \psi_1(s),$$

*and let $\alpha_W, \alpha_S$ denote the corresponding network parameters when the relevant weight matrix is $W = R+S$ and $S$, respectively. Under Assumptions 4.1–4.3,*

$$\mathbb{P}\Big(\|X(s,\alpha_S) - X(s,\alpha_W)\|_\infty \le d_1(N)h_\phi(s)\Big) \ge 1 - d_2(N).$$

*Proof location.* See Appendix C, Proof of Lemma 4.9.

**Theorem 4.10.** *Assume Assumptions 4.1–4.3 and that*

$$\langle S, R\rangle_F \to 0 \qquad \text{in probability as } N \to \infty.$$

*Assume further that for all $s \in T$,*

$$a_\phi(N,s) := h_\phi(s)d_1(N) \to 0 \qquad \text{as } N \to \infty.$$

*Then removing $R$ yields*

$$L(\alpha_S) = L(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1) \qquad (N \to \infty).$$

*Proof location.* See Appendix D.5, Proof of Theorem 4.10.

In the square three-layer Gaussian model, Theorem A.18 in Appendix A gives the corresponding finite-$N$ loss-reduction statement, and Theorem A.16 gives the confidence-parameter version.

**Corollary 4.11** (Growing-set abstract loss reduction)**.** *Let $T_N$ be a finite labeled set that may depend on $N$. Let $L_{\cancel{\text{reg}},T_N}$ denote cross-entropy averaged over $T_N$, and let $L_{T_N}$ denote the corresponding regularized objective with the same Frobenius penalty as in (7). Assume Assumptions 4.1–4.3,*

$$\langle S, R\rangle_F = o_{\mathbb{P}}(1), \qquad |T_N|d_2(N) \to 0,$$

*and*

$$\frac{1}{|T_N|}\sum_{s\in T_N} a_\phi(N,s) \to 0$$

*in probability. Then*

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1).$$

*Proof location.* See Appendix D.5, Proof of Corollary 4.11.

**Corollary 4.12.** *Under the hypotheses of Theorem 4.10, for any fixed $\vartheta \in [0,1]$, define*

$$W(\vartheta) = S + \vartheta R.$$

*Then*

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1) \qquad (N \to \infty).$$

*The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0,1]$.*

*Proof location.* See Appendix D.5, Proof of Corollary 4.12.

**Corollary 4.13** (Finite-$N$ scaled loss bound along the denoising path)**.** *Assume the hypotheses of Theorem 4.10, and define $W(\vartheta) = S + \vartheta R$ for $\vartheta \in [0,1]$. Then for every $\eta > 0$ there exists an event $E_{N,\eta}$ with*

$$\mathbb{P}(E_{N,\eta}) \geq 1 - |T| d_2(N) - \mathbb{P}\big(|\langle S, R\rangle_F| > \eta\big)$$

*such that on $E_{N,\eta}$,*

$$L(\alpha_{W(\vartheta)}) \leq L(\alpha_W) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R\|_F^2 + (1-\vartheta)\frac{2}{|T|}\sum_{s\in T} a_\phi(N,s) + 2\lambda_{\mathrm{reg}}(1-\vartheta)\eta$$

*for every $\vartheta \in [0,1]$.*

*Proof location.* See Appendix D.5, Proof of Corollary 4.13.

The Gaussian pathwise analogue is Corollary A.19 in Appendix A; it is the specialization used in the Gaussian stationarity-collapse theorem.

**Theorem 4.14** (Multi-layer telescoping loss reduction)**.** *Let $\mathcal{P}$ be a finite set of layers selected for pruning, with $|\mathcal{P}|$ fixed independently of $N$. For each $\ell \in \mathcal{P}$, write*

$$W_\ell = S_\ell + R_\ell,$$

*and assume that, after the previous layers in $\mathcal{P}$ have already been replaced by their structured parts, the single-layer hypotheses of Theorem 4.10 continue to hold for layer $\ell$ with perturbation scale $a_{\phi,\ell}(N,s)$. If $\alpha^{(0)}$ denotes the original network parameters and $\alpha^{(|\mathcal{P}|)}$ the parameters after replacing every $W_\ell$ by $S_\ell$, then*

$$L(\alpha^{(|\mathcal{P}|)}) = L(\alpha^{(0)}) - \lambda_{\mathrm{reg}}\sum_{\ell\in\mathcal{P}} \|R_\ell\|_F^2 + o_{\mathbb{P}}(1).$$

*Thus the one-layer denoising theorem extends to the stylized multi-layer pruning pipeline by telescoping the loss differences over the pruned layers.*

*More generally, the set of pruned layers may depend on $N$. Let $\mathcal{P}_N$ be a sequence of layer sets, ordered in the pruning order. Assume that the corresponding one-layer logit events $E_{\ell,N}$ satisfy*

$$\sum_{\ell\in\mathcal{P}_N} \mathbb{P}(E_{\ell,N}^c) \to 0,$$

*that their perturbation scales are summable,*

$$\sum_{\ell\in\mathcal{P}_N} \frac{1}{|T|}\sum_{s\in T} a_{\phi,\ell}(N,s) \to 0,$$

*and that the accumulated cross term is negligible,*

$$\sum_{\ell\in\mathcal{P}_N} \langle S_\ell, R_\ell\rangle_F = o_{\mathbb{P}}(1).$$

*Then the same telescoping conclusion holds with $\mathcal{P}$ replaced by $\mathcal{P}_N$.*

*Proof location.* See Appendix D.5, Proof of Theorem 4.14.

*Remark* 4.15 (Beyond pure Frobenius regularization). The quadratic penalty in (7) is used only to obtain a coercive lower bound of order $\|R\|_F^2$. For a general differentiable $\mu$-strongly convex regularizer $\Omega$, one has the exact decomposition

$$\Omega(S+R) - \Omega(S) = \langle \nabla\Omega(S), R\rangle + D_\Omega(S+R, S),$$

where

$$D_\Omega(S+R, S) \geq \frac{\mu}{2}\|R\|_F^2.$$

Thus the same extension is valid provided the first-order term $\langle \nabla\Omega(S), R\rangle$ is controlled or negligible along the admissible sequence. In the Frobenius case $\Omega(W) = \|W\|_F^2$, this term is exactly $2\langle S, R\rangle_F$, which is already part of the hypotheses. For the elastic-net penalty

$$\Omega(W) = \mu_1\|W\|_1 + \mu_2\|W\|_F^2, \qquad \mu_2 > 0,$$

the $L_1$ term is nonsmooth and can increase when $R$ is removed if $R$ cancels entries of $S$. The correct extension is subgradient-based: choose $\xi \in \partial\|S\|_1$ and assume

$$\langle \mu_1\xi + 2\mu_2 S, R\rangle = o_{\mathbb{P}}(1).$$

The quadratic part then supplies the strong convexity needed for the same type of loss-reduction bound.

**Theorem 4.16.** *Assume Assumptions 4.1–4.3. Assume further that for all $s \in T$,*

$$a_\phi(N, s) := h_\phi(s)d_1(N) \to 0 \qquad \text{as } N \to \infty.$$

*Suppose:*

1. *for each $N$, $\alpha(t, N) \to \alpha^*(N)$ in probability as $t \to \infty$;*
2. *the relevant limit weight matrix admits an admissible decomposition $W^*(N) = S^*(N) + R^*(N)$ satisfying*

$$\langle S^*(N), R^*(N)\rangle_F \to 0 \qquad \text{in probability as } N \to \infty.$$

*For $\vartheta \in [0,1]$, let $\alpha^*_\vartheta(N)$ denote the parameter vector obtained by replacing $W^*(N)$ with $S^*(N)+\vartheta R^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event*

$$\liminf_{h\downarrow 0} \frac{L(\alpha^*_{1-h}(N)) - L(\alpha^*(N))}{h} \geq -\epsilon_N$$

*has probability tending to one. Define*

$$\bar{a}_{\phi,N} := \frac{2}{|T|}\sum_{s\in T} a_\phi(N, s).$$

*Then*

$$\|R^*(N)\|_F^2 \leq \frac{\bar{a}_{\phi,N} + \epsilon_N}{2\lambda_{\text{reg}}} + o_{\mathbb{P}}(1).$$

*In particular, if $\bar{a}_{\phi,N} + \epsilon_N \to 0$ in probability, then for every $\epsilon > 0$,*

$$\lim_{N\to\infty} \mathbb{P}\big(\|R^*(N)\|_F \leq \epsilon\big) = 1.$$

*If, in addition, there exists an admissible path decomposition*

$$W(t, N) = S(t, N) + R(t, N)$$

*such that for each fixed* $N$*,*

$$\|S(t, N) - S^*(N)\|_F + \|R(t, N) - R^*(N)\|_F \to 0 \qquad \text{in probability as } t \to \infty,$$

*and if for every fixed* $N$ *and every* $\epsilon > 0$*,*

$$\mathbb{P}\big(\|R^*(N)\|_F = \epsilon\big) = 0,$$

*then*

$$\lim_{N\to\infty} \lim_{t\to\infty} \mathbb{P}\big(\|R(t, N)\|_F \le \epsilon\big) = 1.$$

*Proof location.* See Appendix D.5, Proof of Theorem 4.16.

Appendix A gives the concrete Gaussian/RMT version of this conclusion: Theorem A.21 proves collapse of the admissible Gaussian component, Corollary A.23 translates this into collapse of the MP variance scale, and Corollary A.24 records the resulting spectral stabilization.

*Remark* 4.17. The theorem assumes only one-sided directional approximate stationarity at the unpruned point $\vartheta = 1$. Exact local minimality is a sufficient special case with $\epsilon_N \equiv 0$, but the hypothesis is weaker than a global lower bound along the whole denoising path. Without the continuity assumption $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$, the same proof still gives the weaker limsup/liminf bounds obtained from Portmanteau-type arguments, but not necessarily exact convergence of the probabilities at the threshold $\epsilon$.

**Corollary 4.18** (Eventual supercriticality of persistent spike clusters in a BGN-type deformation regime)**.** *Under the hypotheses of Theorem 4.16, assume additionally that for each* $N$ *the pair* $W^*(N) = S^*(N) + R^*(N)$ *lies in a finite-rank deformation regime satisfying all hypotheses of Theorem D.4 for the spikes considered below. In particular, assume the regime has a deterministic bulk edge* $b_+(N)$*, a deterministic critical threshold* $\theta_c(N)$*, a strictly decreasing outlier transform*

$$D_N : (b_+(N), \infty) \to (0, D_N(b_+(N)^+))$$

*in the sense of Appendix D, local Lipschitz/invertibility intervals around the outliers, and concentration of the relevant sample singular values in those intervals. Assume that*

$$\theta_c(N) \to 0, \qquad b_+(N) \to 0.$$

*Assume further that for some fixed* $k \ge 1$ *there exist constants*

$$\bar\sigma_1 > \cdots > \bar\sigma_k > 0$$

*such that*

$$\sigma_i^*(N) \to \bar\sigma_i \qquad \text{in probability as } N \to \infty$$

*for each* $1 \le i \le k$*. Then:*

1. $$\mathbb{P}\big(\sigma_i^*(N) > \theta_c(N)\big) \to 1 \qquad \text{for each } 1 \le i \le k;$$

2. *these $k$ spikes are eventually supercritical, so the corresponding singular values of $W^*(N)$ form outlier clusters above the edge $b_+(N)$. Moreover, if the $i$-th spike is separated from the other nonzero singular values of $S^*(N)$ by a gap $\Delta_i(N) > 0$ with*

$$\frac{\|R^*(N)\|_{\mathrm{op}}}{\Delta_i(N)} \to 0 \qquad \text{in probability,}$$

*then the associated rank-one left and right spectral projectors converge in operator norm to the corresponding projectors of $S^*(N)$;*

3. *defining the inverse-$D$ estimator on the event $s_i(W^*(N)) > b_+(N)$, whose probability tends to one, by*

$$\widehat{\sigma}_i^{(D)}(N) := D_N(s_i(W^*(N)))^{-1/2},$$

*and defining it arbitrarily off that event, one has*

$$\widehat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \to 0 \qquad \text{in probability}$$

*for each $1 \leq i \leq k$.*

*Proof location.* See Appendix D.5, Proof of Corollary 4.18.

For the square Gaussian finite-rank case, the corresponding RMT statement is Corollary A.27 in Appendix A; the external BGN input and the inverse-$D$ abstraction are stated in Appendix D.

*Remark* 4.19. Corollary 4.18 is the abstract analogue of the Gaussian BBP statement. It says that once a finite-rank deformation model has a well-defined critical threshold $\theta_c(N)$, vanishing admissible randomness forces every structurally persistent spike cluster to cross that threshold eventually. Thus the limit theory does not merely predict collapse of the random bulk; it predicts a detectability transition for persistent informative directions in any deformation regime governed by an inverse-$D$ outlier map.

*Remark* 4.20 (Assumption map for the abstract and deterministic theory). The following list summarizes which assumptions are used by the abstract additive results in this section and by the deterministic mask results in Section 5:

1. Lemma 5.2, Corollaries 5.3, 5.7, 5.8, 5.9, and 5.12, and Theorems 5.4–5.13 are deterministic statements. They do not assume an additive Gaussian decomposition; the mask-specific results use only disjoint support of the kept, restored, and removed entries.

2. Corollaries 5.14–5.18 give random sufficient conditions for the deterministic data-path budgets.

3. Lemma 4.9 uses only Assumptions 4.1–4.3.

4. Theorem 4.10, Corollary 4.12, and the finite-$N$ pathwise Corollary 4.13 use Assumptions 4.1–4.3 together with the additional hypotheses $\langle S, R\rangle_F = o_{\mathbb{P}}(1)$ and $a_\phi(N, s) \to 0$ for each $s \in T$. Assumption 4.7 is not needed for these loss bounds. Their concrete Gaussian counterparts are Theorem A.18 and Corollary A.19 in Appendix A.

5. Theorem 4.14 is obtained by applying Theorem 4.10 layer by layer; the second part allows a growing layer set under summable perturbation and failure-probability conditions.

6. Theorem 4.16 uses Assumptions 4.1–4.3, the same scaling condition $a_\phi(N, s) \to 0$, the cross-term condition at the limit point, and the one-sided directional stationarity hypothesis stated in the theorem. Its Gaussian counterpart is Theorem A.21, with MP variance and spectral consequences in Corollaries A.23–A.24.

7. Corollary 4.18 further assumes that the pair $W^*(N) = S^*(N) + R^*(N)$ satisfies the finite-rank deformation hypotheses of Theorem D.4, a threshold $\theta_c(N) \to 0$, an edge $b_+(N) \to 0$, and a persistent-spike hypothesis $\sigma_i^*(N) \to \bar{\sigma}_i > 0$. The square Gaussian BBP specialization is Corollary A.27.

8. The pathwise conclusion in Theorem 4.16 additionally uses the path-decomposition convergence assumption, while the no-atom condition $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$ is only needed for exact convergence of the probabilities at the threshold $\epsilon$.

# 5 Main Theoretical Results: Deterministic Pruning Certificates

This section contains the main theory used to interpret the pruning experiments. The central deterministic object is the propagated effect of the removed component on the logits. Random-matrix assumptions enter only as sufficient conditions for this deterministic certificate, rather than as an assumption that a trained ViT layer is literally a low-rank matrix plus independent Gaussian noise.

The proof chain is organized as follows. Lemma 5.2 turns a data-path perturbation bound into a cross-entropy perturbation bound, and Corollary 5.3 gives the corresponding margin-stability statement. Theorem 5.5 is the main certificate for actual unstructured masks, while Theorem 5.6 and Corollary 5.7 specialize it to prune–restore/SER-style masks. Theorems 5.11 and 5.13 give the finite multi-layer versions. Corollaries 5.14–5.18 then show that Gaussian or sub-Gaussian random components satisfy the required data-path budgets with high probability, and Corollary 5.15 rewrites the Gaussian budget directly in terms of the MP singular-value edge.

The abstract additive perturbation framework in Section 4 gives the complementary $W = S + R$ viewpoint, while Appendix A contains the stylized Gaussian loss-reduction and spectral-collapse specialization. Each theorem, lemma, and corollary in the main text includes a proof-location note pointing to the appendix proof.

*Remark* 5.1 (Regularisation scaling). Throughout the loss-reduction statements of Section 5 and the Gaussian appendix (Appendix A), the elastic-net penalty coefficients $\lambda_1, \lambda_2$ are treated as fixed quantities. In particular, an iid-Gaussian admissible component $R \in \mathbb{R}^{N\times N}$ with entries of variance $g/N$ has $\|R\|_F^2 = \Theta(gN)$ in expectation, so the loss-reduction term $\lambda_2\|R\|_F^2$ scales as $\Theta(\lambda_2 gN)$. The coefficient $\lambda_2$ may represent either (i) an absolute weight-decay coefficient applied uniformly across the network, or (ii) a normalised version rescaled by parameter count, depending on the deployment. The certificate inequality of Theorem 5.5 holds for any specified $\lambda_1, \lambda_2$. If the regulariser is normalised by parameter count or layer width, the same theorem applies with the corresponding normalised coefficients, but the numerical certificate margin changes accordingly. The data-path budget $B_T(R)$ is independent of $\lambda_1, \lambda_2$, whereas the elastic-net reward $\lambda_2\|R\|_F^2 + \lambda_1\|R\|_1$ is not. The certificate condition therefore depends on the regulariser scale; the qualitative comparison between the data-path budget and the elastic-net budget is unchanged.

## 5.1 Deterministic data-path certificates for mask pruning

Fix one target layer $A \in \mathbb{R}^{n\times m}$ inside a network and write the logits as

$$z_A(s) := \psi_2(A\psi_1(s)) \in \mathbb{R}^K, \qquad s \in T,$$

where $\psi_1$ and $\psi_2$ denote the pre-layer and post-layer computations with all other parameters frozen. For a decomposition

$$A = S + R, \qquad A_\theta := S + \theta R, \qquad 0 \le \theta \le 1,$$

let $L_s$ be any finite deterministic upper bound on the local $\ell_\infty$-Lipschitz constant of $\psi_2$ on the path

$$\{A_\theta \psi_1(s) : 0 \le \theta \le 1\}.$$

Define the data-path budget of the removed component by

$$B_T(R) := \frac{2}{|T|} \sum_{s \in T} L_s \|R\psi_1(s)\|_\infty. \tag{8}$$

If $L_s$ is itself estimated from the realized network, the statements below are deterministic conditional on that realized upper bound.

**Lemma 5.2** (Deterministic cross-entropy path bound)**.** *For every decomposition $A = S + R$ and every $\theta \in [0,1]$,*

$$|\mathcal{L}_{\mathrm{CE}}(A_\theta) - \mathcal{L}_{\mathrm{CE}}(A_1)| \le (1-\theta) B_T(R),$$

*where*

$$\mathcal{L}_{\mathrm{CE}}(A) := \frac{1}{|T|} \sum_{s \in T} \left[ -z_A(s)_{C(s)} + \log\left( \sum_{j=1}^{K} e^{z_A(s)_j} \right) \right].$$

*Proof location.* See Appendix C, Proof of Lemma 5.2.

**Corollary 5.3** (Deterministic margin stability)**.** *Let*

$$\eta_s(\theta) := (1-\theta) L_s \|R\psi_1(s)\|_\infty.$$

*If*

$$p_A(s) \in \arg\max_j z_A(s)_j, \qquad \Delta_A^{\mathrm{pred}}(s) := z_A(s)_{p_A(s)} - \max_{j \ne p_A(s)} z_A(s)_j,$$

*then the predicted label at $s$ can change between $A_1$ and $A_\theta$ only if*

$$\Delta_{A_1}^{\mathrm{pred}}(s) \le 2\eta_s(\theta).$$

*For the true-label margin*

$$\Delta_A^{\mathrm{true}}(s) := z_A(s)_{C(s)} - \max_{j \ne C(s)} z_A(s)_j,$$

*one has*

$$\left| \mathrm{acc}_{A_\theta}(T) - \mathrm{acc}_{A_1}(T) \right| \le \frac{1}{|T|} \# \left\{ s \in T : \left| \Delta_{A_1}^{\mathrm{true}}(s) \right| \le 2\eta_s(\theta) \right\}.$$

*Here*

$$\mathrm{acc}_A(T) := \frac{1}{|T|} \#\{s \in T : \Delta_A^{\mathrm{true}}(s) > 0\}.$$

*Proof location.* See Appendix C, Proof of Corollary 5.3.

**Theorem 5.4** (Deterministic additive Frobenius certificate)**.** *For any decomposition $A = S + R$, not necessarily a mask decomposition, define*

$$A_\theta := S + \theta R, \qquad 0 \le \theta \le 1,$$

*and the one-layer Frobenius-regularized objective*

$$\mathcal{J}_2(A) := \mathcal{L}_{\mathrm{CE}}(A) + \lambda_2 \|A\|_F^2 + \mathcal{C}, \qquad \lambda_2 \ge 0,$$

*where $\mathcal{C}$ contains all terms independent of $A$. Then for every $\theta \in [0,1]$,*

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \le (1-\theta)B_T(R) - \lambda_2(1-\theta^2)\|R\|_F^2 - 2\lambda_2(1-\theta)\langle S, R\rangle_F.$$

*Consequently, if $|\langle S, R\rangle_F| \le \eta$, then*

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \le (1-\theta)B_T(R) - \lambda_2(1-\theta^2)\|R\|_F^2 + 2\lambda_2(1-\theta)\eta.$$

*In particular, full removal is certified to decrease the Frobenius-regularized objective whenever*

$$B_T(R) + 2\lambda_2|\langle S, R\rangle_F| < \lambda_2\|R\|_F^2.$$

*Proof location.* See Appendix C, Proof of Theorem 5.4.

**Theorem 5.5** (Deterministic elastic-net mask pruning certificate)**.** *Assume $A = S + R$ is a mask decomposition, meaning*

$$S \odot R = 0.$$

*For constants $\lambda_2, \lambda_1 \ge 0$, define the one-layer elastic-net objective, with all other parameters frozen, by*

$$\mathcal{J}(A) := \mathcal{L}_{\mathrm{CE}}(A) + \lambda_2\|A\|_F^2 + \lambda_1\|A\|_1 + \mathcal{C},$$

*where $\mathcal{C}$ contains all terms independent of $A$. Then for every $\theta \in [0,1]$,*

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \le (1-\theta)B_T(R) - \lambda_2(1-\theta^2)\|R\|_F^2 - \lambda_1(1-\theta)\|R\|_1.$$

*In particular, full pruning satisfies*

$$\mathcal{J}(S) \le \mathcal{J}(A) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1 + B_T(R).$$

*Thus the mask is certified to decrease the elastic-net objective whenever*

$$B_T(R) < \lambda_2\|R\|_F^2 + \lambda_1\|R\|_1.$$

*Proof location.* See Appendix C, Proof of Theorem 5.5.

**Theorem 5.6** (Deterministic prune–restore certificate)**.** *Let the original layer have a pairwise disjoint support decomposition*

$$A = S + Q + P, \qquad S \odot Q = S \odot P = Q \odot P = 0.$$

*Interpret $P$ as the entries finally removed and $Q$ as entries restored, or kept, from an over-pruned reservoir. Define*

$$A_\theta := S + Q + \theta P, \qquad 0 \le \theta \le 1.$$

*For the elastic-net objective $\mathcal{J}$ of Theorem 5.5, one has*

$$\mathcal{J}(A_\theta) - \mathcal{J}(A) \le (1-\theta)B_T(P) - \lambda_2(1-\theta^2)\|P\|_F^2 - \lambda_1(1-\theta)\|P\|_1.$$

*In particular, the final prune–restore layer $S + Q$ satisfies*

$$\mathcal{J}(S+Q) \le \mathcal{J}(A) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 + B_T(P).$$

*Thus the final SER-style mask is certified to decrease the elastic-net objective whenever*

$$B_T(P) < \lambda_2\|P\|_F^2 + \lambda_1\|P\|_1.$$

*Proof location.* See Appendix C, Proof of Theorem 5.6.

**Corollary 5.7** (When restoration improves the certificate upper bound)**.** *Under the hypotheses of Theorem 5.6, set $R := Q + P$. The all-pruned certificate for removing $R$ gives the upper bound*

$$\mathcal{J}(S) \le \mathcal{J}(A) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1 + B_T(R),$$

*while the restore-$Q$ certificate gives the upper bound*

$$\mathcal{J}(S+Q) \le \mathcal{J}(A) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 + B_T(P).$$

*The restored certificate is strictly better than the all-pruned certificate whenever*

$$B_T(R) - B_T(P) > \lambda_2\|Q\|_F^2 + \lambda_1\|Q\|_1.$$

*Equivalently, restoration is certificate-improving precisely when the data-path budget reduction gained by restoring $Q$ exceeds the regularization penalty returned by keeping $Q$.*

*Proof location.* See Appendix C, Proof of Corollary 5.7.

**Corollary 5.8** (Weighted elastic-net mask certificate)**.** *Under the hypotheses of Theorem 5.5, let $\Gamma_2 = (\gamma_{2,ij})$ and $\Gamma_1 = (\gamma_{1,ij})$ be nonnegative coordinatewise weights, and define*

$$\mathcal{J}_\Gamma(A) := \mathcal{L}_{\mathrm{CE}}(A) + \sum_{i,j}\gamma_{2,ij}A_{ij}^2 + \sum_{i,j}\gamma_{1,ij}|A_{ij}| + \mathcal{C}.$$

*Then, for every $\theta \in [0,1]$,*

$$\mathcal{J}_\Gamma(A_\theta) - \mathcal{J}_\Gamma(A_1) \le (1-\theta)B_T(R) - (1-\theta^2)\sum_{i,j}\gamma_{2,ij}R_{ij}^2 - (1-\theta)\sum_{i,j}\gamma_{1,ij}|R_{ij}|.$$

*In particular, full pruning is certified to decrease the weighted elastic-net objective whenever*

$$B_T(R) < \sum_{i,j}\gamma_{2,ij}R_{ij}^2 + \sum_{i,j}\gamma_{1,ij}|R_{ij}|.$$

*Proof location.* See Appendix C, Proof of Corollary 5.8.

**Corollary 5.9** (Weighted stationarity controls removable masks)**.** *Under the hypotheses of Corollary 5.8, assume the one-sided directional stationarity condition*

$$\liminf_{h\downarrow 0}\frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \ge -\varepsilon$$

*holds for some $\varepsilon \ge 0$. Then*

$$2\sum_{i,j}\gamma_{2,ij}R_{ij}^2 + \sum_{i,j}\gamma_{1,ij}|R_{ij}| \le B_T(R) + \varepsilon.$$

*More generally, if*

$$\mathcal{J}_\Gamma(A_\theta) \ge \mathcal{J}_\Gamma(A_1) - \varepsilon(1-\theta) \qquad \forall\theta \in [0,1],$$

*then for every $\theta < 1$,*

$$(1+\theta)\sum_{i,j}\gamma_{2,ij}R_{ij}^2 + \sum_{i,j}\gamma_{1,ij}|R_{ij}| \le B_T(R) + \varepsilon.$$

*Proof location.* See Appendix C, Proof of Corollary 5.9.

**Theorem 5.10** (One-sided stationarity controls removable masks). *Under the hypotheses of Theorem 5.5, assume the one-sided directional stationarity condition holds for some $\varepsilon \geq 0$:*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(A_{1-h}) - \mathcal{J}(A_1)}{h} \geq -\varepsilon.$$

*Then*

$$2\lambda_2 \|R\|_F^2 + \lambda_1 \|R\|_1 \leq B_T(R) + \varepsilon.$$

*More generally, if*

$$\mathcal{J}(A_\theta) \geq \mathcal{J}(A_1) - \varepsilon(1-\theta) \qquad \forall \theta \in [0,1],$$

*then for every $\theta < 1$,*

$$\lambda_2(1+\theta)\|R\|_F^2 + \lambda_1 \|R\|_1 \leq B_T(R) + \varepsilon.$$

*Proof location.* See Appendix C, Proof of Theorem 5.10.

**Theorem 5.11** (Deterministic multi-layer mask certificate). *Let $\mathcal{J}(\alpha)$ denote the full cross-entropy plus layerwise elastic-net objective. Consider an ordered finite sequence of mask-pruning operations on layers $j = 1, \ldots, M$. At step $j$, let*

$$A_j = S_j + R_j, \qquad S_j \odot R_j = 0,$$

*be the current layer decomposition after the previous $j-1$ pruning operations have already been applied, and let $B_j(R_j)$ be the corresponding data-path budget computed in that intermediate network. If $\alpha^{(0)}$ and $\alpha^{(M)}$ denote the parameters before and after the $M$ mask removals, then*

$$\mathcal{J}(\alpha^{(M)}) \leq \mathcal{J}(\alpha^{(0)}) - \sum_{j=1}^{M} \lambda_{2,j} \|R_j\|_F^2 - \sum_{j=1}^{M} \lambda_{1,j} \|R_j\|_1 + \sum_{j=1}^{M} B_j(R_j),$$

*where $\lambda_{2,j}$ and $\lambda_{1,j}$ are the elastic-net weights applied to layer $j$.*

*Proof location.* See Appendix C, Proof of Theorem 5.11.

**Corollary 5.12** (Deterministic multi-layer margin stability). *Use the ordered pruning sequence and notation of Theorem 5.11. Let $z^{(j)}(s)$ be the logit vector after the $j$-th mask operation, with $z^{(0)}$ denoting the original network and $z^{(M)}$ the final pruned network. Suppose that for each $s \in T$ and each step $j$ there is a deterministic bound $\eta_{j,s} \geq 0$ such that*

$$\|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \leq \eta_{j,s}.$$

*Define $\eta_s^{\mathrm{tot}} := \sum_{j=1}^{M} \eta_{j,s}$. Then*

$$\left|\mathcal{L}_{\mathrm{CE}}(\alpha^{(M)}) - \mathcal{L}_{\mathrm{CE}}(\alpha^{(0)})\right| \leq \frac{2}{|T|} \sum_{s \in T} \eta_s^{\mathrm{tot}}.$$

*Moreover, if $p_0(s) \in \arg\max_j z_j^{(0)}(s)$ and*

$$\Delta_0^{\mathrm{pred}}(s) := z_{p_0(s)}^{(0)}(s) - \max_{j \neq p_0(s)} z_j^{(0)}(s),$$

*then the predicted label at $s$ can change between the original and final networks only if*

$$\Delta_0^{\mathrm{pred}}(s) \le 2\eta_s^{\mathrm{tot}}.$$

*For the true-label margin*

$$\Delta_0^{\mathrm{true}}(s) := z^{(0)}_{C(s)}(s) - \max_{j \neq C(s)} z_j^{(0)}(s),$$

*the accuracy change obeys*

$$\left| \mathrm{acc}_{\alpha^{(M)}}(T) - \mathrm{acc}_{\alpha^{(0)}}(T) \right| \le \frac{1}{|T|} \#\{s \in T : |\Delta_0^{\mathrm{true}}(s)| \le 2\eta_s^{\mathrm{tot}}\}.$$

*Proof location.* See Appendix C, Proof of Corollary 5.12.

**Theorem 5.13** (Multi-layer stationarity controls jointly removable masks)**.** *Use the notation of Theorem 5.11. For $h \in [0,1]$, let $\alpha_h$ be the network obtained by applying the same ordered pruning operations, but replacing each current layer decomposition $A_j = S_j + R_j$ by the fractional mask $S_j + (1-h)R_j$. Assume that, for each $j$, the budget $B_j(R_j)$ is valid for this fractional step in the intermediate network, uniformly for $h \in [0,1]$, so that the cross-entropy increase at step $j$ is at most $hB_j(R_j)$. If the one-sided directional stationarity condition*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)})}{h} \ge -\varepsilon$$

*holds for some $\varepsilon \ge 0$, then*

$$2\sum_{j=1}^{M} \lambda_{2,j} \|R_j\|_F^2 + \sum_{j=1}^{M} \lambda_{1,j} \|R_j\|_1 \le \sum_{j=1}^{M} B_j(R_j) + \varepsilon.$$

*More generally, if*

$$\mathcal{J}(\alpha_h) \ge \mathcal{J}(\alpha^{(0)}) - \varepsilon h \qquad \forall h \in [0,1],$$

*then for every $h \in (0,1]$,*

$$\sum_{j=1}^{M} \lambda_{2,j}(2-h) \|R_j\|_F^2 + \sum_{j=1}^{M} \lambda_{1,j} \|R_j\|_1 \le \sum_{j=1}^{M} B_j(R_j) + \varepsilon.$$

*Proof location.* See Appendix C, Proof of Theorem 5.13.

**Corollary 5.14** (Gaussian sufficient condition for the data-path budget)**.** *Let $R \in \mathbb{R}^{n\times m}$ have independent entries $R_{ij} \sim N(0, g/n)$, independent of the collection $v_s := \psi_1(s)$, $s \in T$. Assume $L_s$ are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of $R$. Then for every finite $T$ and every $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$B_T(R) \le \frac{2}{|T|} \sum_{s \in T} L_s \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}}.$$

*Proof location.* See Appendix C, Proof of Corollary 5.14.

This is the one-layer random-matrix sufficient condition for the deterministic certificates above. In the concrete three-layer Gaussian model, the same concentration mechanism appears as Lemma A.13 and Theorem A.16 in Appendix A.

**Corollary 5.15** (MP-edge form of the Gaussian data-path budget)**.** *Under the hypotheses of Corollary 5.14, define the nominal rectangular MP singular-value edge of $R \in \mathbb{R}^{n\times m}$ by*

$$\sigma_{+,R} := \sqrt{g}\left(1+\sqrt{\frac{m}{n}}\right).$$

*Then, with probability at least $1-\delta$,*

$$B_T(R) \le \frac{2\sigma_{+,R}}{1+\sqrt{m/n}}\sqrt{\frac{2}{n}\log\frac{2n|T|}{\delta}}\,\frac{1}{|T|}\sum_{s\in T} L_s\|v_s\|_2.$$

*Thus, in the Gaussian model, the fitted MP edge controls the same data-path budget that appears in the deterministic pruning certificates, after normalization by the layer aspect ratio and by the data-dependent factors $L_s\|v_s\|_2$.*

*Proof location.* See Appendix C, Proof of Corollary 5.15.

This corollary is the direct mathematical bridge between the fitted MP edge used by the pruning algorithms and the deterministic data-path budget $B_T(R)$. The stylized spectral picture behind this edge is developed in Proposition A.30 and Corollaries A.23–A.27 in Appendix A.

**Corollary 5.16** (Sub-Gaussian sufficient condition for the data-path budget)**.** *Let $v_s = \psi_1(s)$, and assume $L_s$ are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of $R$. Let the rows $r_1,\dots,r_n$ of $R$ be independent and centered conditionally on those variables, and suppose that for some $c, g > 0$,*

$$\mathbb{P}(|\langle r_i, v\rangle| > t \mid v) \le 2\exp\left(-\frac{cnt^2}{g\|v\|_2^2}\right)$$

*for all $i$, $v$, and $t > 0$. Then, with probability at least $1-\delta$,*

$$B_T(R) \le \frac{2}{|T|}\sum_{s\in T} L_s\|v_s\|_2\sqrt{\frac{g}{cn}\log\frac{2n|T|}{\delta}}.$$

*Proof location.* See Appendix C, Proof of Corollary 5.16.

**Corollary 5.17** (Simultaneous multi-layer Gaussian data-path budget)**.** *Consider the ordered finite pruning sequence in Theorem 5.11. At step $j$, let the removed component be modeled by $R_j \in \mathbb{R}^{n_j\times m_j}$ with independent $N(0, g_j/n_j)$ entries, conditionally independent of the intermediate pre-layer vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. Let $\delta_1,\dots,\delta_M > 0$ satisfy $\sum_{j=1}^M \delta_j \le \delta$. Then, with probability at least $1-\delta$, all layerwise budgets satisfy*

$$B_j(R_j) \le \frac{2}{|T|}\sum_{s\in T} L_{j,s}\|v_{j,s}\|_2\sqrt{\frac{2g_j}{n_j}\log\frac{2n_j|T|}{\delta_j}} \qquad (1 \le j \le M).$$

*Consequently, on the same event, the deterministic multi-layer mask certificate gives*

$$\mathcal{J}(\alpha^{(M)}) \le \mathcal{J}(\alpha^{(0)}) - \sum_{j=1}^M \lambda_{2,j}\|R_j\|_F^2 - \sum_{j=1}^M \lambda_{1,j}\|R_j\|_1 + \sum_{j=1}^M \frac{2}{|T|}\sum_{s\in T} L_{j,s}\|v_{j,s}\|_2\sqrt{\frac{2g_j}{n_j}\log\frac{2n_j|T|}{\delta_j}}.$$

*Proof location.* See Appendix C, Proof of Corollary 5.17.

**Corollary 5.18** (Simultaneous multi-layer sub-Gaussian data-path budget)**.** *Consider the ordered finite pruning sequence in Theorem 5.11. At step $j$, let $R_j \in \mathbb{R}^{n_j \times m_j}$ have independent centered rows $r_{j,1}, \dots, r_{j,n_j}$, conditionally on the intermediate pre-layer vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. Suppose that for constants $c_j, g_j > 0$,*

$$\mathbb{P}(|\langle r_{j,i}, v\rangle| > t \mid v) \leq 2 \exp\left(-\frac{c_j n_j t^2}{g_j \|v\|_2^2}\right)$$

*for every $j, i, v, t > 0$. Let $\delta_1, \dots, \delta_M > 0$ satisfy $\sum_{j=1}^{M} \delta_j \leq \delta$. Then, with probability at least $1 - \delta$,*

$$B_j(R_j) \leq \frac{2}{|T|} \sum_{s \in T} L_{j,s} \|v_{j,s}\|_2 \sqrt{\frac{g_j}{c_j n_j} \log \frac{2n_j|T|}{\delta_j}} \qquad (1 \leq j \leq M).$$

*Consequently, the deterministic multi-layer mask certificate holds on the same event with the sum of these displayed bounds replacing $\sum_j B_j(R_j)$.*

*Proof location.* See Appendix C, Proof of Corollary 5.18.

*Remark* 5.19 (Mask decompositions versus additive random decompositions). The mask certificate treats the actual unstructured pruning operation:

$$S = M \odot A, \qquad R = (1 - M) \odot A,$$

so $S \odot R = 0$ by construction and no independence or Gaussianity is required. This is distinct from the additive Gaussian model $A = S + R$ with full i.i.d. noise. In the additive Gaussian model, disjoint support generally fails, and the relevant replacement for the mask identity is a cross-term condition such as $\langle S, R\rangle_F = o_{\mathbb{P}}(1)$.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*Remark* .20 (Local Lipschitz constants $L_s$). The data-path budget $B_T(R)$ in Eq. (8) carries per-sample post-layer Lipschitz constants $L_s$ that bound the change in the logit at sample $s$ induced by a perturbation in the projected layers output. For ReLU MLPs these constants are products of operator norms of subsequent weight matrices and are easily bounded. For modern Vision Transformers with LayerNorm, attention, GELU, and residual connections, the local Lipschitz constant is finite at any non-degenerate operating point but is not estimated in this paper. The framework treats $\{L_s\}$ as an architecture-dependent quantity to be controlled; this paper does not numerically certify it. The empirical 282-cell audit (the methodology paper) reports the operator-norm proxy $\hat{B}_T = \|R\|_{\mathrm{op}} \cdot \overline{\|\psi\|_\infty}$ as a within-architecture predictor of the realised top-1 drop; this proxy correlates strongly with the post-FT accuracy gap ($r = +0.91$ pooled across 18 architectures), but is not itself an upper bound on $B_T(R)$.

# A Gaussian Specialization and Spectral Interpretation

This appendix records the Gaussian specialization and spectral interpretation. The deterministic mask certificates in Section 5 are the theory closest to the implemented unstructured pruning operation; the Gaussian results below provide sufficient-condition and interpretation results for admissible random-like components.

## A.1 Assumptions for the Gaussian specialization

Throughout the asymptotic statements in this section, the training set $T$ is finite and fixed independently of $N$, unless a result explicitly states a different finite set or uses a confidence parameter.

Recall the induced $\ell_\infty$ operator norm of a matrix $W = (w_{ij}) \in \mathbb{R}^{n\times m}$:

$$\|W\|_\infty := \max_{1\le i\le n} \sum_{j=1}^{m} |w_{ij}|.$$

Whenever $\lambda_+$ denotes the right MP edge for the eigenvalues of

$$X_\ell = \frac{1}{N} W_\ell^\top W_\ell,$$

we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding edge on the singular-value scale.

**Assumption A.1** (Architecture and outer-layer scaling)**.** We consider the simplified three-layer network

$$X(\alpha, s) = \lambda \circ W_3 \circ \lambda \circ W_2 \circ \lambda \circ W_1 s, \qquad s \in T, \tag{9}$$

$$\phi(\alpha, s) = \rho \circ X(\alpha, s), \tag{10}$$

where $\lambda$ is the coordinatewise absolute value or ReLU, and $W_1, W_2, W_3$ are the weight matrices. Biases are omitted for simplicity.

For fixed training time $t$, define

$$a(N, s) := \|W_3\|_\infty \, \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}}. \tag{11}$$

We assume that for every $s \in T$, $a(N, s) \to 0$ as $N \to \infty$. The same assumption is imposed at any admissible limit point considered below.

*Remark* A.2 (Interpretation of Assumption A.1)*.* This is the concrete version of the abstract factorization Assumption 4.1. The target layer is $W_2$; the computations before $W_2$ play the role of $\psi_1$, and the computations after $W_2$ play the role of $\psi_2$. The quantity $a(N, s)$ is the Gaussian-model version of $a_\phi(N, s) = h_\phi(s) d_1(N)$: it is the logit perturbation scale predicted by multiplying an $N^{-1/2}\sqrt{\log N}$-size random effect through the outer layer. The assumption $a(N, s) \to 0$ is therefore the concrete data-path smallness condition needed for the loss and margin bounds.

**Assumption A.3** (Admissible Gaussian decomposition of the middle layer)**.** For each training time $t$, the middle layer $W_2(t) \in \mathbb{R}^{N\times N}$ admits a decomposition

$$W_2(t) = R_2(t) + S_2(t),$$

where:

1. $R_2(t)$ has i.i.d. centered Gaussian entries

$$R_2(t)_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g(t)}{N}\right), \qquad 0 \le g(t) \le 1;$$

   where $g(t)$ is a deterministic variance scale;

2. $R_2(t)$ is independent of $S_2(t)$;

3. $R_2(t)$ is also independent of the outer-layer parameters $W_1(t)$ and $W_3(t)$.

*Remark* A.4. The variance scaling $g(t)/N$ ensures that $R_2(t)$ has bounded operator norm as $N \to \infty$. Rectangular variants $W_2(t) \in \mathbb{R}^{N\times M}$ with $M/N \to c \in (0, \infty)$ can be treated similarly.

*Remark* A.5 (Interpretation of Assumption A.3). This is a stylized additive-noise model, not a claim that the implemented ViT mask literally samples an independent Gaussian matrix. It is used to prove a sufficient condition: if the removed component behaves like an independent variance-$g/N$ random matrix along the data path, then it satisfies the perturbation bounds required by the abstract theory. In the notation of Assumption 4.3, it gives $J(v) = \|v\|_2$, $d_1(N) \asymp \sqrt{\log N/N}$, and $d_2(N) \asymp 1/N$. The same variance scale has MP singular-value edge $2\sqrt{g}$ in the square case, which is why the fitted MP edge is a natural diagnostic in the algorithms.

**Assumption A.6** (Sub-root-$N$ structured component for the loss theorems)**.** For each training time $t$, the structured component $S_2(t)$ satisfies

$$\|S_2(t)\|_F \le B_N, \qquad B_N = o(\sqrt{N}),$$

for some positive deterministic sequence $B_N$ independent of $t$. Equivalently, the normalized size condition $\|S_2(t)\|_F^2/N \to 0$ is enough. The loss-reduction theorems use this assumption only to prove

$$\langle S_2(t), R_2(t)\rangle_F = o_{\mathbb{P}}(1);$$

one may replace it directly by this cross-term condition whenever it is known by another argument.

*Remark* A.7 (Interpretation of Assumption A.6). This assumption is a convenient Gaussian sufficient condition for the cross-term bound $\langle S_2, R_2\rangle_F = o_{\mathbb{P}}(1)$. Conditional on $S_2$,

$$\mathrm{Var}(\langle S_2, R_2\rangle_F \mid S_2) = \frac{g(t)}{N}\|S_2\|_F^2.$$

Thus $B_N = o(\sqrt{N})$ forces the variance to vanish. The loss theorems would remain valid if this cross-term condition were verified by some other argument. This is the Gaussian counterpart of the explicit cross-term hypothesis used in the companion methodology paper.

**Assumption A.8** (Approximate low rank for the spectral corollaries)**.** Whenever a spectral conclusion is invoked, we additionally assume that the limit structured component $S_2^*(N)$ is approximately low rank in the sense that there exists a sequence $k_N = o(N)$ such that

$$\sum_{i>k_N} s_i\big(S_2^*(N)\big)^2 \to 0 \qquad \text{in probability as } N \to \infty.$$

The exact fixed-rank regime is the special case $k_N \equiv r$.

*Remark* A.9 (Interpretation of Assumption A.8). This is the Gaussian section's version of Assumption 4.7. It is needed only for statements about the surviving singular spectrum and subspaces of $S_2^*$, not for the loss-reduction theorem. If $S_2^*(N)$ has exact finite rank $r$ independent of $N$, take $k_N = r$; then $k_N = o(N)$ and the tail sum is exactly zero. If the rank grows but remains $o(N)$, take $k_N = \mathrm{rank}(S_2^*(N))$. If $S_2^*(N)$ is not exactly low rank, the assumption allows it as long as the squared singular-value tail beyond $k_N$ tends to zero.

*Remark* A.10 (No spectral-separation assumption). Assumptions A.6–A.8 do *not* require the informative singular values of $S_2$ to lie above the MP edge $\sigma_+$. Thus signal directions may still be embedded in the bulk created by $R_2$.

**Definition A.11** (Admissible decomposition)**.** A decomposition $W_2 = S_2 + R_2$ is called *admissible* if it satisfies Assumptions A.3–A.6. At a limit point $W_2^*(N)$, an admissible decomposition $W_2^*(N) =$

$S_2^*(N) + R_2^*(N)$ additionally requires that the outer-layer scaling condition from Assumption A.1 holds for the corresponding limit outer layers $W_1^*(N), W_3^*(N)$, i.e.

$$a_*(N, s) := \|W_3^*(N)\|_\infty \|W_1^*(N)s\|_2 \sqrt{\frac{\log(2N)}{N}} \to 0 \qquad \text{for each } s \in T.$$

A *spectrally admissible* decomposition is an admissible decomposition that also satisfies Assumption A.8.

**Definition A.12** (Local-path convergence of an admissible decomposition)**.** An admissible path decomposition

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

is said to converge locally to an admissible limit decomposition

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

if, for each fixed $N$,

$$\|S_2(t, N) - S_2^*(N)\|_F + \|R_2(t, N) - R_2^*(N)\|_F \to 0 \qquad \text{in probability as } t \to \infty.$$

## A.2 Key perturbation lemma

**Lemma A.13.** *Suppose Assumptions A.1 and A.3 hold. Define*

$$\tau_N(s) := 2\sqrt{2}\, \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2}\, a(N, s).$$

*If we replace $W_2 = S_2 + R_2$ by $S_2$, then*

$$\mathbb{P}(\|X(s, \alpha_{S_2}) - X(s, \alpha_{W_2})\|_\infty \le \tau_N(s)) \ge 1 - \frac{1}{N}.$$

*Proof location.* See Appendix C, Proof of Lemma A.13.

For later use, define the classification margin on a finite test set $T'$:

$$\delta X(s, \alpha) := X_{C(s)}(s, \alpha) - \max_{j \ne C(s)} X_j(s, \alpha),$$

and the corresponding accuracy

$$\mathrm{acc}_\alpha(T') := \frac{1}{|T'|} \#\{s \in T' : \delta X(s, \alpha) > 0\}.$$

**Corollary A.14** (Margin-sensitive label stability)**.** *Suppose Assumptions A.1 and A.3 hold. Let $T'$ be a finite test set. Then*

$$\mathbb{P}\Big( \big| \mathrm{acc}_{\alpha_{W_2}}(T') - \mathrm{acc}_{\alpha_{S_2}}(T') \big| \le \frac{1}{|T'|} \# \Big\{ s \in T' : |\delta X(s, \alpha_{W_2})| \le 2\tau_N(s) \Big\} \Big) \ge 1 - \frac{|T'|}{N}.$$

*In particular, if*

$$\min_{s \in T'} |\delta X(s, \alpha_{W_2})| > 2 \max_{s \in T'} \tau_N(s),$$

*then*

$$\mathrm{acc}_{\alpha_{W_2}}(T') = \mathrm{acc}_{\alpha_{S_2}}(T')$$

*with probability at least $1 - \frac{|T'|}{N}$.*

*Proof location.* See Appendix D, Proof of Corollary A.14.

## A.3 A theorem on how removing randomness reduces the regularized loss

We work with the $\ell_2$-regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|}\sum_{s\in T}\log\big(\phi_{C(s)}(s,\alpha(t))\big) + \lambda_{\text{reg}}\sum_{i=1}^{L}\|W_i(t)\|_F^2, \qquad \lambda_{\text{reg}} > 0. \tag{12}$$

We first isolate the unregularized perturbation term.

**Lemma A.15.** *Suppose Assumptions A.1–A.3 hold. Define the unregularized cross-entropy*

$$L_{\cancel{\text{reg}}}(\alpha) := -\frac{1}{|T|}\sum_{s\in T}\log\big(\phi_{C(s)}(s,\alpha)\big).$$

*Then there exists $N_0$ such that for all $N \geq N_0$,*

$$\mathbb{P}\left(\big|L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})\big| \leq \frac{4\sqrt{2}}{|T|}\sum_{s\in T} a(N,s)\right) \geq 1 - \frac{|T|}{N}.$$

*In particular,*

$$L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2}) \to 0 \qquad \text{in probability as } N \to \infty.$$

*Proof location.* See Appendix C, Proof of Lemma A.15.

**Theorem A.16** (Confidence-parameter finite-sample perturbation bound)**.** *Suppose Assumptions A.1 and A.3 hold, and fix a finite set $T_0$ and $\delta \in (0,1)$. Define*

$$\tau_{N,\delta}(s) := \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{2g(t)\log(2N|T_0|/\delta)}{N}}.$$

*Then, with probability at least $1-\delta$,*

$$\|X(s,\alpha_{S_2}) - X(s,\alpha_{W_2})\|_\infty \leq \tau_{N,\delta}(s) \qquad \forall s \in T_0.$$

*Consequently, for $T_0 = T$,*

$$\big|L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})\big| \leq \frac{2}{|T|}\sum_{s\in T}\tau_{N,\delta}(s)$$

*with probability at least $1-\delta$. If Assumption A.6 also holds, then for every $\eta > 0$,*

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}}\|R_2\|_F^2 + \frac{2}{|T|}\sum_{s\in T}\tau_{N,\delta}(s) + 2\lambda_{\text{reg}}\eta\right) \geq 1 - \delta - 2e^{-N\eta^2/(2B_N^2)}.$$

*Proof location.* See Appendix C, Proof of Theorem A.16.

**Corollary A.17** (Growing-set Gaussian loss bound)**.** *Suppose Assumptions A.1–A.3 hold with a finite labeled set $T_N$ that may depend on $N$, and let $\delta_N \in (0,1)$ with $\delta_N \to 0$. Let $L_{\cancel{\text{reg}},T_N}$ denote cross-entropy averaged over $T_N$, and let $L_{T_N}$ denote the corresponding regularized objective with the same Frobenius penalty as in (12). Define*

$$b_{N,\delta_N}(T_N) := \frac{2}{|T_N|}\sum_{s\in T_N}\|W_3\|_\infty\|W_1 s\|_2\sqrt{\frac{2g(t)\log(2N|T_N|/\delta_N)}{N}}.$$

*If $b_{N,\delta_N}(T_N) \to 0$ in probability, then the cross-entropy loss averaged over $T_N$ satisfies*

$$L_{\cancel{\mathrm{reg}},T_N}(\alpha_{S_2}) - L_{\cancel{\mathrm{reg}},T_N}(\alpha_{W_2}) \to 0 \qquad \textit{in probability.}$$

*If Assumption A.6 also holds, then*

$$L_{T_N}(\alpha_{S_2}) \leq L_{T_N}(\alpha_{W_2}) - \lambda_{\mathrm{reg}}\|R_2\|_F^2 + o_{\mathbb{P}}(1).$$

*Proof location.* The bound follows by union-bounding Lemma A.13 over the growing set $T_N$ and invoking the Gaussian concentration estimate of Theorem A.16 under the growing-set hypothesis $|T_N|/N \to 0$.

**Theorem A.18.** *Suppose Assumptions A.1–A.6 hold. If $W_2 = R_2 + S_2$ and we replace $W_2$ by $S_2$, then for every fixed training time $t$ and every $\eta > 0$,*

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\mathrm{reg}}\|R_2\|_F^2 + \frac{4\sqrt{2}}{|T|}\sum_{s\in T} a(N,s) + 2\lambda_{\mathrm{reg}}\eta\right) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}.$$

*In particular,*

$$L(\alpha_{S_2}) = L(\alpha_{W_2}) - \lambda_{\mathrm{reg}}\|R_2\|_F^2 + o_{\mathbb{P}}(1) \qquad (N \to \infty).$$

*Proof location.* See Appendix D, Proof of Theorem A.18.

**Corollary A.19.** *Suppose Assumptions A.1–A.6 hold. For any $\vartheta \in [0,1]$, define*

$$W_2(\vartheta) = S_2 + \vartheta R_2.$$

*Then for every fixed training time $t$ and every $\eta > 0$, there is an event $E_{N,\eta}$ with*

$$\mathbb{P}(E_{N,\eta}) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}$$

*such that on $E_{N,\eta}$, for every $\vartheta \in [0,1]$,*

$$L(\alpha_{W_2(\vartheta)}) \leq L(\alpha_{W_2}) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R_2\|_F^2 + \frac{4\sqrt{2}(1-\vartheta)}{|T|}\sum_{s\in T} a(N,s) + 2\lambda_{\mathrm{reg}}(1-\vartheta)\eta.$$

*In particular, for each fixed $\vartheta \in [0,1]$,*

$$L(\alpha_{W_2(\vartheta)}) = L(\alpha_{W_2}) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R_2\|_F^2 + o_{\mathbb{P}}(1) \qquad (N \to \infty).$$

*The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0,1]$.*

*Proof location.* See Appendix D, Proof of Corollary A.19.

**Theorem A.20** (Entrywise thresholding in a sparse signal-plus-noise model)**.** *Let $W_N = S_N + R_N \in \mathbb{R}^{N\times M_N}$, where $R_{ij}$ are independent $N(0, g_N/N)$ random variables, $g_N \geq 0$, and $S_N$ is deterministic or independent of $R_N$. Fix $\delta \in (0,1)$ and set*

$$t_{N,\delta} := \sqrt{\frac{2g_N \log(2NM_N/\delta)}{N}}.$$

*Define the entrywise hard-thresholded matrix*

$$\mathcal{H}_t(W)_{ij} := W_{ij}\,\mathbf{1}_{\{|W_{ij}|>t\}}.$$

*Then, with probability at least $1-\delta$,*

$$\{(i,j) : |(S_N)_{ij}| > 2t_{N,\delta}\} \subseteq \operatorname{supp}\left(\mathcal{H}_{t_{N,\delta}}(W_N)\right) \subseteq \operatorname{supp}(S_N).$$

*Equivalently, thresholding creates no false positives and can miss only entries of $S_N$ whose magnitude is at most $2t_{N,\delta}$:*

$$\operatorname{supp}(S_N) \setminus \operatorname{supp}\left(\mathcal{H}_{t_{N,\delta}}(W_N)\right) \subseteq \{(i,j) : 0 < |(S_N)_{ij}| \le 2t_{N,\delta}\}.$$

*Let $\mathcal{B}_{N,\delta}$ be the beta-min event*

$$\mathcal{B}_{N,\delta} := \left\{ \min_{(i,j)\in\operatorname{supp}(S_N)} |(S_N)_{ij}| > 2t_{N,\delta} \right\},$$

*with the condition interpreted as vacuous if $\operatorname{supp}(S_N) = \emptyset$. If $\mathbb{P}(\mathcal{B}_{N,\delta}) \ge 1-\gamma$, then entrywise thresholding $W_N$ at level $t_{N,\delta}$ recovers the support of $S_N$ exactly with probability at least $1-\delta-\gamma$:*

$$\operatorname{supp}\left(\mathcal{H}_{t_{N,\delta}}(W_N)\right) = \operatorname{supp}(S_N).$$

*In particular, if $\mathcal{B}_{N,\delta}$ holds deterministically, or almost surely when $S_N$ is random, the recovery probability is at least $1-\delta$. This result is a stylized support-recovery statement. Without beta-min separation it still gives a no-false-positive and large-signal-retention certificate, but exact support recovery requires all nonzero structured entries to clear the $2t_{N,\delta}$ noise buffer. It is not a proof that the ViT pruning masks identify the true structured support.*

*Proof location.* See Appendix C, Proof of Theorem A.20.

## A.4 A theorem on the asymptotic magnitude of the admissible noise

We now replace exact local minimality by a weaker one-sided stationarity condition along the denoising direction $R_2^*(N)$.

**Theorem A.21.** *Suppose Assumptions A.1–A.6 hold along training. Assume that:*

1. *for each $N$, the parameter vector $\alpha(t,N)$ converges in probability as $t \to \infty$ to a limit $\alpha^*(N)$;*
2. *the limiting middle layer $W_2^*(N)$ admits an admissible decomposition*

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

*in the sense of Definition A.11.*

*For $\vartheta \in [0,1]$, let $\alpha_\vartheta^*(N)$ denote the parameter vector obtained from $\alpha^*(N)$ by replacing $W_2^*(N)$ with $S_2^*(N) + \vartheta R_2^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event*

$$\liminf_{h\downarrow 0} \frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \ge -\epsilon_N$$

*has probability tending to one. Define*

$$\bar{a}_N := \frac{4\sqrt{2}}{|T|} \sum_{s\in T} a_*(N,s).$$

*Then*

$$\|R_2^*(N)\|_F^2 \le \frac{\bar{a}_N + \epsilon_N}{2\lambda_{\mathrm{reg}}} + o_{\mathbb{P}}(1).$$

*In particular, if $\bar{a}_N + \epsilon_N \to 0$ in probability, then for every $\epsilon > 0$,*

$$\lim_{N\to\infty} \mathbb{P}\big(\|R_2^*(N)\|_F \le \epsilon\big) = 1.$$

*If, in addition, there exists an admissible path decomposition*

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

*which converges locally to the limit decomposition in the sense of Definition A.12, then for every $\epsilon > 0$,*

$$\lim_{N\to\infty} \lim_{t\to\infty} \mathbb{P}\big(\|R_2(t, N)\|_F \le \epsilon\big) = 1.$$

*Proof location.* See Appendix D, Proof of Theorem A.21.

*Remark* A.22. Theorem A.21 uses only a one-sided directional condition: a genuine local minimum is a sufficient special case with $\epsilon_N \equiv 0$. The hypothesis is local and one-sided at $\vartheta = 1$, rather than a global lower bound along the full denoising path. The second statement extends the limit-point conclusion to the full training path for a chosen admissible decomposition that converges to the limit decomposition. The Gaussian hypothesis ensures that $\|R_2^*(N)\|_F$ has no atom at any positive threshold $\epsilon$, so convergence in probability along the path implies convergence of the corresponding distribution functions at $\epsilon > 0$.

**Corollary A.23** (Variance-scale collapse at the limit point)**.** *Under the hypotheses of Theorem A.21, write the admissible Gaussian limit perturbation as*

$$R_2^*(N)_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

*Here $g_*(N)$ is the deterministic variance scale of the limit perturbation. Then*

$$g_*(N)\, N \to 0.$$

*In particular, in the square case the MP singular-value edge of the admissible limit perturbation satisfies*

$$\sigma_{+,*}^{\mathrm{bulk}}(N) = 2\sqrt{g_*(N)} \to 0.$$

*For a rectangular $N \times M_N$ perturbation with entries $N(0, g_*(N)/N)$, the corresponding Frobenius collapse condition is $g_*(N)M_N \to 0$; when $M_N/N \to c \in (0, \infty)$, this is equivalent to $g_*(N)N \to 0$.*

*Proof location.* See Appendix D, Proof of Corollary A.23.

**Corollary A.24** (Bulk collapse and spike stabilization at the limit point)**.** *Under the hypotheses of Theorem A.21, assume additionally that the limit decomposition is spectrally admissible in the sense of Definition A.11. Let*

$$s_1\big(W_2^*(N)\big) \ge \cdots \ge s_N\big(W_2^*(N)\big)$$

*denote the singular values of $W_2^*(N)$, and let*

$$\sigma_1^*(N) \ge \sigma_2^*(N) \ge \cdots \ge 0$$

*denote the singular values of $S_2^*(N)$. Let $k_N = o(N)$ be as in Assumption A.8. Then:*

1. $\|R_2^*(N)\|_{\mathrm{op}} \to 0$ *in probability;*

2. *for each fixed* $i \geq 1$,
$$\left|s_i(W_2^*(N)) - \sigma_i^*(N)\right| \to 0 \qquad \textit{in probability};$$

3. $$\sum_{i>k_N} s_i(W_2^*(N))^2 \to 0$$
*in probability;*

4. *for every* $\epsilon > 0$,
$$\nu_{W_2^*(N)}((\epsilon, \infty)) \to 0 \qquad \textit{in probability},$$
*where* $\nu_{W_2^*(N)}$ *is the singular-value empirical spectral distribution.*

*If, in addition,* $\sigma_i^*(N) \to \bar{\sigma}_i$ *for some constants* $\bar{\sigma}_i \geq 0$, *then*
$$s_i(W_2^*(N)) \to \bar{\sigma}_i \qquad \textit{in probability for each fixed } i \geq 1.$$

*Proof location.* See Appendix D, Proof of Corollary A.24.

**Corollary A.25** (Singular-subspace stabilization under an asymptotic gap condition)**.** *Under the hypotheses of Theorem A.21, let*
$$\sigma_1^*(N) \geq \sigma_2^*(N) \geq \cdots \geq 0$$
*denote the singular values of* $S_2^*(N)$. *Fix* $k \geq 1$, *and define*
$$\delta_k(N) := \sigma_k^*(N) - \sigma_{k+1}^*(N),$$
*with the convention* $\sigma_j^*(N) = 0$ *for* $j$ *beyond the rank of* $S_2^*(N)$. *Assume that* $\mathbb{P}(\delta_k(N) > 0) \to 1$ *and that, after setting the ratio below to* $+\infty$ *on the event* $\{\delta_k(N) = 0\}$,
$$\frac{\|R_2^*(N)\|_{\mathrm{op}}}{\delta_k(N)} \to 0 \qquad \textit{in probability as } N \to \infty.$$
*Let* $U_{*,k}(N)$ *and* $V_{*,k}(N)$ *be orthonormal bases for the top-*$k$ *left and right singular subspaces of* $S_2^*(N)$, *respectively. Let* $\widehat{U}_k(N), \widehat{V}_k(N) \in \mathbb{R}^{N\times k}$ *be matrices whose columns are orthonormal left and right singular vectors corresponding to the top* $k$ *singular values of* $W_2^*(N)$, *and define the orthogonal projectors*
$$P_{U_*,k}(N) := U_{*,k}(N)U_{*,k}(N)^\top, \qquad P_{V_*,k}(N) := V_{*,k}(N)V_{*,k}(N)^\top,$$
$$\widehat{P}_{U,k}(N) := \widehat{U}_k(N)\widehat{U}_k(N)^\top, \qquad \widehat{P}_{V,k}(N) := \widehat{V}_k(N)\widehat{V}_k(N)^\top.$$
*Then*
$$\|\widehat{P}_{U,k}(N) - P_{U_*,k}(N)\|_{\mathrm{op}} \to 0, \qquad \|\widehat{P}_{V,k}(N) - P_{V_*,k}(N)\|_{\mathrm{op}} \to 0$$
*in probability as* $N \to \infty$.

*Proof location.* See Appendix D, Proof of Corollary A.25.

*Remark* A.26. Corollary A.25 converts spectral shrinkage into geometric stability. Once the admissible random component is asymptotically smaller than the relevant spectral gap, the corresponding left/right singular subspaces of the learned layer become asymptotically identifiable. This is strictly weaker than requiring a fixed positive spike floor: the structured gap $\delta_k(N)$ may itself tend to 0, provided it dominates the residual operator-norm noise. For pruning, the result concerns both the amount of surviving spectrum and the directions that survive.

**Corollary A.27** (Eventual Gaussian supercriticality of persistent spikes)**.** *Under the hypotheses of Theorem A.21, write*

$$R_2^*(N)_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

*Assume additionally that, for this corollary only, the admissible limit decomposition lies in the square Gaussian finite-rank deformation regime of Appendix D; in particular, $S_2^*(N)$ has rank at most $r_0$ for some deterministic $r_0$ independent of $N$. Assume that for some fixed $k \geq 1$ there exist constants*

$$\bar{\sigma}_1 > \cdots > \bar{\sigma}_k > 0$$

*such that*

$$\sigma_i^*(N) \to \bar{\sigma}_i \qquad \text{in probability as } N \to \infty$$

*for each $1 \leq i \leq k$. Then:*

1. $$\mathbb{P}\big(\sigma_i^*(N) > \sqrt{g_*(N)}\big) \to 1 \qquad \text{for each } 1 \leq i \leq k;$$

2. *in the sense of Theorem D.3, these $k$ spikes are eventually supercritical, so the corresponding singular values of $W_2^*(N)$ form BBP outlier clusters above the Gaussian bulk edge $2\sqrt{g_*(N)}$. Moreover, if the $i$-th spike is separated from the other nonzero singular values of $S_2^*(N)$ by a gap $\Delta_i(N) > 0$ with*
$$\frac{\|R_2^*(N)\|_{\text{op}}}{\Delta_i(N)} \to 0 \qquad \text{in probability,}$$
*then the rank-one left and right spectral projectors of $W_2^*(N)$ associated with this spike converge in operator norm to the corresponding projectors of $S_2^*(N)$;*

3. *defining the inverse-BBP estimator on the event $s_i(W_2^*(N))^2 > 4g_*(N)$, whose probability tends to one, by*
$$\widehat{\sigma}_i^{\text{BBP}}(N) := \frac{1}{2}\left(s_i(W_2^*(N)) + \sqrt{s_i(W_2^*(N))^2 - 4g_*(N)}\right),$$
*and defining it arbitrarily off that event, one has*
$$\widehat{\sigma}_i^{\text{BBP}}(N) - \sigma_i^*(N) \to 0 \qquad \text{in probability}$$
*for each $1 \leq i \leq k$.*

*Proof location.* See Appendix D, Proof of Corollary A.27.

*Remark* A.28. Corollary A.27 strengthens the vanishing-randomness theorem beyond bulk collapse. When an informative direction stays macroscopically nonzero while the admissible Gaussian variance decays, the direction is preserved in the limit and eventually crosses the BBP threshold, becoming statistically identifiable as an outlier before the noise has fully disappeared. The explicit inverse-BBP map then gives a principled bias-corrected estimate of the underlying spike strength.

*Remark* A.29 (Assumption map for the Gaussian section). Among the theorem-like results proved in the Gaussian section, the dependence on the hypotheses is as follows:

1. Theorem 2.2 and Proposition A.30 are standalone random-matrix statements and do not rely on Assumptions A.1–A.6.

2. Lemma A.13 and Corollary A.14 use only Assumptions A.1–A.3.

3. The first part of Theorem A.16 uses only Assumptions A.1–A.3.
4. Theorem A.18, Corollary A.19, and the regularized part of Theorem A.16 use Assumptions A.1–A.6; Assumption A.6 is used only to control $\langle S_2, R_2\rangle_F$.
5. Theorem A.20 is a standalone sparse signal-plus-noise statement.
6. Theorem A.21 uses Assumptions A.1–A.6 together with the one-sided directional stationarity hypothesis stated in the theorem; its pathwise conclusion additionally uses Definition A.12.
7. Corollary A.23 is a consequence of Theorem A.21.
8. Corollary A.24 additionally uses Assumption A.8.
9. Corollary A.25 is a further consequence of Theorem A.21 plus the asymptotic gap condition

$$\|R_2^*(N)\|_{\mathrm{op}}/\delta_k(N) \to 0 \qquad \text{in probability.}$$

10. Corollary A.27 uses Theorem A.21 together with a persistent-spike assumption $\sigma_i^*(N) \to \bar{\sigma}_i > 0$ and the additional exact finite-rank deformation hypothesis stated in the corollary itself.

**Proposition A.30** (Spectral consequence of an MP-scale admissible random component)**.** *Let $M_N/N \to c \in (0,\infty)$. Let*

$$W_N = R_N + S_N,$$

*where $R_N \in \mathbb{R}^{N\times M_N}$ has i.i.d. $\mathcal{N}(0, g_N/N)$ entries with $g_N \to g_\infty > 0$, and $S_N$ is independent of $R_N$ with*

$$\operatorname{rank}(S_N) = o(\min\{N, M_N\}).$$

*Then:*

1. *the singular-value ESD $\nu_{R_N}$, defined as in Definition 2.1 over the* $\min\{N, M_N\}$ *singular values of $R_N$, converges almost surely to the pushforward of the* nonzero covariance-spectrum part *of $\mu_{\mathrm{MP}}^{c,g_\infty}$ under $x \mapsto \sqrt{x}$; equivalently, when $c > 1$ one first removes the MP atom at $0$ and renormalizes. Its support is*

$$\Big[\sqrt{g_\infty}\,|1-\sqrt{c}|,\ \sqrt{g_\infty}(1+\sqrt{c})\Big];$$

2. *the singular-value ESD of $W_N$ has the same weak limit as $\nu_{R_N}$, because*

$$\sup_{x\in\mathbb{R}} |\nu_{W_N}((-\infty,x]) - \nu_{R_N}((-\infty,x])| \le \frac{\operatorname{rank}(S_N)}{\min\{N, M_N\}} \to 0;$$

3. *suppose, in addition, that $S_N$ has fixed rank $r$, its nonzero singular values $\theta_i(N)$ converge to positive limits, and the pair $(R_N, S_N)$ satisfies the finite-rank deformation hypotheses of Theorem D.4 with deterministic edge $b_+(N)$, critical threshold $\theta_c(N)$, and inverse-D outlier map. Then every spike for which $\mathbb{P}(\theta_i(N) > \theta_c(N) + \gamma) \to 1$ for some $\gamma > 0$ produces an outlier singular value $s_i(W_N)$ satisfying*

$$s_i(W_N) - D_N^{-1}\left(\frac{1}{\theta_i(N)^2}\right) \to 0 \qquad \textit{in probability,}$$

*and the inverse-D estimator from Theorem D.4 recovers $\theta_i(N)$ in probability. In the square Gaussian case $c = 1$, this reduces to the usual BBP outlier transition.*

*Thus any admissible random component that remains at MP scale leaves a persistent MP-type bulk in the singular spectrum.*

*Proof location.* See Appendix D, Proof of Proposition A.30.

The Gaussian theory starts from a decomposition $W_2 = S_2 + R_2$. If $R_2$ remains at MP scale, it produces a persistent MP-type bulk in the spectrum of $W_2$; removing $R_2$ collapses that bulk and lowers the regularized objective by the term $\lambda_{\mathrm{reg}}\|R_2\|_F^2$, up to the perturbation bounds above.

*Remark* A.31. Theorem A.21 is stronger than the purely spectral statement. A visible MP bulk rules out only *MP-scale* admissible noise. The theorem, together with local-path convergence, rules out even admissible sub-MP noise at the level of the Frobenius norm.

# B Supplementary Numerical Results

The supplementary numerical material covers MP-based pruning of fully connected DNNs on Fashion MNIST, outer-layer scaling diagnostics, and the noise-perturbation experiment. The extended version appears in the companion *methodology and extended-numerics* paper, `cast_2e/methodology.pdf` (LaTeX source: `cast_2e/methodology.tex`). The repository root is `https://github.com/yspennstate/RMT_based_pruning_in_deep_learning`.

## B.1 Interpretation of the ViT-B/16 sweep

For ViT pruning, the classical RMT cycle is used primarily for layerwise budget allocation rather than direct removal of sub-edge singular components, which is brittle for modern transformers. MP-like layers, especially MLP projections, absorb more pruning, while attention projections require more protection. This matches the deterministic theory in Section 5: once a mask is fixed, the relevant object is the perturbation it induces on the data path, and the fitted MP edge $\sigma_+(\ell)$ acts as a layerwise proxy for where that perturbation is more likely to be admissible. The full layer-by-layer interpretation is in the methodology paper.

## B.2 Multi-architecture cert connection

**Why the slope steepens with sparsity.** The fitted log-parameter slope steepens monotonically from $-0.73$ pp/decade at $s = 0.50$ to $-4.18$ pp/decade at $s = 0.70$ ($R \approx -0.4$ across the 17 rows). This larger-model, smaller-drop ordering is consistent with the random-matrix view of SER: the MP bulk grows with width while the spike count is roughly architecture-fixed, so a larger model can be pushed deeper into the bulk before the protocol crosses the spike-detection threshold. The full discussion and the per-architecture certificate audit are in the methodology paper.

**Connecting the deterministic certificates to the multi-architecture numerics.** The deterministic mask certificate (Theorem 5.5) certifies that pruning a removed component $R$ from a layer $A = S + R$ decreases the elastic-net objective whenever the data-path budget satisfies

$$B_T(R) \;<\; \lambda_2\|R\|_F^2 + \lambda_1\|R\|_1, \tag{13}$$

where $B_T(R)$ is the data-path budget (see the companion methodology paper for the full per-architecture audit).

In the Gaussian sufficient-condition setting of Corollary 5.15, the fitted MP edge $\sigma_+(\ell)$ is the leading-order term of a high-probability upper bound on $B_T(R)$ under iid weights, motivating the $\sigma_+$-budget allocation used in Hybrid Magnitude–SER. The prune–restore extension (Theorem 5.6) splits the removed mass into a kept reservoir $Q$ and a finally-removed component $P$, so that restoring $Q$ can improve the certificate upper bound under the conditions of Corollary 5.7. This mechanism underlies CAST's free-restoration step; see the companion methodology paper.

The empirical certification audit (see the companion methodology paper) reports three connections between these results and the multi-architecture numerics: (i) within-arch Pearson correlation between summed $B_T$ and post-projection top-1 ($r = +0.91$ pooled, $+0.51$ to $+0.92$ per-arch, with ConvNeXt-Base $+0.51$ and ResNet50.tv $+0.59$ at the low end); (ii) zero violations of the operator-norm proxy across 282 audited cells (the proxy $\hat{B}_T = \|R\|_{\text{op}}\overline{\|\psi\|_\infty}$ is not itself an upper bound on $B_T(R)$, so this is an empirical observation rather than a verification of the literal Lemma 5.2 inequality); (iii) architecture-stratified $\sigma_+ \to B$-proxy correlation, strongest in transformers ($r$ up to $+0.83$ for DeiT-Tiny) and weakest for depthwise architectures ($r = -0.29$ for ConvNeXt-Base). Detailed per-architecture comparisons against Table 2 and Figure 2 are reported in the companion methodology paper.

# C Proofs of the perturbation lemmas

**Lemma C.1** (Slack removal)**.** *Let $X_N$ and $Y_N$ be real random variables. Suppose that for every fixed $\eta > 0$,*

$$\big(X_N - Y_N - \eta\big)_+ \to 0 \qquad \text{in probability.}$$

*Then*

$$\big(X_N - Y_N\big)_+ \to 0 \qquad \text{in probability.}$$

*Equivalently, if $X_N \le Y_N + \eta + o_{\mathbb{P}}(1)$ for every fixed $\eta > 0$, then $X_N \le Y_N + o_{\mathbb{P}}(1)$.*

*Proof.* Fix $\delta > 0$ and set $\eta = \delta/2$. Then

$$\mathbb{P}(X_N - Y_N > \delta) \le \mathbb{P}(X_N - Y_N - \eta > \delta - \eta) \le \mathbb{P}\big((X_N - Y_N - \eta)_+ > \delta/2\big),$$

and the last probability tends to 0 by assumption. □

## C.1 Proofs of the deterministic pruning certificates

*Proof of Lemma 5.2.* For each $s \in T$, the definition of $L_s$ gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \le \|\psi_2(A_\theta\psi_1(s)) - \psi_2(A_1\psi_1(s))\|_\infty \le (1-\theta)L_s\|R\psi_1(s)\|_\infty.$$

For the single-sample cross-entropy

$$\ell_s(z) := -z_{C(s)} + \log\left(\sum_{j=1}^{K} e^{z_j}\right),$$

we have $\nabla\ell_s(z) = \rho(z) - e_{C(s)}$, hence $\|\nabla\ell_s(z)\|_1 \le 2$. Therefore $\ell_s$ is 2-Lipschitz with respect to $\|\cdot\|_\infty$. Averaging over $s \in T$ yields

$$|\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1)| \le \frac{2}{|T|}\sum_{s\in T}(1-\theta)L_s\|R\psi_1(s)\|_\infty = (1-\theta)B_T(R).$$

□

*Proof of Corollary 5.3.* The logit perturbation bound in the proof of Lemma 5.2 gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \le \eta_s(\theta).$$

If the predicted label changes, then the original top logit and some competing logit must be separated by at most $2\eta_s(\theta)$; otherwise every competing logit remains below the original top logit after perturbation. This proves the prediction-margin claim.

For accuracy, correctness is determined by the sign of the true-label margin. The true class logit can move by at most $\eta_s(\theta)$, and the maximum competing logit can move by at most $\eta_s(\theta)$, so the true-label margin can change by at most $2\eta_s(\theta)$. Therefore the sign of $\Delta^{\text{true}}_{A_1}(s)$ can change only if $|\Delta^{\text{true}}_{A_1}(s)| \le 2\eta_s(\theta)$. Counting such samples gives the displayed bound. □

*Proof of Theorem 5.4.* By Lemma 5.2,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \le (1-\theta)B_T(R).$$

For the Frobenius term,

$$\|S + \theta R\|_F^2 - \|S + R\|_F^2 = -(1-\theta^2)\|R\|_F^2 - 2(1-\theta)\langle S, R\rangle_F.$$

Multiplying this identity by $\lambda_2$ and adding the cross-entropy bound gives the displayed path inequality. The $\eta$-version follows from

$$-2\lambda_2(1-\theta)\langle S, R\rangle_F \le 2\lambda_2(1-\theta)|\langle S, R\rangle_F|.$$

The full-removal sufficient condition is the case $\theta = 0$. □

*Proof of Theorem 5.5.* By Lemma 5.2,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \le (1-\theta)B_T(R).$$

Since $S \odot R = 0$,

$$\|S + \theta R\|_F^2 = \|S\|_F^2 + \theta^2\|R\|_F^2, \qquad \|S + \theta R\|_1 = \|S\|_1 + \theta\|R\|_1.$$

Similarly,

$$\|A_1\|_F^2 = \|S\|_F^2 + \|R\|_F^2, \qquad \|A_1\|_1 = \|S\|_1 + \|R\|_1.$$

Substituting these identities into $\mathcal{J}(A_\theta) - \mathcal{J}(A_1)$ gives

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \le (1-\theta)B_T(R) - \lambda_2(1-\theta^2)\|R\|_F^2 - \lambda_1(1-\theta)\|R\|_1.$$

Taking $\theta = 0$ proves the full-pruning statement and the sufficient decrease condition follows immediately. □

*Proof of Theorem 5.6.* Apply Theorem 5.5 to the mask decomposition

$$A = (S + Q) + P.$$

The pairwise disjoint-support assumption implies $(S + Q) \odot P = 0$, so the theorem applies with structured part $S + Q$ and removed part $P$. This gives the displayed path inequality and the full prune–restore statement by taking $\theta = 0$. The sufficient decrease condition follows immediately. □

*Proof of Corollary 5.7.* The all-pruned upper bound is Theorem 5.5 applied to

$$A = S + R, \qquad R = Q + P.$$

The restored upper bound is Theorem 5.6. Since $Q \odot P = 0$,

$$\|R\|_F^2 = \|Q\|_F^2 + \|P\|_F^2, \qquad \|R\|_1 = \|Q\|_1 + \|P\|_1.$$

Therefore the restored upper bound is smaller than the all-pruned upper bound exactly when

$$B_T(P) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 < B_T(R) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1,$$

which is equivalent to

$$B_T(R) - B_T(P) > \lambda_2\|Q\|_F^2 + \lambda_1\|Q\|_1.$$

□

*Proof of Corollary 5.8.* The cross-entropy part is the same deterministic bound as in Lemma 5.2. Since $S \odot R = 0$, each coordinate satisfies either $S_{ij} = 0$ or $R_{ij} = 0$. Hence

$$(S_{ij} + \theta R_{ij})^2 = (S_{ij})^2 + \theta^2 (R_{ij})^2, \qquad |S_{ij} + \theta R_{ij}| = |S_{ij}| + \theta |R_{ij}|.$$

Multiplying by the nonnegative weights and summing over $(i, j)$ gives

$$\sum_{i,j} \gamma_{2,ij}\big((A_\theta)_{ij}^2 - (A_1)_{ij}^2\big) = -(1-\theta^2)\sum_{i,j} \gamma_{2,ij} R_{ij}^2,$$

and

$$\sum_{i,j} \gamma_{1,ij}\big(|(A_\theta)_{ij}| - |(A_1)_{ij}|\big) = -(1-\theta)\sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Combining these two identities with the cross-entropy bound proves the displayed inequality. The full-pruning sufficient condition is the case $\theta = 0$. □

*Proof of Corollary 5.9.* Set $\theta = 1 - h$ in Corollary 5.8 and divide by $h > 0$. This gives

$$\frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \le B_T(R) - (2-h)\sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Taking $h \downarrow 0$ and using one-sided stationarity yields

$$-\varepsilon \le B_T(R) - 2\sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|,$$

which is the directional claim. If the pathwise lower bound holds for every $\theta$, combine it with Corollary 5.8 and divide by $1 - \theta$ to obtain the finite-$\theta$ inequality. □

*Proof of Theorem 5.10.* Set $\theta = 1 - h$ in Theorem 5.5 and divide by $h > 0$. This gives

$$\frac{\mathcal{J}(A_{1-h}) - \mathcal{J}(A_1)}{h} \le B_T(R) - \lambda_2(2-h)\|R\|_F^2 - \lambda_1\|R\|_1.$$

Taking $h \downarrow 0$ and using the one-sided stationarity hypothesis yields

$$-\varepsilon \le B_T(R) - 2\lambda_2\|R\|_F^2 - \lambda_1\|R\|_1,$$

which is the first claim. If the stronger pathwise stationarity condition holds for every $\theta$, then combining it with Theorem 5.5 and dividing by $1 - \theta$ gives

$$\lambda_2(1+\theta)\|R\|_F^2 + \lambda_1\|R\|_1 \le B_T(R) + \varepsilon$$

for every $\theta < 1$. □

*Proof of Theorem 5.11.* Apply Theorem 5.5 with $\theta = 0$ at each pruning step in the network obtained after the previous $j-1$ pruning operations. If $\alpha^{(j)}$ denotes the parameters after step $j$, then

$$\mathcal{J}(\alpha^{(j)}) - \mathcal{J}(\alpha^{(j-1)}) \le -\lambda_{2,j}\|R_j\|_F^2 - \lambda_{1,j}\|R_j\|_1 + B_j(R_j).$$

Summing over $j = 1, \dots, M$ telescopes the left-hand side and proves the displayed multi-layer bound. $\square$

*Proof of Corollary 5.12.* For each $s \in T$, telescoping the logits gives

$$\|z^{(M)}(s) - z^{(0)}(s)\|_\infty \le \sum_{j=1}^{M} \|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \le \eta_s^{\mathrm{tot}}.$$

Since single-sample cross-entropy is 2-Lipschitz in the logit $\ell_\infty$ norm, averaging over $T$ gives the displayed cross-entropy bound. The top predicted logit and the largest competing logit can each move by at most $\eta_s^{\mathrm{tot}}$, so the prediction can change only if the original prediction margin is at most $2\eta_s^{\mathrm{tot}}$. The same argument applied to the true-label logit and the largest competing true-label logit shows that the sign of the true-label margin can change only if $|\Delta_0^{\mathrm{true}}(s)| \le 2\eta_s^{\mathrm{tot}}$. Counting those samples proves the accuracy bound. $\square$

*Proof of Theorem 5.13.* Fix $h \in (0, 1]$, and let $\alpha_h^{(j)}$ be the network after the first $j$ fractional pruning steps, with $\alpha_h^{(0)} = \alpha^{(0)}$ and $\alpha_h^{(M)} = \alpha_h$. At step $j$, apply the one-layer deterministic certificate with $\theta = 1 - h$ in the intermediate network $\alpha_h^{(j-1)}$. By the assumed validity of the fractional budget $B_j(R_j)$,

$$\mathcal{J}(\alpha_h^{(j)}) - \mathcal{J}(\alpha_h^{(j-1)}) \le hB_j(R_j) - \lambda_{2,j}(2h - h^2)\|R_j\|_F^2 - \lambda_{1,j}h\|R_j\|_1.$$

Summing over $j = 1, \dots, M$ gives

$$\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)}) \le h\sum_{j=1}^{M} B_j(R_j) - h(2-h)\sum_{j=1}^{M} \lambda_{2,j}\|R_j\|_F^2 - h\sum_{j=1}^{M} \lambda_{1,j}\|R_j\|_1.$$

After division by $h$, the one-sided directional stationarity assumption and the limit $h \downarrow 0$ imply

$$-\varepsilon \le \sum_{j=1}^{M} B_j(R_j) - 2\sum_{j=1}^{M} \lambda_{2,j}\|R_j\|_F^2 - \sum_{j=1}^{M} \lambda_{1,j}\|R_j\|_1,$$

which is the first claim. If the stronger pathwise lower bound holds for every $h \in [0, 1]$, combine it with the displayed upper bound and divide by $h$ to obtain the finite-$h$ inequality. $\square$

*Proof of Corollary 5.14.* Condition on the vectors $v_s = \psi_1(s)$ and on the bounds $L_s$. For each $s \in T$ and row $i$,

$$\langle r_i, v_s\rangle \sim N\left(0, \frac{g}{n}\|v_s\|_2^2\right).$$

Thus

$$\mathbb{P}\left(|\langle r_i, v_s\rangle| > \|v_s\|_2\sqrt{\frac{2g}{n}\log\frac{2n|T|}{\delta}}\right) \le \frac{\delta}{n|T|}.$$

A union bound over $i = 1, \dots, n$ and $s \in T$ gives, with probability at least $1 - \delta$,

$$\|Rv_s\|_\infty \le \|v_s\|_2\sqrt{\frac{2g}{n}\log\frac{2n|T|}{\delta}} \qquad \forall s \in T.$$

Substitution into (8) proves the claim. $\square$

*Proof of Corollary 5.15.* For an $n \times m$ Gaussian matrix with entries $N(0, g/n)$, the nominal MP singular-value edge is

$$\sigma_{+,R} = \sqrt{g}\left(1 + \sqrt{\frac{m}{n}}\right).$$

Equivalently,

$$\sqrt{g} = \frac{\sigma_{+,R}}{1 + \sqrt{m/n}}.$$

Substituting this identity into the bound of Corollary 5.14 gives the displayed inequality. □

*Proof of Corollary 5.17.* Apply Corollary 5.14 at pruning step $j$ with failure probability $\delta_j$, conditional on the intermediate vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. This gives the displayed bound for $B_j(R_j)$ with conditional probability at least $1 - \delta_j$. A union bound over $j = 1, \ldots, M$ gives simultaneous validity with probability at least $1 - \sum_j \delta_j \geq 1 - \delta$. Substituting these simultaneous budget bounds into Theorem 5.11 proves the objective inequality. □

*Proof of Corollary 5.18.* Apply Corollary 5.16 at pruning step $j$ with failure probability $\delta_j$, conditional on the intermediate vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. This gives the displayed bound for $B_j(R_j)$ with conditional probability at least $1 - \delta_j$. A union bound over the $M$ pruning steps gives simultaneous validity with probability at least $1 - \sum_j \delta_j \geq 1 - \delta$. The final objective statement is obtained by substituting those bounds into Theorem 5.11. □

*Proof of Corollary 5.16.* Condition on the variables with respect to which $v_s$ and $L_s$ are measurable. Apply the assumed tail bound with

$$t_s := \|v_s\|_2 \sqrt{\frac{g}{cn} \log \frac{2n|T|}{\delta}}.$$

Then

$$\mathbb{P}(|\langle r_i, v_s\rangle| > t_s \mid v_s) \leq \frac{\delta}{n|T|}.$$

The same union bound over rows and samples gives the simultaneous bound

$$\|Rv_s\|_\infty \leq t_s \qquad \forall s \in T$$

with probability at least $1 - \delta$. Substituting into (8) proves the corollary. □

### C.2 Proof of Lemma A.13

*Proof.* Let

$$\nu := \lambda(W_1 s).$$

Since $\lambda$ is either absolute value or ReLU, it is coordinatewise 1-Lipschitz and satisfies $\|\nu\|_2 \leq \|W_1 s\|_2$. Write

$$X(s, \alpha_{W_2}) = \lambda\big(W_3 \lambda((S_2 + R_2)\nu)\big), \qquad X(s, \alpha_{S_2}) = \lambda\big(W_3 \lambda(S_2 \nu)\big).$$

Using the 1-Lipschitz property of $\lambda$ and $\|Av\|_\infty \leq \|A\|_\infty \|v\|_\infty$, we obtain

$$\big\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\big\|_\infty \leq \|W_3\|_\infty \, \|R_2 \nu\|_\infty.$$

So it suffices to bound $\|R_2 \nu\|_\infty$.

By Assumption A.3, $R_2$ is independent of $W_1$, hence independent of $\nu$. Conditioned on $\nu$, each coordinate $(R_2\nu)_j$ is centered Gaussian with variance

$$\mathrm{Var}\big((R_2\nu)_j \mid \nu\big) = \frac{g(t)}{N}\|\nu\|_2^2 \le \frac{\|W_1 s\|_2^2}{N}.$$

Set

$$u_N := 2\sqrt{2}\,\|W_1 s\|_2\sqrt{\frac{\log(2N)}{N}}.$$

If $\|W_1 s\|_2 = 0$, then $\nu = 0$ and the claim is immediate. Otherwise, the elementary Gaussian tail bound gives, conditionally on $\nu$,

$$\mathbb{P}(|(R_2\nu)_j| > u_N \mid \nu) \le 2\exp\left(-\frac{u_N^2}{2\|W_1 s\|_2^2/N}\right) = 2(2N)^{-4}.$$

A union bound over $j = 1, \dots, N$ gives

$$\mathbb{P}\left(\|R_2\nu\|_\infty > \; 2\sqrt{2}\,\|W_1 s\|_2\sqrt{\frac{\log(2N)}{N}}\,\middle|\, \nu\right) \le \frac{1}{8N^3} \le \frac{1}{N}.$$

The same bound holds unconditionally after averaging over $\nu$. Multiplying by $\|W_3\|_\infty$ yields the desired estimate. $\square$

### C.3 Proof of Lemma 4.9

*Proof.* Let $u := \psi_1(s)$. Since

$$X(s, \alpha_W) = \psi_2((R+S)u), \qquad X(s, \alpha_S) = \psi_2(Su),$$

Assumption 4.1 gives

$$\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \le L_{\psi_2}(s)\,\|Ru\|_\infty.$$

Applying Assumption 4.3 with $v = u$ and then averaging over the conditioning variables gives

$$\mathbb{P}\big(\|Ru\|_\infty > d_1(N)J(u)\big) \le d_2(N).$$

Therefore

$$\mathbb{P}(\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \le d_1(N)J(\psi_1(s))\,L_{\psi_2}(s)) \ge 1 - d_2(N),$$

which is the claim. $\square$

### C.4 Justification of Example 4.6

The standard sub-Gaussian tail bound gives absolute constants $c, C > 0$ such that, conditionally on $S, \psi_1, \psi_2$,

$$\mathbb{P}(|\langle r_j, v\rangle| > u \mid S, \psi_1, \psi_2) \le 2\exp\left(-c\frac{Nu^2}{\kappa^2\|v\|_2^2}\right).$$

Taking

$$u = C\kappa\|v\|_2\sqrt{\frac{\log(2N/\delta)}{N}}$$

with $C$ large enough and applying a union bound over $j = 1, \dots, N$ yields the displayed $\ell_\infty$ concentration bound.

## C.5 Proof of Lemma A.15

*Proof.* For $s \in T$, let $X(\alpha, s) \in \mathbb{R}^K$ denote the vector of pre-softmax logits. By Lemma A.13, for each fixed $s \in T$,

$$\mathbb{P}\Big(\|X(\alpha_{S_2}, s) - X(\alpha_{W_2}, s)\|_\infty \leq \tau_N(s)\Big) \geq 1 - \frac{1}{N},$$

where

$$\tau_N(s) = \; 2\sqrt{2}\,\|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2}\,a(N, s).$$

Now define the single-sample cross-entropy

$$\ell_s(z) := -\log\left(\frac{e^{z_{C(s)}}}{\sum_{j=1}^K e^{z_j}}\right) = -z_{C(s)} + \log\left(\sum_{j=1}^K e^{z_j}\right), \qquad z \in \mathbb{R}^K.$$

Its gradient is $\nabla \ell_s(z) = \rho(z) - e_{C(s)}$, so $\|\nabla \ell_s(z)\|_1 \leq 2$. Hence $\ell_s$ is 2-Lipschitz with respect to $\|\cdot\|_\infty$:

$$|\ell_s(z) - \ell_s(z')| \leq 2\|z - z'\|_\infty.$$

Applying this with $z = X(\alpha_{S_2}, s)$ and $z' = X(\alpha_{W_2}, s)$, we get

$$|\ell_s(X(\alpha_{S_2}, s)) - \ell_s(X(\alpha_{W_2}, s))| \leq 4\sqrt{2}\,a(N, s)$$

on the high-probability event from Lemma A.13. Since $T$ is finite, a union bound over $s \in T$ yields

$$\mathbb{P}\left(|L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|}\sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N},$$

which proves the lemma. $\square$

## C.6 Proof of Theorem A.16

*Proof.* For each $s \in T_0$, set $\nu_s = \lambda(W_1 s)$. Conditional on all $\nu_s$'s, each coordinate $(R_2 \nu_s)_j$ is centered Gaussian with variance

$$\frac{g(t)}{N}\|\nu_s\|_2^2 \leq \frac{g(t)}{N}\|W_1 s\|_2^2.$$

Therefore, for fixed $s$ and $j$,

$$\mathbb{P}\left(|(R_2\nu_s)_j| > \|W_1 s\|_2 \sqrt{\frac{2g(t)\log(2N|T_0|/\delta)}{N}} \;\middle|\; (\nu_u)_{u \in T_0}\right) \leq \frac{\delta}{N|T_0|}.$$

A union bound over $j = 1, \ldots, N$ and $s \in T_0$ gives simultaneous control of all $\|R_2\nu_s\|_\infty$'s with conditional probability at least $1 - \delta$, hence also unconditionally. Multiplying by $\|W_3\|_\infty$ and using the vector estimate in the proof of Lemma A.13 proves the displayed logit bound. The cross-entropy bound follows from the 2-Lipschitz property of $\ell_s$ with respect to $\|\cdot\|_\infty$.

For the regularized statement, use the exact identity

$$L(\alpha_{S_2}) - L(\alpha_{W_2}) = L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2}) - \lambda_{\text{reg}}\|R_2\|_F^2 - 2\lambda_{\text{reg}}\langle S_2, R_2\rangle_F.$$

Conditional on $S_2$,

$$\langle S_2, R_2\rangle_F \sim \mathcal{N}\left(0, \frac{g(t)}{N}\|S_2\|_F^2\right),$$

and Assumption A.6 gives

$$\mathbb{P}\big(|\langle S_2, R_2\rangle_F| > \eta \mid S_2\big) \leq 2e^{-N\eta^2/(2B_N^2)}.$$

Intersecting the logit event with $\{|\langle S_2, R_2\rangle_F| \leq \eta\}$ gives the claimed probability and inequality. $\square$

### C.7 Proof of Theorem A.20

*Proof.* Condition on $S_N$ when it is random; the remaining probability is over $R_N$. If $g_N = 0$, then $R_N = 0$ almost surely and the claim is immediate. Assume $g_N > 0$. For every entry,

$$\mathbb{P}(|R_{ij}| > t_{N,\delta}) \le 2\exp\left(-\frac{Nt_{N,\delta}^2}{2g_N}\right) = \frac{\delta}{NM_N},$$

and a union bound gives

$$\mathbb{P}\left(\max_{i,j} |R_{ij}| \le t_{N,\delta}\right) \ge 1 - \delta.$$

On this event, if $(i,j) \notin \operatorname{supp}(S_N)$, then $|W_{ij}| = |R_{ij}| \le t_{N,\delta}$, so $\mathcal{H}_{t_{N,\delta}}$ removes that entry. Thus thresholding creates no false positives. If $|(S_N)_{ij}| > 2t_{N,\delta}$, then

$$|W_{ij}| \ge |(S_N)_{ij}| - |R_{ij}| > 2t_{N,\delta} - t_{N,\delta} = t_{N,\delta},$$

so thresholding keeps that entry. This proves the support sandwich and the missed-support inclusion. On the beta-min event $\mathcal{B}_{N,\delta}$, every element of $\operatorname{supp}(S_N)$ has magnitude larger than $2t_{N,\delta}$, so the support sandwich collapses to exact support recovery. Unconditionally, the exact-recovery failure probability is at most $\delta + \mathbb{P}(\mathcal{B}_{N,\delta}^c) \le \delta + \gamma$. □

# D Low-rank deformations and proofs of the main theorems

### D.1 External random-matrix inputs quoted in the manuscript

The Marchenko–Pastur theorem in Theorem 2.2 and the square Gaussian finite-rank deformation theorem in Theorem D.1 are used as standard random-matrix inputs. The statements are classical; see, for example, [32, 62] for Marchenko–Pastur theory and [2] for the Benaych-Georges–Nadakuditi outlier theorem. All paper-specific consequences stated after these inputs are proved below.

### D.2 Finite-rank deformations of random matrices

**Theorem D.1** (Benaych-Georges–Nadakuditi, square Gaussian case [2])**.** *Let $W_N = R_N + S_N$, where $R_N$ is an $N \times N$ random matrix with i.i.d. $N(0, 1/N)$ entries and $S_N$ is deterministic, or independent of $R_N$, with fixed rank $r$ and nonzero singular values $\sigma_1 > \cdots > \sigma_r > 0$ independent of $N$. Then the $i$-th largest singular value $s_i(W_N)$ converges almost surely to*

$$s_i(W_N) \to \begin{cases} \sigma_i + \sigma_i^{-1}, & \sigma_i > 1, \\ 2, & \sigma_i \le 1. \end{cases}$$

*In particular, subcritical spikes merge with the bulk edge* 2.

*Proof location.* This is a classical cited input; see the source note in Appendix D.1. The variance-$g$ and varying-scale consequences are proved immediately below.

**Corollary D.2** (Square Gaussian case with variance $g/N$)**.** *Let $W_N = R_N + S_N$, where $R_N$ is an $N \times N$ random matrix with i.i.d. $N(0, g/N)$ entries for some fixed $g > 0$, and $S_N$ is deterministic, or independent of $R_N$, with fixed rank $r$ and distinct nonzero singular values $\sigma_1 > \cdots > \sigma_r > 0$. Then*

$$s_i(W_N) \to \begin{cases} \sigma_i + g/\sigma_i, & \sigma_i > \sqrt{g}, \\ 2\sqrt{g}, & \sigma_i \le \sqrt{g}. \end{cases}$$

*Proof.* Write

$$W_N = \sqrt{g}\,(\widetilde{R}_N + \widetilde{S}_N), \qquad \widetilde{R}_N := R_N/\sqrt{g}, \qquad \widetilde{S}_N := S_N/\sqrt{g}.$$

Then $\widetilde{R}_N$ has i.i.d. $N(0, 1/N)$ entries, and the nonzero singular values of $\widetilde{S}_N$ are $\sigma_i/\sqrt{g}$. Apply Theorem D.1 and multiply by $\sqrt{g}$. $\square$

**Theorem D.3** (Square Gaussian outlier map with varying spike and noise scales)**.** *Let $W_N = R_N + S_N$, where $R_N$ is an $N \times N$ random matrix with i.i.d. $N(0, g_N/N)$ entries and $g_N \to g_\infty \in [0, \infty)$. Assume $S_N$ has fixed rank $r$. Write its nonzero singular values in decreasing order,*

$$\sigma_1(S_N) \geq \cdots \geq \sigma_r(S_N) > 0,$$

*and assume that for each fixed $1 \leq i \leq r$,*

$$\sigma_i(S_N) \to \bar{\sigma}_i \in (0, \infty) \qquad \text{in probability}$$

*where $\bar{\sigma}_1 \geq \cdots \geq \bar{\sigma}_r > 0$, and that $S_N$ is independent of $R_N$. Then for each fixed $1 \leq i \leq r$,*

$$s_i(W_N) \to \begin{cases} \bar{\sigma}_i + g_\infty/\bar{\sigma}_i, & \bar{\sigma}_i > \sqrt{g_\infty}, \\ 2\sqrt{g_\infty}, & \bar{\sigma}_i \leq \sqrt{g_\infty}. \end{cases} \qquad \text{in probability.}$$

*Moreover, when $g_\infty > 0$, the usual BGN alignment conclusions hold for simple separated spikes: supercritical spikes have asymptotically nonzero projection onto the corresponding signal directions, while subcritical spikes have vanishing projection. For repeated or clustered spikes, these statements are understood as subspace-alignment statements. When $g_\infty = 0$, any separated spike cluster has convergent spectral projectors under the Wedin gap condition $\|R_N\|_{\mathrm{op}}/\mathrm{gap}_N \to 0$.*

*Proof.* Condition on $S_N$. First suppose $g_\infty > 0$. Then $g_N > 0$ for all large $N$, and after dividing by $\sqrt{g_N}$ one reduces to the unit-variance square Gaussian finite-rank deformation theorem of Benaych-Georges and Nadakuditi [2]. On events whose probability tends to one, the conditioned spike values are arbitrarily close to their deterministic limits; the BGN outlier map is continuous away from the threshold, and at the threshold both branches give the edge. Thus the conditional deterministic-spike result and Slutsky's theorem give the displayed convergence in probability. The limiting outlier map in the unit-variance scaling is

$$\theta \mapsto \theta + \theta^{-1},$$

so in the original scaling it becomes

$$\sigma \mapsto \sigma + \frac{g_\infty}{\sigma},$$

with critical threshold $\sqrt{g_\infty}$ and edge $2\sqrt{g_\infty}$. The singular-vector and spike-subspace alignment statements in the positive-noise case are exactly the corresponding supercritical/subcritical BGN conclusions.

Now suppose $g_\infty = 0$. Write $R_N = \sqrt{g_N}\, Z_N$, where $Z_N$ has i.i.d. $N(0, 1/N)$ entries. By the standard Gaussian operator-norm bound, $\|Z_N\|_{\mathrm{op}} = O_{\mathbb{P}}(1)$, hence

$$\|R_N\|_{\mathrm{op}} \to 0 \qquad \text{in probability.}$$

Weyl's singular-value inequality gives, for each fixed $i$,

$$|s_i(W_N) - s_i(S_N)| \leq \|R_N\|_{\mathrm{op}} \to 0 \qquad \text{in probability.}$$

Since $s_i(S_N) \to \bar{\sigma}_i$, we obtain $s_i(W_N) \to \bar{\sigma}_i$, which is the asserted formula with $g_\infty = 0$. The stated projector convergence in the separated-cluster case follows from Wedin's theorem. $\square$

**Theorem D.4** (Abstract inverse-$D$ spike recovery)**.** *Let $W_N = R_N + S_N$. Assume that for each fixed $1 \le i \le r$, the $i$-th nonzero singular value of $S_N$ is $\theta_i(N) > 0$. Assume further that there exist deterministic quantities $b_+(N)$, $\theta_c(N)$, and a strictly decreasing map*

$$D_N : (b_+(N), \infty) \to (0, D_N(b_+(N)^+))$$

*with the following properties:*

1. *on supercritical spike sequences, meaning $\mathbb{P}(\theta_i(N) > \theta_c(N)) \to 1$, one has*

$$s_i(W_N) - \rho_N(\theta_i(N)) \to 0 \qquad \textit{in probability,}$$

*where*

$$\rho_N(\theta) := D_N^{-1}\left(\frac{1}{\theta^2}\right);$$

2. *on subcritical spike sequences, meaning $\mathbb{P}(\theta_i(N) \le \theta_c(N)) \to 1$, one has*

$$s_i(W_N) - b_+(N) \to 0 \qquad \textit{in probability;}$$

3. *in the supercritical case, the singular vectors associated with $s_i(W_N)$ have asymptotically nonvanishing projection onto the corresponding singular directions of $S_N$;*

4. *for each fixed supercritical spike under consideration, $\mathbb{P}(\theta_i(N) > \theta_c(N)) \to 1$, $\theta_i(N) \to \bar{\theta}_i \in (0, \infty)$, and there exist deterministic intervals*

$$I_{i,N} \subset (b_+(N), \infty)$$

*and deterministic constants $L_{i,N} < \infty$ such that $D_N$ is $L_{i,N}$-Lipschitz on $I_{i,N}$ and*

$$\begin{aligned} &\mathbb{P}\big(\rho_N(\theta_i(N)) \in I_{i,N}\big) \to 1, \qquad \mathbb{P}\big(s_i(W_N) \in I_{i,N}\big) \to 1, \\ &L_{i,N}\big|s_i(W_N) - \rho_N(\theta_i(N))\big| \to 0 \quad \textit{in probability.} \end{aligned}$$

*Define the inverse-$D$ estimator on the event $s_i(W_N) > b_+(N)$, whose probability tends to one for every fixed supercritical spike under consideration, by*

$$\widehat{\theta}_i^{(D)}(N) := D_N(s_i(W_N))^{-1/2},$$

*and define it arbitrarily off this event. Then for every fixed supercritical spike,*

$$\widehat{\theta}_i^{(D)}(N) - \theta_i(N) \to 0 \qquad \textit{in probability.}$$

*Proof.* On the event $\{s_i(W_N) \in I_{i,N},\ \rho_N(\theta_i(N)) \in I_{i,N}\}$, whose probability tends to one, the Lipschitz assumption gives

$$\big|D_N(s_i(W_N)) - D_N(\rho_N(\theta_i(N)))\big| \le L_{i,N}\big|s_i(W_N) - \rho_N(\theta_i(N))\big|.$$

Therefore

$$D_N(s_i(W_N)) = D_N(\rho_N(\theta_i(N))) + o_{\mathbb{P}}(1) = \frac{1}{\theta_i(N)^2} + o_{\mathbb{P}}(1).$$

Since $\theta_i(N) \to \bar{\theta}_i \in (0, \infty)$, the quantity $\theta_i(N)^{-2}$ stays in a compact subinterval of $(0, \infty)$ for all large $N$. Applying the continuous map $x \mapsto x^{-1/2}$ on that compact interval yields

$$\widehat{\theta}_i^{(D)}(N) - \theta_i(N) \to 0 \qquad \text{in probability.}$$

This is the inverse-outlier-map viewpoint underlying $D$-transform shrinkage methods such as OptShrink [38]. $\square$

## D.3 Proof of Proposition A.30

*Proof.* Part (i) is the MP law on the covariance scale together with the pushforward $x \mapsto \sqrt{x}$ to the singular-value scale. Because $\nu_{R_N}$ averages over only $\min\{N, M_N\}$ singular values, when $c > 1$ one removes the covariance-side zero atom of the full MP law before pushing forward and renormalizes the remaining mass. Part (ii) follows from the rank inequality for empirical spectral distributions: if $A, B$ are matrices of the same dimensions, then

$$\sup_x |\nu_A((-\infty, x]) - \nu_B((-\infty, x])| \leq \frac{\operatorname{rank}(A - B)}{\min\{N, M_N\}}.$$

Applying this with $A = W_N$, $B = R_N$, and $A - B = S_N$ gives the claimed bound. Part (iii) is the rectangular outlier principle encoded by the inverse-$D$ transform in Theorem D.4; the square Gaussian BBP statement follows as the special case $c = 1$. $\square$

## D.4 Proofs of Theorem A.18, Corollary A.19, Corollary A.14, and Theorem A.21 (Gaussian instances)

*Proof of Theorem A.18.* Conditional on $S_2(t)$, Assumption A.3 gives

$$\langle S_2(t), R_2(t)\rangle_F = \operatorname{tr}(S_2(t)^\top R_2(t)) \sim \mathcal{N}\left(0, \frac{g(t)}{N}\|S_2(t)\|_F^2\right).$$

Since $\|S_2(t)\|_F \leq B_N$ and $g(t) \leq 1$, for every $\eta > 0$,

$$\mathbb{P}(|\langle S_2(t), R_2(t)\rangle_F| > \eta \,|\, S_2(t)) \leq 2e^{-N\eta^2/(2B_N^2)}. \tag{14}$$

Hence

$$\langle S_2(t), R_2(t)\rangle_F \to 0 \qquad \text{in probability as } N \to \infty. \tag{15}$$

Write

$$L(\alpha) = L_{\cancel{\text{reg}}}(\alpha) + \lambda_{\text{reg}} \sum_{i=1}^{L} \|W_i\|_F^2.$$

If $W_2 = R_2 + S_2$ and we replace $W_2$ by $S_2$, then

$$\begin{aligned} L(\alpha_{S_2}) - L(\alpha_{W_2}) &= \big(L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})\big) - \lambda_{\text{reg}}\big(\|W_2\|_F^2 - \|S_2\|_F^2\big) \\ &= \big(L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})\big) - \lambda_{\text{reg}}\|R_2\|_F^2 - 2\lambda_{\text{reg}}\langle S_2, R_2\rangle_F. \end{aligned} \tag{16}$$

By Lemma A.15,

$$\mathbb{P}\left(|L_{\cancel{\text{reg}}}(\alpha_{S_2}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|}\sum_{s\in T} a(N, s)\right) \geq 1 - \frac{|T|}{N}.$$

Intersecting this event with $|\langle S_2, R_2\rangle_F| \leq \eta$, whose probability is bounded below by $1 - 2e^{-N\eta^2/(2B_N^2)}$, yields the finite-$N$ inequality in the theorem. Letting $N \to \infty$ and using (15) gives the asymptotic expansion. $\square$

*Proof of Corollary A.19.* Let $W_2(\vartheta) = S_2 + \vartheta R_2$. Then

$$\begin{aligned} L(\alpha_{W_2(\vartheta)}) - L(\alpha_{W_2}) = &\big(L_{\cancel{\text{reg}}}(\alpha_{W_2(\vartheta)}) - L_{\cancel{\text{reg}}}(\alpha_{W_2})\big) \\ &- \lambda_{\text{reg}}\Big((1 - \vartheta^2)\|R_2\|_F^2 + 2(1 - \vartheta)\langle S_2, R_2\rangle_F\Big). \end{aligned} \tag{17}$$

For each $s \in T$, write $\nu_s = \lambda(W_1 s)$. Let $E_N^{\mathrm{logit}}$ be the event on which

$$\|W_3\|_\infty \|R_2 \nu_s\|_\infty \le \tau_N(s) \qquad \forall s \in T.$$

By the same samplewise Gaussian estimate as in Lemma A.13 and a union bound over $T$,

$$\mathbb{P}(E_N^{\mathrm{logit}}) \ge 1 - \frac{|T|}{N}.$$

On $E_N^{\mathrm{logit}}$, for every $\vartheta \in [0,1]$ and every $s \in T$,

$$\left\|X(s, \alpha_{W_2(\vartheta)}) - X(s, \alpha_{W_2})\right\|_\infty \le (1-\vartheta)\|W_3\|_\infty \|R_2 \nu_s\|_\infty \le (1-\vartheta)\tau_N(s).$$

The 2-Lipschitz property of the cross-entropy therefore gives, on the same event,

$$\left|L_{\cancel{\mathrm{reg}}}(\alpha_{W_2(\vartheta)}) - L_{\cancel{\mathrm{reg}}}(\alpha_{W_2})\right| \le \frac{4\sqrt{2}(1-\vartheta)}{|T|} \sum_{s \in T} a(N, s)$$

for every $\vartheta \in [0,1]$. Combining $E_N^{\mathrm{logit}}$ with $|\langle S_2, R_2 \rangle_F| \le \eta$ proves the finite-$N$ bound. The asymptotic expansion follows from (17) and (15). Since the finite-$N$ event is uniform in $\vartheta$, the same expansion also holds for deterministic sequences $\vartheta_N \in [0,1]$. □

*Proof of Corollary A.14.* By Lemma A.13, for each fixed $s \in T'$,

$$\mathbb{P}(\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\|_\infty \le \tau_N(s)) \ge 1 - \frac{1}{N}.$$

A union bound over $s \in T'$ gives simultaneous vector control with probability at least $1 - \frac{|T'|}{N}$. On that event,

$$|\delta X(s, \alpha_{W_2}) - \delta X(s, \alpha_{S_2})| \le 2\tau_N(s) \qquad \forall s \in T'.$$

Therefore labels can only change on samples with $|\delta X(s, \alpha_{W_2})| \le 2\tau_N(s)$, which yields the claim. □

*Proof of Theorem A.21.* Set

$$b_N := \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a_*(N, s) \to 0.$$

Choose a deterministic sequence $\eta_N \downarrow 0$ such that

$$\frac{N \eta_N^2}{B_N^2} \to \infty;$$

such a sequence exists because $B_N = o(\sqrt{N})$. By Corollary A.19 applied at the limit point, there exists an event $E_N$ with $\mathbb{P}(E_N) \to 1$ such that on $E_N$, for every $\vartheta \in [0,1]$,

$$L(\alpha_\vartheta^*(N)) - L(\alpha^*(N)) \le -\lambda_{\mathrm{reg}}(1-\vartheta^2)\|R_2^*(N)\|_F^2 + (1-\vartheta) b_N + 2\lambda_{\mathrm{reg}}(1-\vartheta)\eta_N$$

Write $\vartheta = 1 - h$. Dividing by $h > 0$ gives

$$\frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \le -\lambda_{\mathrm{reg}}(2-h)\|R_2^*(N)\|_F^2 + b_N + 2\lambda_{\mathrm{reg}}\eta_N$$

for every $h \in (0,1]$. On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h \downarrow 0}$ yields

$$2\lambda_{\mathrm{reg}}\|R_2^*(N)\|_F^2 \le b_N + \epsilon_N + 2\lambda_{\mathrm{reg}}\eta_N.$$

Therefore

$$\|R_2^*(N)\|_F^2 \le \frac{b_N + \epsilon_N}{2\lambda_{\text{reg}}} + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \to 0$, this proves the quantitative estimate in the theorem. The vanishing-in-probability statement follows immediately when $b_N + \epsilon_N \to 0$ in probability.

Now assume the local-path convergence hypothesis from Definition A.12. Fix $\epsilon > 0$. For each fixed $N$,

$$\|R_2(t, N)\|_F \to \|R_2^*(N)\|_F \qquad \text{in probability as } t \to \infty$$

by the continuous mapping theorem. Since $R_2^*(N)$ is Gaussian (possibly degenerate if $g_*(N) = 0$), the random variable $\|R_2^*(N)\|_F$ has no atom at any positive $\epsilon$. Hence

$$\mathbb{P}\big(\|R_2(t, N)\|_F \le \epsilon\big) \to \mathbb{P}\big(\|R_2^*(N)\|_F \le \epsilon\big) \qquad \text{as } t \to \infty.$$

Letting $N \to \infty$ and using the limit-point statement proves the theorem. $\square$

*Proof of Corollary A.23.* Suppose, toward a contradiction, that there exist $\eta_0 > 0$ and a subsequence such that $g_*(N)N \ge \eta_0$. Since

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i,j=1}^{N} Z_{ij}^2, \qquad Z_{ij} \overset{\text{i.i.d.}}{\sim} N(0, 1),$$

we have

$$\frac{\|R_2^*(N)\|_F^2}{g_*(N)N} = \frac{1}{N^2} \sum_{i,j=1}^{N} Z_{ij}^2 \to 1 \qquad \text{in probability.}$$

Therefore

$$\mathbb{P}\big(\|R_2^*(N)\|_F^2 \ge \eta_0/2\big) \to 1,$$

contradicting Theorem A.21. Hence $g_*(N)N \to 0$. The edge statement follows from the square-case MP singular-value edge $2\sqrt{g_*(N)}$. In the rectangular $N \times M_N$ case,

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i=1}^{N} \sum_{j=1}^{M_N} Z_{ij}^2, \qquad \frac{1}{NM_N} \sum_{i,j} Z_{ij}^2 \to 1$$

in probability. Thus $\|R_2^*(N)\|_F \to 0$ forces $g_*(N)M_N \to 0$, and if $M_N/N \to c \in (0, \infty)$ this is equivalent to $g_*(N)N \to 0$. $\square$

*Proof of Corollary A.24.* Since $\|R_2^*(N)\|_{\text{op}} \le \|R_2^*(N)\|_F$, Theorem A.21 immediately gives

$$\|R_2^*(N)\|_{\text{op}} \to 0 \qquad \text{in probability.}$$

This proves (i).

For (ii), singular-value Weyl perturbation gives

$$\big|s_i(W_2^*(N)) - s_i(S_2^*(N))\big| \le \|R_2^*(N)\|_{\text{op}} \qquad \forall i.$$

Hence for each fixed $i \ge 1$,

$$\big|s_i(W_2^*(N)) - \sigma_i^*(N)\big| \le \|R_2^*(N)\|_{\text{op}} \to 0$$

in probability.

Let $S^*_{2,\le k_N}(N)$ denote the best rank-$k_N$ approximation of $S^*_2(N)$. By the Eckart–Young theorem,

$$\sum_{i>k_N} s_i(W^*_2(N))^2 = \inf_{\mathrm{rank}(A)\le k_N} \|W^*_2(N) - A\|_F^2 \le \|W^*_2(N) - S^*_{2,\le k_N}(N)\|_F^2$$

$$= \|R^*_2(N) + (S^*_2(N) - S^*_{2,\le k_N}(N))\|_F^2 \le 2\|R^*_2(N)\|_F^2 + 2\sum_{i>k_N} \sigma^*_i(N)^2,$$

which proves (iii) using Theorem A.21 and Assumption A.8.

For (iv), fix $\epsilon > 0$. Since $\|W^*_2(N)\|_F \le \|S^*_2(N)\|_F + \|R^*_2(N)\|_F$, we have

$$\nu_{W^*_2(N)}((\epsilon,\infty)) \le \frac{\|W^*_2(N)\|_F^2}{N\epsilon^2} \le \frac{2B_N^2 + 2\|R^*_2(N)\|_F^2}{N\epsilon^2},$$

which converges to 0 in probability because $B_N = o(\sqrt{N})$ and $\|R^*_2(N)\|_F \to 0$ in probability. The final claim follows from (ii). □

*Proof of Corollary A.25.* By Corollary A.24(i),

$$\|R^*_2(N)\|_{\mathrm{op}} \to 0 \qquad \text{in probability.}$$

Hence the event

$$F_N := \left\{ \|R^*_2(N)\|_{\mathrm{op}} \le \frac{\delta_k(N)}{4} \right\}$$

satisfies $\mathbb{P}(F_N) \to 1$.

On $F_N$, Weyl's singular-value inequality gives

$$s_k(W^*_2(N)) \ge \sigma^*_k(N) - \|R^*_2(N)\|_{\mathrm{op}},$$

while

$$s_{k+1}(W^*_2(N)) \le \sigma^*_{k+1}(N) + \|R^*_2(N)\|_{\mathrm{op}},$$

with the convention $\sigma^*_j(N) = 0$ beyond the rank of $S^*_2(N)$. Therefore,

$$s_k(W^*_2(N)) - s_{k+1}(W^*_2(N)) \ge \delta_k(N) - 2\|R^*_2(N)\|_{\mathrm{op}} \ge \frac{\delta_k(N)}{2}$$

on $F_N$. Thus on $F_N$, the top-$k$ singular subspace of $W^*_2(N)$ is separated from the rest by a positive gap.

Applying Wedin's $\sin\Theta$ theorem for singular subspaces [63, 53] to

$$W^*_2(N) = S^*_2(N) + R^*_2(N),$$

with target rank $k$, yields a universal constant $C > 0$ such that on $F_N$,

$$\|\widehat{P}_{U,k}(N) - P_{U_*,k}(N)\|_{\mathrm{op}} \le C\frac{\|R^*_2(N)\|_{\mathrm{op}}}{\delta_k(N)}, \qquad \|\widehat{P}_{V,k}(N) - P_{V_*,k}(N)\|_{\mathrm{op}} \le C\frac{\|R^*_2(N)\|_{\mathrm{op}}}{\delta_k(N)}.$$

Since $\|R^*_2(N)\|_{\mathrm{op}}/\delta_k(N) \to 0$ in probability and $\mathbb{P}(F_N) \to 1$, both projector differences converge to 0 in probability. □

*Proof of Corollary A.27.* By Corollary A.23,

$$g_*(N)N \to 0,$$

hence $g_*(N) \to 0$. Since $\sigma_i^*(N) \to \bar{\sigma}_i > 0$ in probability for each $1 \le i \le k$,

$$\mathbb{P}\big(\sigma_i^*(N) > \sqrt{g_*(N)}\big) \to 1,$$

which proves (i).

By Corollary A.24(i),

$$\|R_2^*(N)\|_{\mathrm{op}} \to 0 \qquad \text{in probability.}$$

Hence Weyl's inequality gives

$$s_i(W_2^*(N)) - \sigma_i^*(N) \to 0 \qquad \text{in probability}$$

for each fixed $1 \le i \le k$. Since $\sigma_i^*(N) \to \bar{\sigma}_i > 0$ and $2\sqrt{g_*(N)} \to 0$, these singular values lie above the Gaussian bulk edge $2\sqrt{g_*(N)}$ with probability tending to one. Equivalently, in the exact finite-rank Gaussian deformation language of Theorem D.3, the persistent spikes are eventually supercritical. If the additional gap condition in (ii) holds for a spike, Wedin's theorem applied to $W_2^*(N) = S_2^*(N) + R_2^*(N)$ gives, for the associated left and right rank-one projectors,

$$\|\widehat{P}_i(N) - P_i^*(N)\|_{\mathrm{op}} \le C\frac{\|R_2^*(N)\|_{\mathrm{op}}}{\Delta_i(N)} \to 0 \qquad \text{in probability,}$$

and similarly for the right projector. This proves the strengthened projector statement in (ii).

Set $y_N = s_i(W_2^*(N))$. The preceding Weyl estimate gives $y_N - \sigma_i^*(N) \to 0$ in probability, hence $y_N \to \bar{\sigma}_i > 0$. Also $g_*(N) \to 0$, so the event $y_N^2 > 4g_*(N)$ has probability tending to one. On this event,

$$0 \le y_N - \widehat{\sigma}_i^{\mathrm{BBP}}(N) = \frac{1}{2}\left(y_N - \sqrt{y_N^2 - 4g_*(N)}\right) = \frac{2g_*(N)}{y_N + \sqrt{y_N^2 - 4g_*(N)}} \to 0$$

in probability. Therefore

$$\widehat{\sigma}_i^{\mathrm{BBP}}(N) - \sigma_i^*(N) \to 0 \qquad \text{in probability,}$$

which proves (iii). □

## D.5 Proofs of the abstract-generalization corollaries and theorems

The results below are the abstract additive-perturbation analogues of the Gaussian-instance proofs of Appendix D. Each is obtained by substituting the abstract logit-perturbation lemma (Lemma 4.9) for the Gaussian logit perturbation bound (Lemma A.13) and applying the abstract Assumptions 4.1–4.7 in place of Gaussian Assumptions A.3–A.6.

*Proof of Theorem 4.10.* For $s \in T$, let $X(\alpha, s)$ denote the vector of pre-softmax logits. By Lemma 4.9, for each fixed $s \in T$,

$$\mathbb{P}\big(\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \le a_\phi(N, s)\big) \ge 1 - d_2(N),$$

where $a_\phi(N, s) = h_\phi(s)d_1(N)$. The single-sample cross-entropy is 2-Lipschitz in $\|\cdot\|_\infty$, so on this event $|\ell_s(X(\alpha_S, s)) - \ell_s(X(\alpha_W, s))| \le 2a_\phi(N, s)$. A union bound over $s \in T$ gives

$$\mathbb{P}\left(|L_{\cancel{\mathrm{reg}}}(\alpha_S) - L_{\cancel{\mathrm{reg}}}(\alpha_W)| \le \tfrac{2}{|T|}\sum_{s \in T} a_\phi(N, s)\right) \ge 1 - |T|d_2(N).$$

Since $a_\phi(N,s) \to 0$, $L_{\cancel{\mathrm{reg}}}(\alpha_S) - L_{\cancel{\mathrm{reg}}}(\alpha_W) \to 0$ in probability. Combined with the exact regularizer identity $\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R\rangle_F = \|R\|_F^2 + o_{\mathbb{P}}(1)$, this gives $L(\alpha_S) = L(\alpha_W) - \lambda_{\mathrm{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1)$. □

*Proof of Corollary 4.13.* Let $E_N^{\mathrm{logit}}$ be the samplewise logit event $\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \le a_\phi(N,s)$ for all $s \in T$. By Lemma 4.9 and a union bound, $\mathbb{P}(E_N^{\mathrm{logit}}) \ge 1 - |T| d_2(N)$. On this event the samplewise perturbation argument with $(1-\vartheta)R$ in place of $R$ gives, for every $\vartheta \in [0,1]$,

$$|L_{\cancel{\mathrm{reg}}}(\alpha_{W(\vartheta)}) - L_{\cancel{\mathrm{reg}}}(\alpha_W)| \le (1-\vartheta)\tfrac{2}{|T|}\sum_{s\in T} a_\phi(N,s).$$

Let $E_N^{\mathrm{cross}}(\eta) := \{|\langle S, R\rangle_F| \le \eta\}$ and $E_{N,\eta} := E_N^{\mathrm{logit}} \cap E_N^{\mathrm{cross}}(\eta)$. Then $\mathbb{P}(E_{N,\eta}) \ge 1 - |T| d_2(N) - \mathbb{P}(|\langle S, R\rangle_F| > \eta)$. On $E_{N,\eta}$, the exact regularizer identity $\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1-\vartheta^2)\|R\|_F^2 + 2(1-\vartheta)\langle S, R\rangle_F$ implies

$$L(\alpha_{W(\vartheta)}) \le L(\alpha_W) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R\|_F^2 + (1-\vartheta)\tfrac{2}{|T|}\sum_{s\in T} a_\phi(N,s) + 2\lambda_{\mathrm{reg}}(1-\vartheta)\eta$$

for every $\vartheta \in [0,1]$, which is exactly the claim. □

*Proof of Corollary 4.18.* Since $\theta_c(N) \to 0$ and $\sigma_i^*(N) \to \bar\sigma_i > 0$ in probability, $\mathbb{P}(\sigma_i^*(N) > \theta_c(N)) \to 1$ for each $1 \le i \le k$, which proves (i). On the same high-probability event, the assumed BGN-type deformation regime places the $i$-th spike above the critical threshold, so the corresponding singular value cluster of $W^*(N)$ lies above the bulk edge. If the gap condition in (ii) holds for a spike, Wedin's theorem gives $\|\widehat{P}_i(N) - P_i^*(N)\|_{\mathrm{op}} \le C\|R^*(N)\|_{\mathrm{op}}/\Delta_i(N) \to 0$ in probability, for the left rank-one projectors (and similarly for the right projectors), proving (ii). Applying Theorem D.4 with $\theta_i(N) = \sigma_i^*(N)$ then gives $\widehat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \to 0$ in probability for each fixed $1 \le i \le k$, which is (iii). □

*Proof of Corollary 4.11.* By Lemma 4.9 and a union bound over $s \in T_N$, with probability at least $1 - |T_N| d_2(N)$,

$$\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \le a_\phi(N,s) \qquad \forall s \in T_N.$$

On this event, the 2-Lipschitz property of cross-entropy gives

$$\left|L_{\cancel{\mathrm{reg}},T_N}(\alpha_S) - L_{\cancel{\mathrm{reg}},T_N}(\alpha_W)\right| \le \frac{2}{|T_N|}\sum_{s\in T_N} a_\phi(N,s).$$

The right-hand side is $o_{\mathbb{P}}(1)$, and the event has probability tending to one because $|T_N| d_2(N) \to 0$. The regularized identity

$$\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R\rangle_F$$

and the cross-term hypothesis then yield

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\mathrm{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1).$$

□

*Proof of Corollary 4.12.* Set $W(\vartheta) = S + \vartheta R$. Repeating the argument above with perturbation $(1-\vartheta)R$ in place of $R$, we get

$$L_{\cancel{\mathrm{reg}}}(\alpha_{W(\vartheta)}) - L_{\cancel{\mathrm{reg}}}(\alpha_W) = o_{\mathbb{P}}(1).$$

Also,

$$\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1-\vartheta^2)\|R\|_F^2 + 2(1-\vartheta)\langle S, R\rangle_F = (1-\vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

Hence

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\mathrm{reg}}(1-\vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

The same argument, or the uniform finite-$N$ pathwise bound of Corollary 4.13, gives the statement along any deterministic sequence $\vartheta_N \in [0,1]$. □

*Proof of Theorem 4.14.* Order the pruned layers as $\mathcal{P} = \{\ell_1, \ldots, \ell_m\}$. Let $\alpha^{(j)}$ denote the network parameters after the first $j$ replacements, so $\alpha^{(0)}$ is the original network and $\alpha^{(m)} = \alpha^{(|\mathcal{P}|)}$. By assumption, at stage $j$ the hypotheses of Theorem 4.10 apply to layer $\ell_j$, hence

$$L(\alpha^{(j)}) = L(\alpha^{(j-1)}) - \lambda_{\mathrm{reg}}\|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1).$$

Summing these identities over $j = 1, \ldots, m$ and using that $m = |\mathcal{P}|$ is fixed gives

$$L(\alpha^{(m)}) = L(\alpha^{(0)}) - \lambda_{\mathrm{reg}} \sum_{j=1}^{m} \|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1),$$

which is the claimed finite-layer telescoping formula.

For the growing-layer version, use the finite-$N$ single-layer bound on the intersection $\bigcap_{\ell\in\mathcal{P}_N} E_{\ell,N}$, whose probability tends to one by the assumed summability of the failure probabilities. Summing the exact loss identities over the ordered layers gives the accumulated regularization decrease $-\lambda_{\mathrm{reg}} \sum_{\ell\in\mathcal{P}_N} \|R_\ell\|_F^2$ plus the sum of the cross-entropy perturbation errors and the accumulated cross term. The two summability hypotheses make these remainders $o_{\mathbb{P}}(1)$, yielding the same formula with $\mathcal{P}_N$. □

*Proof of Theorem 4.16.* Let $b_N := (2/|T|)\sum_{s\in T} a_\phi(N,s) \to 0$. Since $\langle S^*(N), R^*(N)\rangle_F = o_{\mathbb{P}}(1)$, choose a deterministic sequence $\eta_N \downarrow 0$ such that $\mathbb{P}(|\langle S^*(N), R^*(N)\rangle_F| > \eta_N) \to 0$. By Corollary 4.13 applied at the limit point, there is an event $E_N$ with $\mathbb{P}(E_N) \to 1$ such that on $E_N$, for every $\vartheta \in [0,1]$,

$$L(\alpha^*_\vartheta(N)) - L(\alpha^*(N)) \le -\lambda_{\mathrm{reg}}(1-\vartheta^2)\|R^*(N)\|_F^2 + (1-\vartheta)b_N + 2\lambda_{\mathrm{reg}}(1-\vartheta)\eta_N.$$

Write $\vartheta = 1-h$ and divide by $h > 0$:

$$\frac{L(\alpha^*_{1-h}(N)) - L(\alpha^*(N))}{h} \le -\lambda_{\mathrm{reg}}(2-h)\|R^*(N)\|_F^2 + b_N + 2\lambda_{\mathrm{reg}}\eta_N \quad \forall h \in (0,1].$$

On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h\downarrow 0}$ yields $2\lambda_{\mathrm{reg}}\|R^*(N)\|_F^2 \le b_N + \epsilon_N + 2\lambda_{\mathrm{reg}}\eta_N$, so

$$\|R^*(N)\|_F^2 \le (b_N + \epsilon_N)/(2\lambda_{\mathrm{reg}}) + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \to 0$, this proves the quantitative estimate; the vanishing-in-probability statement follows whenever $b_N + \epsilon_N \to 0$ in probability. The pathwise conclusion under the convergence/continuity hypotheses follows because $\|R(t,N)\|_F \to \|R^*(N)\|_F$ in probability and $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$. □

# E Algorithms for RMT-based pruning diagnostics

Detailed specifications of the BEMA edge estimator, spectral-fit test, and cycle-based pruning algorithms for the classical RMT baseline are provided in the companion *methodology and extended-numerics* paper, `cast_2e/methodology.pdf` (LaTeX source: `cast_2e/methodology.tex`). The repository root is `https://github.com/yspennstate/RMT_based_pruning_in_deep_learning`.

# F Spectral Edge Budgeting and hybrid RMT pruning protocols

Detailed specifications of the operational SEB and SER protocols, four-regime strategy, layer-type extension, SEB final comparison, Haar-SV preprocessing sweep, Hybrid Magnitude–SER protocol, and checkpoint validation ledgers used in Table 2 are provided in the companion *methodology and extended-numerics* paper, `cast_2e/methodology.pdf` (LaTeX source: `cast_2e/methodology.tex`). The repository root is `https://github.com/yspennstate/RMT_based_pruning_in_deep_learning`.

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**Algorithm 1 Spectral Edge Reallocation (SER):** prune–restore step with MP-budget allocation. Used inside Hybrid Magnitude–SER for $s \geq 0.25$. The full protocol with hyperparameter selection is in the companion methodology paper.

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**Require:** Trained network with weights $\{W_\ell\}_{\ell=1}^{L}$; target sparsity $s \in (0,1)$; over-prune ratio $\rho \in (1, 1+\epsilon]$; layerwise MP edge signal $\{\sigma_+(\ell)\}_{\ell=1}^{L}$; budget-allocation parameters $(\beta, s_{\mathrm{d}}, p)$; short FT schedule $\mathcal{T}_{\mathrm{FT}}$.

1: **function** SER_PRUNE_RESTORE(network, $s$, $\rho$, $\sigma_+$, $\beta, s_{\mathrm{d}}, p$)

2: Compute layerwise weights $w_\ell(s) = \max(0.1, 1 + \beta\, d(s)\, z_\ell)$ where $z_\ell = (\sigma_+(\ell) - \bar{\sigma}_+)/\mathrm{sd}(\sigma_+)$ and $d(s) = (1 - s/s_{\mathrm{d}})_+^p$. ▷ Section 3

3: Score each parameter $|W_\ell|_{ij}/w_\ell(s)$ and prune the smallest $\rho \cdot s$ fraction globally. ▷ Over-prune

4: Define the reservoir $\mathcal{R} = \{(\ell, i, j) : (W_\ell)_{ij} \text{ pruned}\}$.

5: With kept weights frozen, run a short selection phase (one ImageNet-1k epoch at the same optimiser/LR/batch as the eventual fine-tune) that allows only the reservoir entries to receive gradients. Record the absolute movement $|\Delta_{ij}^{(\ell)}|$ of each reservoir entry from its pre-selection value.

6: Rank reservoir entries by $|\Delta_{ij}^{(\ell)}| \cdot \rho_\ell(s)$ where $\rho_\ell(s) = w_\ell(s)$ is the layer's $\sigma_+$-derived budget weight from Step 1, recomputed at the post-prune sparsity. ▷ RMT-ranked reinsertion

7: Restore the top entries until realised post-restoration sparsity equals $s$.

8: Freeze the resulting zero mask $M$ on all layers. Run $\mathcal{T}_{\mathrm{FT}}$ (one fine-tuning epoch in our schedule) with $W_\ell \leftarrow W_\ell \odot M$ re-applied at each step.

9: **return** restored network and final mask $M$.

10: **end function**

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# G CAST: certificate-aware 2:4 + ToMe conversion

Detailed specifications of the original CAST 2:4 + ToMe pipeline, the generalised CAST-2E $k{:}n$ pipeline, permutation alignment, and inline-FT runner are provided in the companion *methodology and extended-numerics* paper, `cast_2e/methodology.pdf` (LaTeX source: `cast_2e/methodology.tex`). The repository root is `https://github.com/yspennstate/RMT_based_pruning_in_deep_learning`.

**Justification of Algorithm 2.** The cost $c_{i,g}(m)$ is the squared $\ell^2$ form of the removed-mass contribution to the output channel $i$ of layer $\ell$, evaluated on the calibration set. By Lemma 5.2, a small $\|R\,\psi_1\|_\infty$ bounds $|\Delta\,\mathrm{CE}|$, so minimising the $\ell^2$ surrogate of $R = W - W'$ on $\mathcal{C}$ minimises a pointwise upper bound on the certificate budget under the data distribution captured by $C_g$. The free-restoration step materialises the slot-value rule of Theorem 5.6: slots that the unstructured mask had zeroed but the chosen $k{:}n$ pattern keeps are filled from the dense checkpoint at zero sparse-execution FLOP cost, strictly improving the certificate bound. The $\alpha_{\mathrm{ser}}$ prior pulls the chosen pattern towards the RMT-allocated SER mask used in the ViT-B experiments.

**Algorithm 2 CAST $k{:}n$ cert-aware projection with free restoration.** Reads an unstructured Hybrid Magnitude–SER source checkpoint and produces a row-wise $k{:}n$ structured mask. Stage 1 only; Stage 2 (frozen-mask distillation) is a standard 3-epoch label-smoothed distillation with mask re-zeroing every step. The $\alpha_{\text{ser}}$ Hamming prior, the Cin permutation alignment, and the sampled large-$n$ variant are documented in the companion methodology paper.

**Require:** Linear layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with $d_{\text{in}} = nG$; SER source mask $M^{\text{SER}} \in \{0,1\}^{d_{\text{out}} \times d_{\text{in}}}$ and SER weight $W^{\text{SER}}$; dense pretrained $W^{\text{dense}}$; calibration set $\mathcal{C}$ (256 ImageNet train images); $k{:}n$ pattern; flag `free_restoration`; $\alpha_{\text{ser}} \geq 0$.

**function** CAST_PROJECT_KN($W$, $M^{\text{SER}}$, $W^{\text{SER}}$, $W^{\text{dense}}$, $\mathcal{C}$, $k$, $n$, $\alpha_{\text{ser}}$, `free_restoration`)
  Reshape rows of $W$ into $G$ contiguous $n$-tuples; let $W_{i,g,k}$ index the $k$-th weight of group $g$ in output row $i$.
  For each input-channel group $g$, capture activations $h_g(x) \in \mathbb{R}^n$ over $\mathcal{C}$ and form within-group covariance $C_g = \frac{1}{|\mathcal{C}|} \sum_{x \in \mathcal{C}} h_g(x) h_g(x)^\top$.
  **for** each output row $i = 1, \ldots, d_{\text{out}}$ and group $g = 1, \ldots, G$ in parallel **do**
    **for** each candidate keep pattern $m \in \{0,1\}^n$ with $|m|_0 = k$ **do**
      Define source tuple $u_{i,g,k} = \begin{cases} W^{\text{SER}}_{i,g,k} & \text{if } M^{\text{SER}}_{i,g,k} = 1, \\ W^{\text{dense}}_{i,g,k} & \text{if } M^{\text{SER}}_{i,g,k} = 0 \text{ and } \texttt{free\_restoration}, \\ 0 & \text{otherwise.} \end{cases}$
      Materialise retained tuple $v_{i,g,k}(m) = m_k \cdot u_{i,g,k}$.
      Compute removed tuple $r_{i,g}(m) = (1 - m) \odot u_{i,g}$ ▷ entries not retained
      Compute cost $c_{i,g}(m) = r_{i,g}(m)^\top C_g \, r_{i,g}(m) + \alpha_{\text{ser}} \, \bar{c}_g \, |m - M^{\text{SER}}_{i,g,:}|_1$,
  where $\bar{c}_g = \text{mean}_{i,k} \, u^2_{i,g,k} \cdot \mathbb{E}_x[h_{g,k}(x)^2]$ normalises the prior to the cost magnitude ▷ $\bar{c}_g$ normalises the prior to the cost magnitude
    **end for**
    $m^*_{i,g} \leftarrow \arg\min_{m \in \mathcal{M}_{k:n}} c_{i,g}(m)$.
    Set the new weight tuple $W'_{i,g,:} \leftarrow v_{i,g,:}(m^*_{i,g})$.
  **end for**
  **return** projected layer $W'$ and the per-row $k{:}n$ mask $M^{\text{CAST}}_{i,g,k} = m^*_{i,g}[k]$.
**end function**